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वार्षिक प्रतिवेदन Annual Report 2024

NAARM



भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
राजेन्द्रनगर, हैदराबाद - 500 030, तेलंगाना, भारत
ICAR-National Academy of Agricultural Research Management
Rajendranagar, Hyderabad-500 030, Telangana, India





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ICAR-National Academy of Agricultural Research Management

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MESSAGE FROM THE DIRECTOR'S DESK



It gives me immense pleasure to present the Annual Report 2024 of ICAR-NAARM, highlighting our continuous strides in strengthening agricultural research, education and extension systems management in India. The year 2024 was marked by significant achievements across our core areas of capacity building, academic and research programs, policy initiatives and innovation-led entrepreneurship development. The Academy remained committed to enhancing leadership and governance in the National Agricultural Research, Education and Extension Systems (NAREES). The Academy conducted 49 capacity building programs, benefiting 5,421 professionals, including scientists, faculty, academicians and administrators. With a keen focus on technology-driven learning, we successfully introduced Massive Open Online Courses (MOOCs), ensuring wider outreach and accessibility. Academy's a-IDEA has contributed immensely to developing and nurturing innovations and entrepreneurship in the field of agriculture and allied sectors. Farmer Outreach Programs organized under SCSP (19 no. benefitting 1,452 farmers) by the Academy focused on skill development and capacity building for farmers across India. Our research and consultancy initiatives have been pivotal in shaping policies and institutional frameworks for agricultural development. We

have actively contributed to national-level committees, policy dialogues, and strategic management initiatives aimed at fostering an innovation-driven agricultural ecosystem. The Agri-Business Incubation (ABI) Centre at NAARM has continued to empower startups, promoting entrepreneurship and job creation in the agri-sector.

The year 2024 also witnessed creation/enriching of infrastructural facilities at the Academy, including Technology Bhawan, Digital Course Content Development Unit, Farmers' Hostel etc. which will further facilitate research and capacity-building programs. As we look ahead, our focus will remain on driving excellence, sustainability and digital transformation in agricultural research and education. Our multifaceted initiatives, ranging from research and startup incubation to transformative capacity-building programs, have positively impacted a broad spectrum of stakeholders, including students, scientists, faculty, academicians, entrepreneurs and farmers.

I take this opportunity to extend my sincere gratitude to our esteemed partners, stakeholders, faculty members, and staff for their unwavering dedication and contributions. I place on record my special appreciation and thanks to the Annual Report Editorial Committee for its painstaking effort in compilation, collation and meticulously editing relevant information covering all activities and accomplishments of the Academy.

R.C. Agrawal
Director, ICAR-NAARM

Date: 05-05-2025
Place: Hyderabad

EXECUTIVE SUMMARY

In 2024, ICAR-National Academy of Agricultural Research Management continued to play a pivotal role in advancing individual and institutional competencies for effective agricultural research, education and extension management by conducting an array of capacity building, educational, policy research, entrepreneurship development and outreach programs, underscoring its commitment to bolstering India's agri-management landscape. The Academy continued to operate under its guiding vision of enabling the National Agricultural Research, Education and Extension Systems (NAREES) to adapt to changes through innovation. With six specialized divisions (Agribusiness Management, Education Systems Management, Human Resource Management, Extension Systems Management, Information & Communication Management, and Research Systems Management), the Academy systematically addressed the capacity-building, policy advocacy and leadership needs of the agricultural ecosystem.

The Academy conducted 49 capacity-building programs (CBPs) in 2024, showcasing a wide array of initiatives designed to enhance skills and knowledge in the agricultural and allied sectors. The programs are categorized into Executive Development Programs (EDPs), Management Development Programs (MDPs), Foundation Courses, Need-based and Sponsored Programs, Training Workshops/ Seminars/ Webinars/ Conferences, and Massive Open Online Courses (MOOCs). These initiatives collectively engaged 5,421 participants, focusing on areas such as academic leadership, digital innovation, agri-startups and advanced technologies like AI,

data analytics, and geospatial tools. The range of CBPs organized underscores Academy's commitment to fostering innovation, leadership and technical expertise in agriculture through diverse, practical and policy-aligned training programs, which not only addressed immediate skill gaps but also aimed to build long-term capacity for sustainable agricultural development. Twenty students' sensitization programs on 'Entrepreneurship development in agriculture and allied sectors' organized by the Academy could create awareness among 12,713 students. When it comes to educational and academic achievements, the 16th Batch of Post Graduate Diploma in Management (Agri-Business Management) was admitted. The passed-out students secured 100% placements with an impressive rise in Average CTC (> ₹10 LPA), reinforcing the Academy's reputation for producing industry-ready agribusiness professionals. Thirty students brought awards/ accolades to the Academy by participating in national competitions.

In 2024, the Academy undertook 30 Institutional research projects and 9 extra-mural projects addressing areas like market reforms, climate resilience, gender mainstreaming in agriculture and agripreneurship. Notable accomplishments include the development of frameworks for gender inclusion in agriculture, establishment of Agri-Business Incubation (ABI) Centers, and analyzing the socio-economic impacts of SC Sub-Plan initiatives. Research in MOOCs, dairy education and digital content production and delivery showed potential in improving learning outcomes, while environmental studies on methane emissions and precision agriculture

could potentially promote sustainable farming. The Academy published 74 research papers, 8 policy/ strategy papers, 6 books and 7 copyrights during 2024 ensuring extensive knowledge dissemination. Through its a-IDEA Technology Business Incubator, the Academy facilitated the growth of agribusiness startups across different domains, hosting programs like Agri Udaan and Aggnite, and conducting FPO immersion programs. The National Conference on Agricultural Startups witnessed participation of over 1,000 stakeholders, including startups, FPOs, and investors.

The Academy organized multiple Farmer Outreach Programs under SCSP benefiting 1,452 farmers. Key initiatives included skill development in horticulture, integrated farming, scientific beekeeping, poultry management, dairy farming, and organic agriculture. Special programs empowered rural women through tailoring, fabric printing, and eco-friendly bag-making. Additionally, demonstrations and distributions of agricultural inputs and farm toolkits were conducted. These programs, held across various Krishi Vigyan Kendras (KVKs)

and institutions, aimed to enhance farmers' knowledge, self-employment opportunities and sustainable agricultural practices, contributing to improvement in rural livelihoods and agrarian resilience.

The Academy established new facilities – one Farmers Hostel, two Custom Hiring Centers, and an upgraded Multi-Input Digital Content Production Unit to enhance effectiveness of knowledge delivery. The inauguration of “Technology Bhawan” added to its infrastructure for advancing agricultural innovation. The Academy promoted sustainable agricultural practices through initiatives like capacity-building programs for natural farming, organic agriculture and waste management techniques. It also celebrated key events such as National Science Day and Kisan Diwas, fostering engagement with students, farmers and stakeholders. With a forward-looking approach, the Academy continues to drive innovation, foster inclusivity and strengthen capacities within NARES, setting upward benchmarks for excellence in agricultural management.

1

OVERVIEW OF THE ACADEMY

NAARM – the National Academy of Agricultural Research Management – located in Rajendranagar, Hyderabad, Telangana was established by the Indian Council of Agricultural Research (ICAR) in 1976 with the following vision, mission and mandates.

ICAR-NAARM operates through six specialized divisions namely (i) Agribusiness Management (ABM) Division, (ii) Education Systems Management (ESM) Division, (iii) Human Resource Management (HRM) Division, (iv) Extension Systems Management (XSM) Division, (v) Information and Communication Management (ICM) Division and (vi) Research Systems Management (RSM) Division. These divisions collectively strengthen India's agricultural research, education and extension ecosystem. ABM Division conducts research and capacity building programs (CBPs) on market

reforms, agricultural enterprise development, and agripreneurship, supporting stakeholders from smallholder farmers to agri-startups. The Division also supports Post Graduate Diploma in Management (Agri Business Management) in a greater way. ESM Division conducts research and CBPs on advancing agricultural higher education through improvement in teaching-learning continuum, digital transformation including application of artificial intelligence in teaching and national education policy implementation. HRM Division operates research projects and CBPs focused on leadership and human resource development for education and research management positions including women leadership. XSM Division implements research projects and CBPs aimed at strengthening agricultural extension services and bridging the researcher-farmer gap. ICM Division runs research projects and CBPs leveraging advanced technologies

Vision

A global knowledge institution enabling National Agricultural Research and Education system (NARES) adapt to change through continuous innovation

Mission

To enhance leadership, governance and innovation capacities of NARES through capacity strengthening, education, research, consultancy and policy support

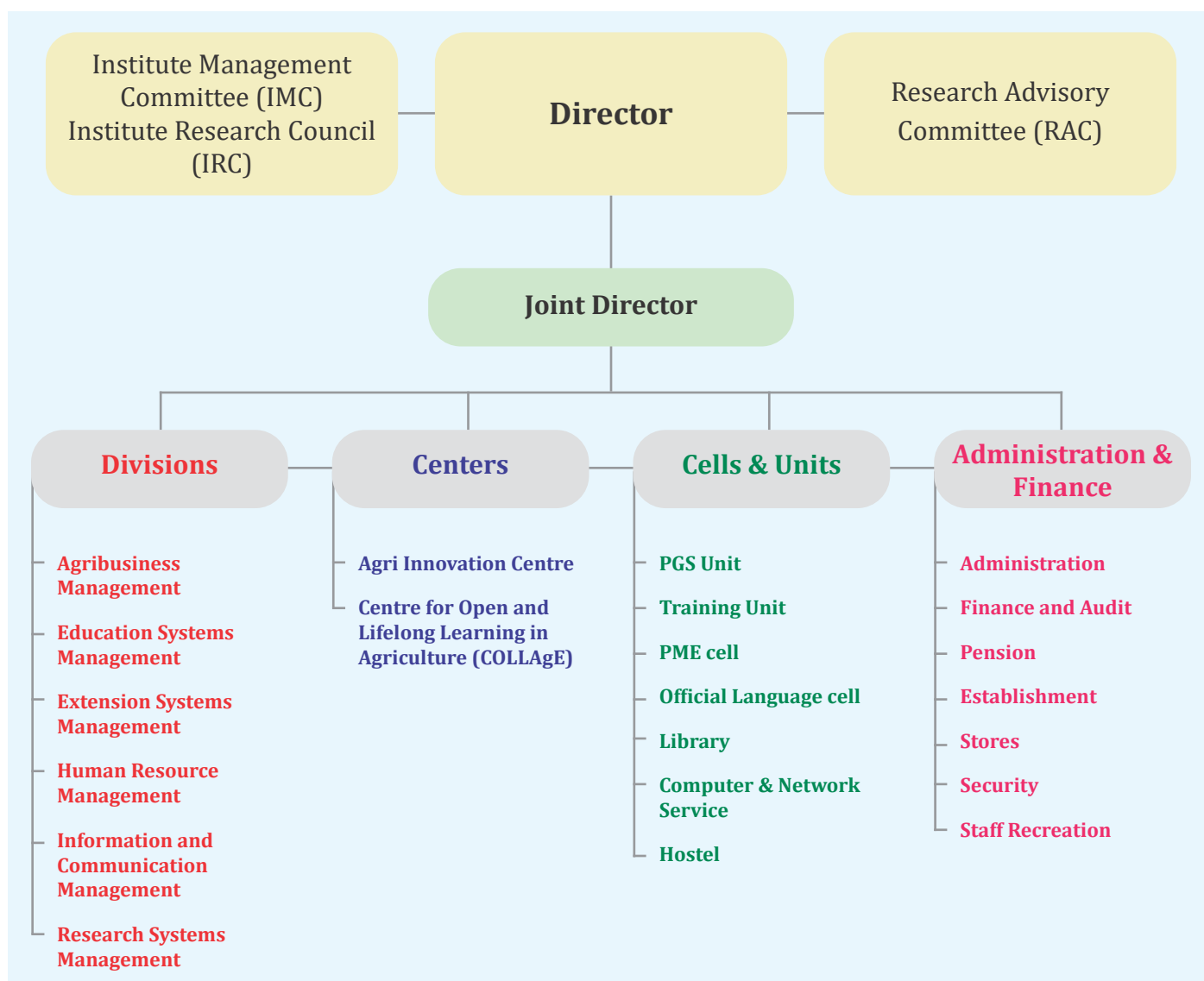
Mandate

- (i) To enhance individual and institutional competencies in managing innovation through capacity development, research and policy advocacy and
- (ii) to serve as a think tank for the NARES and facilitate strategic management of human capital

and data analytics for agricultural research and development. RSM Division executes research projects and CBPs focusing on strategic research planning, technology management and innovation systems in agriculture. Through these integrated activities, NAARM

demonstrates a comprehensive approach to agricultural capacity building, combining research, education, extension, and technology management to advance India's agricultural sector.

Organizational chart



2

CAPACITY BUILDING PROGRAMS

The Academy is known for its pioneering role in capacity-building programs (CBPs) for agricultural professionals. These programs mainly focus on enhancing competencies of scientists/faculty in areas of research and research management, education and extension management, leadership and decision-making and on promoting institutional excellence of NARES. The Academy also addresses emerging areas like agri-entrepreneurship, ICT

in agriculture, and innovation management, ensuring participants are equipped to drive advancements in the agricultural sector. With its holistic approach, the academy empowers individuals and institutions to contribute meaningfully to sustainable agricultural development. Details of various CBPs conducted during 2024 are furnished in Table 2.1.

Table 2.1. Capacity building programs organized by the Academy during 2024

Sl. No.	Type of program	No. of programs	No. of participants
I.	Executive Development Programs (EDPs)	4	76
II.	Management Development Programs (MDPs)	5	131
III.	Foundation Course for Faculty of Agricultural Universities (FOCFAU)	1	45
IV.	Foundation Course for AOs & FAOs	1	61
V.	Need-based and Sponsored Programs	28	874
VI.	Training Workshops/ Seminars/ Webinars/ Conferences	8	2503
VII.	Massive Open Online Courses (MOOCs)	2	1,726
Total		49	5,421

I. EXECUTIVE DEVELOPMENT PROGRAMS (EDPs)

Executive Development Program (EDP) on Academic Leadership and Management

A three-day EDP on Academic Leadership and Management for the Directors of Extension/ Instructions, Deans and Associate Deans of Maharashtra Animal and Fishery Sciences University (MAFSU), Nagpur, Maharashtra was organized during Jan. 8-10, 2024 at the Academy. A total of 15 senior officers of

university were participated in the program. The major objective of the training program was to enhance the academic leadership of top management of the University. The program tried to equip the participants with comprehensive skill set for human resource management, developing leadership qualities, different management strategies, team dynamics and ways to enhance happiness and well-being at workplace. [Program Director: Dr. A. Dhandapani]



Fig. 2.1. Executive Development Program on Academic Leadership and Management (Jan. 8-10, 2024)

Executive Development Program (EDP) on Leadership Excellence (LE)

An EDP on LE was organized during Feb. 12-17, 2024 to develop leadership capacities, competencies and skills to improve organizational efficiency by adapting to change and innovation management in the NARES. The program disseminated knowledge on leadership excellence, attributes, collaboration; leadership lessons from epics, building strength-based organisations; science, management and organisational efficiency, start-up ecosystem, cross-learning from other scientific organisations and digital ecosystem for ICAR. During the program, the leadership experiences and best practices adopted by all the participants were captured and discussed. It was emphasized by the experts that positivity in leadership enhances efficiency of the personnel



Fig. 2.2. Executive Development Program on Leadership Excellence (Feb. 12-17, 2024)

and organisation they work in. A total of 20 newly recruited Research Management Positions (RMPs) participated in the program. [Program Directors: Dr. G. Venkateshwarlu, Dr. D. Thammi Raju and Dr. G.R.K. Murthy]

Executive Development Program for Directors of Extension Education of SAUs

The Academy organized its first-ever Executive Development Program (EDP) for Directors of Extension Education (DEEs) of State Agricultural Universities (SAUs) during Jul. 10-12, 2024. The aim of this program was to enhance the leadership competencies of DEEs by inculcating innovative governance and management strategies for frontline extension. Twenty-eight DEEs representing Agriculture (17), Animal



Fig. 2.3. Executive Development Program for Directors of Extension Education of State Agricultural Universities (Jul. 10-12, 2024)

sciences (4), Horticulture (4) and Fisheries (1) Universities and 16 states participated in the program. Four women DEEs were also among the participants. The program covered strategies for harnessing KVK potential, promoting start-up ecosystem and rural entrepreneurship, technology diffusion through ICT, social

media, farmer-centric systems, innovations in stewarding frontline extension, extension research, women entrepreneurship, public-private partnership in agricultural extension, GMPs in extension and happiness & wellbeing. [Program Directors: Dr. R. Venkattakumar and Dr. P. Venkatesan]

Executive Development Program (EDP) on Leadership Excellence

A 6-day EDP on Leadership Excellence was organized during Aug. 26-31, 2024. Thirteen participants included 5 Directors, 2 Assistant Director Generals and 6 Joint Directors from 8 different ICAR Institutes. The program equipped participants with comprehensive skill set, covering managing polarities, leadership styles, changing management strategies along with empirical exercises on strength-finder, team dynamics and strategies for enhancing



Fig. 2.4. Executive Development Program on Leadership Excellence (Aug. 26-31, 2024)

happiness and well-being. It also included specialized modules on administrative and financial management, integration of startup ecosystems and CSR funding opportunities. Experienced resource persons emphasized on adopting innovative leadership strategies to meet the evolving needs of the organization for enhanced effectiveness and urged participants to think critically, analyze carefully and make well-informed decisions. [Program Directors: Dr. B. Ganesh Kumar and Dr. R. Venkattakumar]

II. MANAGEMENT DEVELOPMENT PROGRAMS (MDPs)

Management Development Program (MDP) on Leadership Development

The 10-working day MDP on leadership development organised during Jun. 3-14, 2024 has benefitted 24 Heads/ Senior Officers (pre-RMPs) including three females. The program aimed at enabling the participants to become an effective and efficient research manager and tackle leadership challenges. It also educated them with important skills such as research



Fig. 2.5. MDP on Leadership Development (Jun. 3-14, 2024)

project management, financial management and emerging technologies. From the feedback, it could be stated that the MDP on leadership development helped many of the participants to switch from individual academicians/ researchers to research manager improving

their abilities to handle different functional roles effectively. [Program Directors: Dr. Ramasubramanian V. and Dr. Ranjit Kumar]

MDP for Newly Recruited Senior Scientists and Heads of KVKs

A two-week long MDP (the 1st phase) was organized at the Academy during Jun. 24 to Jul. 08, 2024. The training module covered organizational and management aspects of Krishi Vikash Kendras (KVKs), personality development, communication skills, time management, media management, statistical tools and impact assessment methodologies. A total of 26 Heads of KVKs representing 8 Agricultural Technology Application Research Institute (ATARI) zones from across the country including five women scientists participated in the program. After concluding the 1st phase at ICAR-NAARM on Jul. 8, 2024, participants underwent experiential learning either at KVK at



Fig. 2.6. Management Development Program for Newly Recruited Senior Scientists and Heads of KVKs (Jun. 24 to 08 Jul., 2024)

Kota, Rajasthan; Ri-Bhoi, Umiam, Meghalaya; Lucknow, Uttar Pradesh; Ranchi, Jharkhand and Pathanamthitta, Kerala during Jul. 11-20, 2024. The 3rd phase was conducted from Jul. 23-27, 2024 in ATARIs situated at Ludhiana, Kanpur, Guwahati, Patna, Kolkata, Bengaluru, Pune or Hyderabad. [Program Directors: Dr. P. Venkatesan, Dr. N.A. Vijay Avinashilingam and Dr. Mahantesh Shirur]

Management Development Program (MDP) on Leadership for Heads of Social Science Division of NARES

An MDP was organised during Oct. 14-18, 2024, which was attended by 23 Heads of Social Science Divisions of ICAR Institutes, SAUs and KVKs from across the country including 8 women. The program was designed to discuss and disseminate updated knowledge on Extension/ Economics/ Statistics covering aspects such as social simulation of adoption process, mainstreaming farmers' innovations, public-private partnership, research foresight, research priorities, impact assessment, science



Fig. 2.7. Management Development Program (MDP) on Leadership for Heads of Social Science Division of NARES (Oct. 14-18, 2024)

communication, Geographic Information System (GIS) and Artificial Intelligence (AI) tools for agriculture. As per the feedback, the participants asserted to have learnt new project formulation and implementation tools, statistical analyses for social science and human resource management methods during the training. [Program Directors: Dr. R. Venkattakumar and Dr. Mahantesh Shirur]

MDP for Newly recruited Senior Scientists and Heads of KVKs

The Academy organized the 1st phase of the Management Development Program (MDP) for the newly recruited Heads of KVKs from Nov. 27 to Dec. 11, 2024. Forty-one Heads of KVKs



Fig. 2.8. MDP for Newly recruited Senior Scientists and Heads of KVKs (Nov. 27 to Dec. 11, 2024)

representing 20 states, 11 ICAR Institutes, 6 Non-Governmental Offices (NGOs) and 3 State Agricultural Universities (SAUs) including five women scientists participated in the program. It aimed to enhance the capacity and skills of KVK

Heads towards efficient technology application. Dr. V. Venkatasubramanian, Director, ICAR-ATARI, Bengaluru emphasized that KVK Heads ought to adopt innovative methods for effective application of farm technologies under field conditions. Dr. Kadirvel Govindaswamy, Director, ICAR-ATARI, Guwahati discussed about the strategies for effective convergence of KVKs with other extension agencies to address the farmers' needs holistically. [Program Directors: Dr. Surya Rathore and Dr. R. Venkattakumar]

MDP on Leadership Development (a pre-RMP Program)

A 10-day MDP on Leadership Development was organized during Dec. 2-13, 2024 in which four Heads of Divisions, 11 Principal Scientists, one Professor from SAU and one industry Manager (17 participants in total) participated. The program focused on enhancing leadership and management skills of the participants. The program coverage ranged from leadership styles, emotional intelligence and research prioritization to cutting-edge technologies such as cloud computing and AI applications in



Fig. 2.9. MDP on Leadership Development (a pre-RMP Program) (Dec. 2-13, 2024)

agriculture. The program emphasized practical tools and theoretical frameworks for strategic planning, team building and management and fostering inclusivity. Overall, the MDP equipped participants with the skills to navigate leadership challenges and drive innovation in agricultural research and development. [Program Directors: Dr. B. Ganesh Kumar and Dr. K. Srinivas]

III. FOUNDATION COURSE FOR FACULTY OF AUs

12th Foundation Course for the Faculty of Agriculture Universities (FOCFAU)

The 12th FOCFAU was organized during Sept. 3-23, 2024 which benefitted 45 participants including 23 women Assistant Professors from ten State Agriculture Universities spread across the country. The foundation course exposed the participants to project management techniques, effective presentation/ communication skills, design thinking in writing research proposals, AI in teaching, New Education Policy, advanced extension strategies, conflict management, IPR issues, curriculum development, digital/ innovative teaching methods, classroom management, psychological principles of teaching-learning, performance evaluation and testing and statistical tools for research. This



Fig. 2.10. 12th Foundation Course for the Faculty of Agriculture Universities (Sept. 3-23, 2024)

foundation course was aimed at enhancing participants' competencies in all aspects of teaching-learning continuum. [Program Directors: Dr. Mahantesh Shirur and Dr. B. Nirmala]

IV. FOUNDATION COURSE FOR AOs and FAOs of ICAR INSTITUTES

The Academy organized a 30-day Foundation course for newly recruited Administrative Officers and Finance and Accounts Officers of ICAR during Mar. 20 to Apr. 19, 2024. A total of 61 participants consisting of 32 AOs and 29 FAOs representing 17 states / UTs including seven female officers were trained. The program included lectures, interaction and case studies on administration and financial aspects of the ICAR system and the Govt of India. The officers got posted in various ICAR Institutes after completion of the training. [Program Directors: Shri Vivek Purwar and Shri Z.H. Khilji]



Fig. 2.11. Foundation Course for AOs and FAOs of ICAR (Mar. 20 to Apr. 19, 2024)

V. NEED-BASED AND SPONSORED PROGRAMS

Online Course Production and Management for MOOCs

A training program was organized during Jan. 1-12, 2024, with an aim to improve competency on digital content production and management, as emphasized by the New



Fig. 2.12. Training program on Online Course Production and Management for MOOCs (Jan. 1-12, 2024)

Education Policy 2020. Sessions covered digital content development practices, course management through learning management system and digital assessment and evaluation in online and self-practice learning mode, which made the program successful. Nineteen (19) trainees including 5 females participated in the program. [Program Director: Dr. G.R.K. Murthy]

Mentoring and Training Program for Agri startups (ABIIGROW 2.0)

Agribusiness Incubation Centre of the Academy organized a 3-day program on Mentoring and Training for Agri startups (ABIIGROW 2.0) during Jan. 22-24, 2024. The program trained the startups on financial and business aspects. Various sessions emphasized the importance of



Fig. 2.13. Mentoring and Training Program for Agri startups (Jan. 22-24, 2024)

networking in agriculture and allied sectors and helped in actual networking by the startups. A total of 33 startups/ entrepreneurs (22 male and 10 female) participated in the program. [Program Director: Dr. S. Senthil Vinayagam]

Induction Training on 'Project Management and Research Methodology' for Scientists of ICFRE-FRI, Dehradun

A 2-week long training program was organized during Feb. 12-25, 2024 to improve the skills of the newly recruited scientists of Forest Research Institute (FRI), Dehradun in project management and research methodology. The program comprised sessions of theoretical as well as hands-on exercises in equal proportions. In addition, participants prepared and presented research project in groups with design thinking methodology discussed during the program. The first week of the program disseminated



Fig. 2.14. Induction Training Program of Newly Recruited Scientists of ICFRE-Forest Research Institute, Dehradun (Feb. 12-25, 2024)

basic and advanced methodologies of project management while in second week, the focus was on research methodology. A total of 38 scientists participated in the program including 12 females. [Program Directors: Dr. Tavva Srinivas and Dr. Mahantesh Shirur]

Data Visualization using R

A 5-day training program was organized during Mar. 3-8, 2024, in which 45 participants including 16 women were provided with theoretical and practical knowledge on basics of R, data wrangling, visualization of different types of data (i.e., numeric/ spatial/ text data) and statistical analyses. [Program Director: Dr. B.S. Yashavanth]



Fig. 2.15. Training on 'Data Visualization using R' (Mar. 3-8, 2024)

Enhancement of Teaching Competencies of Agri. Faculty

A 5-day training program was organized during Mar. 11-15, 2024 with the sole objective of enhancing teaching competencies of agricultural faculty in tune with the core philosophy of National Education Policy (NEP) 2020. The program disseminated knowledge and skills relevant to teaching-learning process, innovative teaching methodologies, NEP 2020,



Fig. 2.16. Training on Enhancement of Teaching Competencies of Agri-Faculty (Mar. 11-15, 2024)

micro-teaching, teaching with technology, creativity and emotional intelligence for teaching effectiveness apart from AL, ML, AR, VR and Mixed Reality. A total of 19 faculty (10 males and 9 females) participated in the program. [Program Director: Dr. D. Thammi Raju]

Innovations in Digital Extension

A training program on 'Innovations in Digital Extension' was organized during Mar. 11-15, 2024. It discussed and disseminated updated knowledge on digital innovations, social media, online survey tools, decision support systems, interactive platforms & presentation softwares, web content design and development, AI & IoT applications and digital video to enhance the capabilities of extension professionals through a blend of lectures, demonstrations, discussions



Fig. 2.17. Training on Innovation in Digital Extension (Mar. 11-15, 2024)

and extensive hands-on activities. Sixteen trainees including two women participated in the program. [Program Director: Dr. N.A. Vijay Avinashilingam]

MDP on Business Plan Development and Accelerating FPOs/FPCs

The MDP organized during May 27-31, 2024 aimed at building business plan and enhance knowledge and skills of farmer producer organizations (FPOs) or farmer producer



Fig. 2.18. MDP on Business Plan Development and Accelerating FPOs/FPCs (May 27-31, 2024)

companies (FPCs) to overcome impending challenges. The program dealt with basic concepts, current status of FPOs in India, formation process of FPOs, governance principles, statutory compliances, taxation for FPOs, framework for selecting business model for FPOs, sustainable business models for FPOs, supply chain management for FPOs, govt. policies and roles of NABARD in FPO promotion, agricultural marketing through social media and branding, strategies for scaling up and accelerating the FPOs. Twenty-eight participants including 12 females were the beneficiaries of the program. [Program Director: Dr. Ranjit Kumar]

Academic Excellence Through Strengthening of Linkages of National Agricultural Education System

A 4-day training program was organised during May 28-31, 2024 for competency enhancement of ICAR Nodal Officers of Agricultural Universities that would help improve the efficiency of Indian Agricultural Education System as a whole through robust linkages between ICAR and Universities. The program disseminated knowledge and skills related to functional and behavioural competencies

leading to individual and organisational effectiveness, the National Institutional Ranking Framework (NIRF), NAAC Accreditation processes, National Education Policy, Massive Open Online Courses (MOOCs) in agricultural education, agripreneurship, time management, team building, positive thinking, strength finder and emotional intelligence. Thirty-one nodal officers including five females participated in the program. [Program Directors: Dr. D. Thammi Raju and Dr. K. N. Yamini]



Fig. 2.19. Training program for ICAR Nodal Officers at Agricultural Universities (May 28-31, 2024)

CBP for Marketing Officers' Resurgence & Excellence (MORE) - Batch I of CIL

A 5-day CBP was organized in collaboration with Coromandel International Ltd. (CIL) during Jun. 3-7, 2024 with an overall objective to address the evolving needs of marketing professionals in



Fig. 2.20. Capability Building Program for Marketing Officers' Resurgence & Excellence (MORE) - Batch I (Jun. 3-7, 2024)

the agroecosystem. It was aimed at empowering participants with essential skills and advanced knowledge for excelling in sales and marketing domains. This program incorporated innovative elements including modules on artificial intelligence, the startup ecosystem, digital agriculture, digital marketing, advertising, and branding. The program emphasized building a robust foundation for marketing officers, facilitating their professional resurgence and excellence. By integrating foundational agroecosystem knowledge with advanced marketing techniques, the program equipped participants to address complex challenges, leverage emerging opportunities and drive

impactful outcomes in their roles. Twenty-one marketing officers of CIL were the beneficiaries of the program. [Program Director: Dr. Navneet Kumar]

Training Program on 'Entrepreneurial Skill Development' for students of ICAR-NDRI, Karnal

A 30-day unique training program organized during Jun. 5 to Jul. 2, 2024 was meant for the students of ICAR-NDRI, Karnal pursuing 3rd year B.Tech. (Dairy Tech.). The training program included technical sessions, group/individual activities, interactions and hands-on exercises on various topics on entrepreneurship and soft skills including educational tours and field visits. Basic objectives were developing skillsets for agripreneurship, business opportunities in agriculture and allied sectors, business plan development, personality quotients, market research for business, essentials of oral communication, business model canvas, costing/pricing for new business, business foresight, marketing in dairy sector, understanding financial statements, artificial intelligence for entrepreneurship, branding and



Fig. 2.21. Training on 'Entrepreneurial Skill Development' for students of ICAR-NDRI, Karnal (Jun. 5 to Jul. 2, 2024)

advertisement in the startup ecosystem, oral, writing and comprehension skills, policy and institutional support to entrepreneurs, dairy value chain and digital marketing, qualitative research methods for entrepreneurs, block chain technology, emotional intelligence and stress management, IPRs in agriculture, export potential of dairy products, exploring career opportunities in the dairy sector. Thirty-four students including five females participated in the training program. [Program Directors: Dr. B.S. Yashavanth, Dr. B. Nirmala and Dr. Tavva Srinivas]

CBP for Young Agri-Business Professionals of CIL

A 5-day CBP was organized in collaboration with Coromandel International Ltd. (CIL) during Jun. 10-14, 2024 with an objective to enhance the proficiency of mid-level and senior officers from CIL. The training program was divided into several modules focusing on different aspects of technical and managerial domains in the agricultural sector including climate change and balanced nutrition, drones, digital agriculture, digital marketing, AI in agriculture, virtual leadership, planning and organizing, decision making, stress management, emotional



Fig. 2.22. CBP for Young Agri-Business Professionals of CIL (Jun. 10-14, 2024)

intelligence and different aspects of marketing. A total of 32 officers representing various states of India participated in the program. [Program Director: Dr. Sona Yengkokpam]

CBP for Young Agri-Business Professionals of CIL

A capability building program was organized during Jun. 17-21, 2024 for young agri-business professionals of Coromandel International Ltd. (CIL). The main aim of the program was to educate the new entrants on basics of



Fig. 2.23. CBP for Young Agri-Business Professionals of CIL (Jun. 17-21, 2024)

agroecosystem, and to upgrade the knowledge and skills in sales and marketing domain. The program had touched various contemporary aspects of leadership development. It also covered topics such as agronomical principles of crop production, market environment for agricultural inputs, rural marketing and communication skills. Thirty-one staff of CIL participated in the training program. [Program Director: Dr. Sivakumar S.]

Generative AI tools for Agriculture

A 3-day program on 'Generative AI tools for Agriculture' was organized by the Academy during Jun. 26-28, 2024. The program, which was a blend of theory and practical sessions, enriched the participants with knowledge on potential applications of generative AI across various domains of agriculture. The sessions,



Fig. 2.24. Training on Generative AI tools for Agriculture (Jun. 26-28, 2024)

dealt with by seasoned academicians and skillful industry personnel catered to diverse audiences, covered aspects like optimizing water usage in drought-prone regions, developing predictive models for crop diseases and preventive measures, creating automated farming systems for urban areas, managing farm and livestock using AI, applying AI for fish and fisheries management, selecting suitable crops with AI and exploring AI applications in education. The workshop witnessed large number of (48 including 13 women) participants representing 21 research disciplines from 21 ICAR Institutions, 3 Agricultural Universities and 6 private organizations. [Program Directors: Dr. M. Balakrishnan and Dr. A. Dhandapani]

Capacity Building Program on Building Successful Incubation Ecosystem

Agribusiness Incubation Centre of the Academy organized a 3-day CBP on 'Building Successful Incubation Ecosystem' for PIs and Co-PIs of ICAR ABI Centers during Jul. 3-5, 2024 with objective to catalyse the growth of ICAR-hosted Agribusiness Incubation Centres. The program covered the funding opportunities for agri-incubator, business strategy case study with startup, business model canvas and disruptive

innovation. Twenty-two investigators including three females participated in the program. [Program Director: Dr. S. Senthil Vinayagam]



Fig. 2.25. Capacity Building Program on Building Successful Incubation Ecosystem (Jul. 3-5, 2024)

Training Program on Innovation Management

A 5-day training program on 'Innovation Management' was organized during Jul. 8-12, 2024. The objectives were to (i) provide an overview of the key concepts of innovation and Intellectual Property (IP) issues, (ii) introduce management framework for development and commercialization of innovative agricultural technologies including financial issues, (iii) provide and understanding of new product development, technology product life cycle and pricing, etc., (iv) develop a practical understanding of principles of innovation



Fig. 2.26. Training Program on Innovation Management (Jul. 8-12, 2024)

management such as market research, commercialization, negotiations, deals cracking and post deal issues and have hands-on experience on IP valuation and pricing. This program consisted of a blend of lectures, self-exploration instruments, case research equivalents (FTE). There were 18 participants representing ICAR, SAUs and industry from 8 states of India. [Program Director: Dr. K. Srinivas]

Training Program on Developing Competency in Econometrics

A first-of-its-kind program on 'Developing Competency in Econometrics' was organized during Jul. 22-26, 2024 which was attended by 26 participants. The program combined knowledge of theory, mathematics and statistics for analysing different economic



Fig. 2.27. Training on Developing Competency in Econometrics (Jul. 22-26, 2024)

happenings with the aim to strengthen the theoretical fundamentals of the researchers and academicians and expose them to a wide range of econometric techniques like ridge regression, stepwise regression, structural break-chow test, Just-Pope model, impact assessment techniques like difference-in-difference method, and economic surplus model. Participants were exposed to various open source econometric softwares viz., R and GRETL. In addition, they had hands-on sessions for analysis of data using

Strata and Tableau Public. [Program Director: Dr. S.M. Feroze]

Online Training Program on Multivariate Data Analysis using R

A training program on 'Multivariate Data Analysis using R' was organized during Jul. 22-26, 2024 which was attended by 79 participants (43% male and 57% female). The objectives of the program were to expose/ educate the enthusiasts on R software, basic data analysis, data visualization, correlation and regression analysis, logistic regression, path analysis, principal component and cluster analysis. [Program Director: Dr. N. Sivaramane]

Advances in Mobile Application Development

A CBP on 'Advances in mobile application development' was organized during Aug. 5-9, 2024 in virtual mode. The objective of the program was to increase awareness and build essential skills for mobile app development. The participants (51 no.) were provided with theoretical and practical sessions on android platform, Java programming, android studio installation, development of android applications, database connectivity with android apps and insight into no-coding technologies for android app development during the training program. [Program Director: Dr. N. Srinivasa Rao]

Training Program on Multivariate Analysis Using R

A 5-day training program on 'Multivariate analysis using R' was organized during Aug. 26-30, 2024, which was attended by 21 participants. In the program, various multivariate techniques were introduced through R programming including creation of data frame and importing



Fig. 2.28. Training on 'Multivariate analysis using R' (Aug. 26-30, 2024)

data from various formats were covered. Multiple linear regression, logistic regression, principal component analysis, cluster analysis, factor analysis, generalized additive and linear models were some of the multivariate techniques covered during the program. Data visualization using ggplot2 library and other libraries was covered. [Program Director: Dr. A. Dhandapani]

Faculty Development Program (FDP)

A 5-day FDP on Competency Enhancement in Agricultural Research and Education for



Fig. 2.29. FDP on Competency Enhancement for the Faculty of RLBCAU (Sept. 9-13, 2024)

the Faculty of Rani Lakshmi Bai Central Agricultural University (RLBCAU), Jhansi was organized during Sept. 9-13, 2024. A total of 40 faculty including 8 women participated in the program. The program highlighted the role of teachers in education and research to improve the overall performance of any university; faculty need to hone their communication skills and grab funding opportunities for research. It also underlined the modern and innovative teaching methods and technologies available for effective teaching. [Program Director: Dr. M. Balakrishnan]

Training Program on Fostering Collaborative Ecosystem for Promoting Agri-Startups

A 3-day program was organized during Sept. 18-20, 2024 for PIs and Co-PIs of ICAR ABI Centres and Nodal officers of ICAR Institutions with whom the Academy has signed MoU. The main objective of the program was to promote holistic development of agri-startups through effective collaboration with partner institutions. The program discussed and deliberated revenue models for ABIs, IPR management and network building with ICAR institutions. Seventeen officials attended the program including five women. [Program Directors: Dr. Dr. S. Senthil Vinayagam and Dr. N.A. Vijay Avinashilingam]



Fig. 2.30. Training on 'Fostering Collaborative Ecosystem for Promoting Agri-Startups' (Sept. 18-20, 2024)

Training program on Agricultural Production System Modelling

A 5-day training program on 'Agricultural Production System Modelling' was organized during Sept. 23-27, 2024 with the primary aim to disseminate knowledge on recent modelling approaches and strategies followed in agriculture. The program delved into effective use of different simulation models such as APSIM, DSSAT, RothC and their application in agriculture to evaluate agricultural management practices and their impact on crop yield, soil and plant processes under possible future climatic scenarios. The program provided hands-on training on simulation models using 'R' and time-series data analysis to sixteen trainees. [Program Directors: Dr. M. Ramesh Naik and Dr. D. Damodar Reddy]

Training program on 'Next Generation Sequencing (NGS) Data Analysis'

The Academy organized a training program on 'Next Generation Sequencing (NGS) Data Analysis' during Oct. 21-25, 2024 which was attended by 44 participants including 22 women representing 14 research disciplines from 13 ICAR Institutions and 11 Agricultural Universities. The program covered tools and techniques for NGS and data analysis in agricultural research and development.



Fig. 2.31. Training program on 'Next Generation Sequencing (NGS) Data Analysis' (Oct. 21-25, 2024)

Participants gained theoretical and hands-on practice using various tools and softwares. Transformative potential of bioinformatics in agriculture including response to challenges posed by climate change and low productivity was emphasized by esteemed faculty. Forty-four trainees including 22 women participated in the program. [Program Directors: Dr. M. Balakrishnan and Dr. P. Supriya]

Training Program on 'Impactful ICT Applications and Technologies in Agriculture'

A training program on 'Impactful ICT Applications and Technologies in Agriculture' was organized online during Nov. 4-8, 2024



Fig. 2.32. Training program on 'Impactful ICT Applications and Technologies in Agriculture' (Nov. 4-8, 2024)

with an aim to sensitize the trainees on ICT tools including AI that have potential applications in agriculture. All the resources required for the program were shared in online mode. The program disseminated concepts of ICTs, Artificial Intelligence, Machine Learning, Deep Learning, IoT and Python programming. The program also elaborated on selected AI tools such as ChatGPT, Claude, Google Gemini, Gamma app, RunwayML and ElevenLabs. A

total of 25 (11 male and 14 female) participants were the beneficiaries of the program. [Program Director: Dr. V.V. Sumanth Kumar]

Training Program on 'Vigilance Perspectives for ICAR Officers'

A 3-day training program on 'Vigilance Perspectives for ICAR Officers' was organised during Nov. 6-8, 2024 as a part of Vigilance Awareness Campaign/ Week with the primary objective to sensitise the participants about vigilance requirements among functionaries of ICAR Institutions. The program deliberated on preventive vigilance, ethics and governance, procurement procedures, cyber hygiene and security, implications of penalty, sexual



Fig. 2.33. Training on Vigilance Perspectives for ICAR Officers (Nov. 6-8, 2024)

harassment at the workplace, conduct rules, issues and challenges in vigilance administration, CCS and CCA rules. A WhatsApp group was created with the participating Vigilance Officers who share their experiences, cases and resources for enhancing efficiency of ICAR officials. Twenty-eight officers including three females participated in the program. [Program Directors: Dr. D. Thammi Raju and Shri Vivek Purwar]

Training Program on Statistical Learning Techniques

A 10-day training program on 'Statistical Learning Techniques' was organized during

Nov. 18-27, 2024 in on-line mode and total of 41 participants completed the program. The program focused on fitting appropriate models based on data through various statistical modelling techniques. Statistical learning technique is a new approach using the different model fitting tools but models are assessed based on how well the model predicts unseen observations rather than the data based on which they were build. This has been done by splitting the data into two parts commonly called training and test datasets wherein the training data set is used for model building and the test dataset is used to assess the model in terms of prediction errors. In case of small datasets, resampling techniques such as k-fold cross validation techniques are used to evaluate the models. The topics covered included concepts of statistical learning; trade-off between prediction and interpretability; introduction to R; visualization techniques to understand the relationships; regression diagnostics, polynomial regression; splines, local regression and generalized additive models; bagging, boosting, random forest and decision trees; artificial neural networks, deep learning and classification techniques. [Program Director: Dr. A. Dhandapani]

From Reads to Revelations: Transcriptome Data Decoded

A 3-day training program on 'From Reads to Revelations: Transcriptome Data Decoded' was organized during Dec. 9-11, 2024 that provided participants with a thorough and in-depth understanding of transcriptomics, from the initial raw RNA sequencing data to the extraction of valuable biological insights. The training focused on hands-on, real-world applications, enabling attendees to gain proficiency in using a wide range of bioinformatics tools



Fig. 2.34. Training program on 'From Reads to Revelations: Transcriptome Data Decoded' (Dec. 9-11, 2024)

such as FastQC, Trimmomatic, Bowtie2 and HISAT2. Additionally, participants learned to apply advanced statistical methods and tools like R-studio, DESeq2 and edgeR for data visualization and analysis for interpreting complex datasets and drawing actionable conclusions that could drive biological discoveries and research advancements. The program was attended by 22 trainees including 16 women. [Program Director: Dr. P. Supriya]

Innovations in Digital Extension

A 5-day training program on 'Innovations in Digital Extension' was organized during Dec. 16-20, 2024. A total of 29 trainees participated in the program including seven women. The



Fig. 2.35. One participant receiving certificate for TP on Innovations in Digital Extension (Dec. 16-20, 2024)

program covered up-to-date and most relevant aspects on digital innovations, decision support systems, web content design and development, AI & IoT applications, social media, online survey tools, interactive platforms, oral and video presentation softwares to enhance the capabilities of extension professionals. The program included a blend of lectures, demonstrations, discussions and hands-on activities. [Program Director: Dr. N.A. Vijay Avinashilingam]

VI. TRAINING WORKSHOPS/ SEMINARS/WEBINARS/CONFERENCES

Content Workshop for Development of MOOC Certificate Course on 'Management of FPOs'

A content development workshop for MOOC Certificate Course on 'Management of FPOs' was organized on May 16, 2024 involving twenty-two external management consultants, agribusiness experts, scientists and faculty of ICAR Institutes. The certificate course will serve as a comprehensive guide to start and manage a farmer producer organization (FPO) efficiently and effectively and hence promote FPO culture in the agricultural sector. The MOOC will be of 2 months duration with video lectures, assignments and assessment/ evaluation during the sessions and after the course. [Program Director: Dr. Sivaramane N.]

Workshop on 'Career Development Centres'

A virtual workshop on Career Development Centres (CDCs) was organized on Jun. 18, 2024 under the chairmanship of Dr. S.K. Soam, CPI, NAHEP to finalize the action plan for the newly established CDCs at three

SAUs. The workshop was attended by Dr. D.R. Salomi Suneetha, Nodal Officer CDC and Dean of Student Affairs (FAC), Dr. YSRHU, Venkataramannagudem, Andhra Pradesh; Dr. Channappagouda S. Birdar, Nodal Officer CDC and Associate Professor, Veterinary Extension, KVAFSU, Bidar, Karnataka; and Dr. Sanjay Guleria, Nodal Officer CDC, SKUAST, Jammu, Co-PIs and RAs of the project. The workshop included deliberations on CDC activities, functionalities, achievements of existing CDCs, and expectations from the newly established ones. Participants also presented their respective universities' action plans during the session. [Program Director: Dr. N. Srinivasa Rao]

Content Development for Educational Technology and Educational Management Courses

A brainstorming workshop was organized on Jul. 18, 2024 with an objective to design two certificate courses – one on 'Educational Technology' and another on 'Educational Management' to be offered by the Education Systems Management Division of the Academy. External and internal experts (20) discussed the content and decided to contribute to developing the courses. The proposed online certificate courses will be useful for Teachers (higher education), Faculty of Universities (Assistant Professors/ Associate Professors/ Professors)



Fig. 2.36. Brainstorming on Content Development for Educational Technology & Educational Management (Jul. 18, 2024)

and also for post-graduate students who are interested in choosing teaching profession. [Program Director: Dr. V.V. Sumanth Kumar]

Webinar on Dynamic Digital Presence

As part of the Vigilance Awareness Campaign, the Academy organized a webinar on 'Dynamic Digital Presence' on Oct. 7, 2024 with the aim to promote transparency, accountability and technological advancement among ICAR employees. The discussion underscored growing importance for enhanced digital

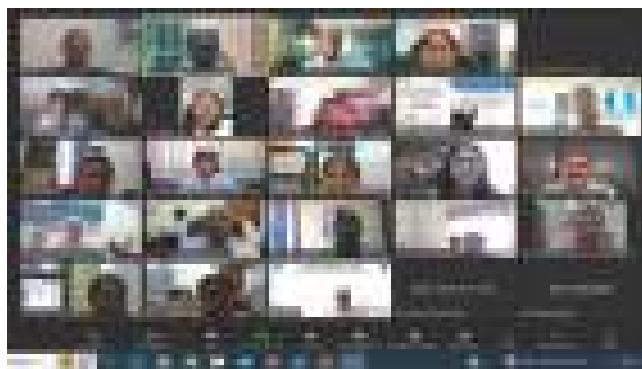


Fig. 2.37. Webinar on Dynamic Digital Presence (Oct. 7, 2024)

vigilance and robust cybersecurity practices to enhance security and efficiency in crucial areas, ensuring a more dynamic and impactful online presence within ICAR organizations. Mr. Ajith M., Senior Project Manager, CDAC, Hyderabad, delivered a lecture on 'Website Testing, Security Auditing, and Certification'. Dr. Anil Rai, Assistant Director General (ICT), ICAR, New Delhi, delivered a talk on 'Dynamic Digital Presence of ICAR: Way Forward'. A total of 82 staff attended the webinar. [Program Director: Dr. D. Thammi Raju]

Enhancing Academic Excellence of Agricultural Higher Education Institutions

A 3-day training workshop on Enhancing Academic Excellence of Agricultural Higher Education Institutions (HEIs) through NIRF



Fig. 2.38. Training workshop on Enhancing Academic excellence of Agricultural HEIs through NIRF Ranking and NAAC Accreditation (Oct. 16-18, 2024)

Ranking and NAAC Accreditation organized during Oct. 16-18, 2024 was attended by 51 participants representing 29 SAUs, 2 Deemed Universities and one CAU (Imphal) and from 15 states and 2 UTs (J&K and Delhi). The program imparted theoretical and practical knowledge regarding NIRF ranking and NAAC accreditation systems in India that are required for enhancing academic standings of the higher agricultural education institutions. Participants were also exposed to quality standards in higher education, other ranking and accreditation systems of India and abroad, data collection, management, monitoring and evaluation, strategies to secure higher rankings and do's and don'ts. [Program Directors: Dr. K. N. Yamini and Dr. Navneet Kumar]

National Learning Week (*Karmayogi Saptah*)

The Academy in collaboration with the Capacity Building Unit of Department of Agricultural Research and Education (CBU-DARE) organized a series of four webinars through Integrated Government Online Training (iGOT) platform during Oct. 22-25, 2024. This week-long celebration was designed to inspire and upskill government employees through continuous learning and professional development. The 1st webinar (22.10.24) was on 'Mission Karmayogi

@ ICAR' by Dr. S.K. Sharma, ADG (HRM) which highlighted the roles and responsibilities of public servants. The 2nd webinar (23.10.24) was on 'Time Management' by Mr. Tirmal Reddy, L&D Professional, Blossum, Hyderabad. The 3rd webinar (24.10.24) was by Dr. Himanshu Pathak, Secretary, DARE & DG, ICAR on 'Viksit Bharat@Agriculture', and the 4th webinar (25.10.24) was on 'Climate change, agriculture and Indian economy' by Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM. A total of 1294 employees of DARE/ICAR HQs/ASRB/NAARM participated in the webinars. [Program Directors: Dr. B. Ganesh Kumar and Dr. Sona Yengkokpam]



Fig. 2.39. National Learning Week (Oct. 22-25, 2024)

National Conference on Building Sustainable Agricultural Startups in India

A 2-day National Conference on 'Building Sustainable Agricultural Start-ups in India' was organized during Nov. 12-13, 2024. The major objectives of the program were to provide a right platform for academicians, entrepreneurs, start-ups, incubators, investors, scientists, developmental agencies, farmers, policy makers, administrators and NGOs to share their expertise and experiences. Distinguished speakers included Dr. Himanshu Pathak, Secretary, DARE & Director General, ICAR; Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM; Dr.



Fig. 2.40. National Conference on 'Building Sustainable Agricultural Start-ups in India' (Nov. 12-13, 2024)

M. Prabhakar Rao, Chairman of Nuziveedu Seeds Ltd; and Dr. Seema Jaggi, ADG (HRD), ICAR. The conference drew over 1,000 participants from 13 states, including representatives from 45 FPOs, 6 ICAR Institute ABIs, 3 Nationalized Banks (SBI, BOB, and UBI), IIT Madras and other ICAR Institutes. Attendees included 421 students, 225 farmers, 100+ FPOs/farmers, agriculture experts, government officials, investors, startups, incubators and other key stakeholders in the startup ecosystem. The event featured five-panel discussions with eminent experts on key topics within the agricultural start-up ecosystem. In parallel, a networking session was organized, allowing participating start-ups to pitch their products, technologies, and services to FPOs. Additionally, 73 start-ups showcased their innovations at dedicated stalls for visitors. [Program Directors: Dr. S. Senthil Vinayagam, Dr. N.A. Vijay Avinashilingam and Dr. Ch. Srinivasa Rao]

Workshop on Geo-Spatial Analytical Tools for Natural Resource Management

A workshop was organized during Dec. 17-19, 2024 which discussed and demonstrated advanced geo-spatial analytical techniques and provided hands-on training for analysis of spatial data classification, map preparation for precision agriculture, crop health monitoring and image processing using drones, area estimation using SAR data, suitability analysis and catchment delineation using open source softwares such as QGIS, SNAP and R. The program was successful in orienting the researchers and students with understanding to apply geospatial technologies and develop a decision support framework for managing agricultural production systems. Fourteen participants (10 males and 4 females) were the beneficiaries of the program. [Program Directors: Dr. P. D. Sreekanth and Dr. A. Dhandapani]



Fig. 2.41. Workshop on 'Geo-Spatial Analytical Tools for Natural Resource Management' (Dec. 17-19, 2024)

VII. MASSIVE OPEN ONLINE COURSES (MOOCs)

MOOC on Artificial Intelligence (AI) in Agriculture

A MOOC was organized during March 1-31, 2024 with the objective of sensitizing participants

about concepts of AI, Machine Learning, IoT and Cloud Computing, and their applications in agriculture. The 4-week MOOC also exposed the trainees to tools and technologies such as ChatGPT, Teachable machine, python software and python IDE. Image processing and classification in agriculture were demonstrated using matlab and opencv packages. In the end, successful startups who implemented AI-based solutions in agriculture and allied sectors were showcased to motivate the participants. A total number 4423 (2632 male and 1791 female) participants registered for the course with 1197 participants successfully completing the certification requirements. [Program Directors: Dr. V.V. Sumanth Kumar and Dr. G.R.K. Murthy]



Fig. 2.42. Massive Open Online Course on Artificial Intelligence in Agriculture (Mar. 1-31, 2024)

MOOC on Digital Assessment and Evaluation Methodologies

A 1-month MOOC was organized during Sept. 1-30, 2024, with an aim to equip participants with modern tools and techniques for designing, implementing, and managing digital assessments in educational setting. The course was delivered online using pre-recorded interactive video lectures, along with learning materials provided weekly to ensure a structured learning experience. A notable innovation introduced in this program was the automated generation of digital certificates embedded with the learners photographs, enhancing personalization and authenticity of the certification process. A total of 1,501 participants (864 males, 637 females) registered, with 529 participants successfully completing the certification requirements. [Program Directors: Dr. G.R.K. Murthy and Dr. Navneet Kumar]

3

ACADEMIC/ EDUCATIONAL PROGRAMS

The academic/ educational programs in 2024 mainly focused on excellence in agribusiness education through the PGDM-ABM program of the Academy. The Academy's PGDM-ABM-2024 (16th batch) admitted 57 students from a competitive pool of applicants maintaining high academic standards and industry relevance. PGDM-ABM students excelled in national competitions, securing top honors in various events. The program achieved 100% placements this year also with students placed in banking, agri-input, supply chain and agritech sectors. Sankalp 9.0, a premier business fest organized by the Academy, saw over 10,500 registrations. Academy's Faculty contributed to agricultural research and education through teaching, research guidance and collaborations as well. Educational visits, workshops and innovative curriculum ensured industry readiness reinforcing Academy's leadership in agribusiness management education.

PGDM-ABM (16th Batch): Admissions

The process of admission (conducted during June 25-29, 2024) was completed for the Post Graduate Diploma in Management - Agri-Business Management (PGDM-ABM) 2024-26 batch. A total of 930 applications were received, out of which 380 candidates were shortlisted for Group Discussions (GD) and interviews. Ultimately, 243 candidates participated in the AWST (Analytical Writing Skill Test), GD and interview rounds. The program admitted 57 students, maintaining its reputation for academic excellence. Approved by the All India Council



Fig. 3.1. Profile and group photo of the 16th batch of PGDM-ABM program of the Academy

for Technical Education (AICTE) and accredited by the National Board of Accreditation (NBA), the PGDM-ABM program has also been granted equivalence to an MBA by the Association of Indian Universities (AIU).

PGDM ABM Students: Outstanding Achievements

Students of the PGDM ABM program achieved remarkable success in various prestigious national competitions during 2024, showcasing their talent and versatility. They secured top honors in diverse events, including visual symphony, B-plan competitions, case studies, marketing challenges, policy challenges, short ad films, photography and sustainable supply chain challenges.

Table 3.1. List of awards/ accolades won by the PGDM-ABM students of the Academy

Name of the Students	Event and Organizer	Position
Nithin S.	Visual Symphony of Imperium 2024, organized by Management Development Institute, Gurgaon during Feb. 9-11, 2024	Winner
Monish Gowda T.P. and Anuradha Jog	Aarambh '24 B-Plan Competition, organized by Institute of Management Technology (IMT), Nagpur during Feb. 10-11, 2024	Winner
Madhusudan M. Sakare and Jupally Kavya Sri	Samiksha - The Case Study Competition of Coalescence'24, organized by Indian Institute of Forest Management (IIFM), Bhopal during Feb. 22-24, 2024	Winner
Monish Gowda T.P. and Anuradha Jog	Marketing Maverick of Coalescence'24, organized by Indian Institute of Forest Management, Bhopal during Feb. 22-24, 2024	Winner
Satyam Purohit and Saurabh Singh Pokhriyal	Inventure - The Startup Challenge of Coalescence'24, organized by Indian Institute of Forest Management (IIFM), Bhopal during Feb. 22-24, 2024	1 st Runner-up
Banda Abhinaya, Saineni Abhigna, and Thukuntla Nithya Sri	Resilient Routes - The Sustainable Supply Chain Challenge of Coalescence'24, organized by Indian Institute of Forest Management (IIFM), Bhopal during Feb. 22-24, 2024	2 nd Runner-up
Vinay Kumar	Policy Challenge competition of Jagriti 2024, organized by Banaras Hindu University, Varanasi during Mar. 1-3, 2024	Winner
Vijay R.	Parchaii - The Photography competition of Meraz 4.0, organized by Tata Institute of Social Sciences on Mar. 16, 2024	2 nd Runner-up
Pooja K.G.	Pixel Snapper - Photography competition of Meraz 4.0, organized by Indian Institute of Technology, Bhilai during Mar. 21-23, 2024	Winner
Mohammad Abdul Raffi and Keerthana M.	Paraspar - Coopathon, a case study competition of Spardha'24, organized by Vamnicom, Pune during Mar. 27-28, 2024	Winner
Nithin S., Vijay R. and Monish Gowda	Drishyamitra - Short ad film competition of Spardha'24, organized by Vamnicom, Pune during Mar. 27-28, 2024	Winner
Satyam Purohit and Saurabh Singh	Abhyudya Business Plan, organized by VAMNICOM, Pune during Mar. 27-28, 2024	Winner
Ayushi	Chitra Lekhna - Painting competition of Spardha'24, organized by VAMNICOM, Pune during Mar. 27-28, 2024	2 nd Runner-up
Pooja K.G.	Pratibimb - Photography competition of Spardha'24, organized by VAMNICOM, Pune during Mar. 27-28, 2024	2 nd Runner-up
Nithin S. and Pooja K.G.	Ad Quest of Empyrean 2024, organized by Indian Institute of Management, Jammu during Mar. 28-29, 2024	Winner
Arjun Prasad D. and Jagan R.	Event held at the Imperial School of Agri Business, Greater Noida, Uttar Pradesh during Dec. 11-16, 2024	Winner
Srivamsi Yekkala and Keerthana M.	Guesstimania of Krishi Chanakya (2024), organized by MANAGE, Hyderabad during Sept. 15-16, 2024	1 st Runner-up
Pooja K.G.	Branding and Design Challenge of The Business Conclave 2024, organized by Shiv Nadar University (SNU), Greater Noida during Nov. 17-18, 2024	1 st Runner-up



Fig. 3.2. PGDM-ABM students receiving award in Agri Startup Competition held at ISAB

REUNIRSE'24

REUNIRSE'24 - the alumni meet of PGDM-ABM of the Academy was organized during Sept. 28-29, 2024. The NAARM PG Alumni Association (NPAA) is 12-year young association of the passed-out students of the Academy which was formed with the motto of 'Growing together'. The event envisaged networking, insightful discussions and collaborative opportunities among current students and the alumni of the Academy. They got engaged in meaningful exchanges, reflecting shared accomplishments

and envisioning future initiatives. Mr Monish Gowda, Secretary, NPAA welcomed guests, faculty and alumni. Dr. Ranjit Kumar, President, NPAA outlined the purpose of the association and presented the initiatives undertaken during last 10-12 years. The event saw participation of 150+ alumni coming from different parts of the country. Next day, a unique program named KreedALL 1.0 (Alumni Legacy League) was organized in which alumni participated in volleyball, throwball and cricket games. The event concluded with prize distribution.



Fig. 3.3. Alumni meet of PGDM-ABM program (REUNIRSE'24) of the Academy organized during Sept. 28-29, 2024

PGDM-ABM: Placements Legacy Continued

The Post Graduate Diploma in Management – Agri-Business Management (PGDM-ABM) Program of the Academy has consistently set a benchmark in management education with its successful placement record. Since its inception, the program has proudly maintained 100% placements, highlighting its credibility, relevance and the industry's unwavering

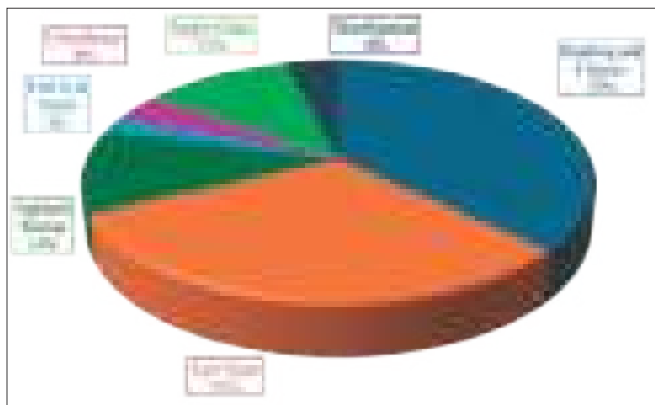


Fig. 3.4. Final placements of PGDM-ABM 2024-26 batch

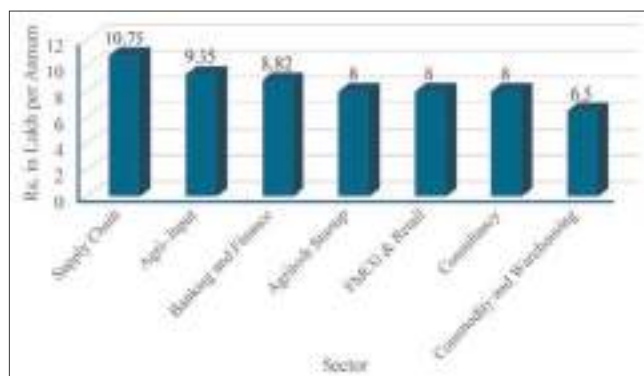


Fig. 3.5. Average Package of PGDM-ABM 2022-24 Batch

trust in its graduates. Over the years, it has emerged as a leading choice for aspiring agri-business professionals, mostly because of its well-designed curriculum, strong industry partnerships and focus on preparing students for real-world challenges. For the 2023-25 batch, 42 students have already been placed in final placements, with the placement process

still underway. Similarly, the 2024-26 batch has seen 35 students securing summer internships, showcasing the program's ability to provide valuable industry exposure to its students. These achievements demonstrate not only the high employability of students but also the trust leading industries place in the program.

A sectoral analysis of the placements reflected program's alignment with industry trends and demands. The Banking and Finance sector continued to be a major recruiter, absorbing 37% of students, followed by Agri-Input (31%), Supply Chain (11%) and Agritech Startups (11%). Additionally, students have secured niche roles in areas such as Development, Consultancy and FMCG. The rise in average annual CTC (Cost to Company) further emphasized the program's success in preparing students for lucrative careers. The 2022-24 batch saw an impressive average CTC of ₹10.7 LPA, with Supply Chain roles offering the highest average package of ₹10.75 LPA. The program's growing reputation was also evident in the sharp increase in number of applications.

Sankalp 9.0

The 9th edition of Sankalp (Business Festival of NAARM) – an annual business fest inviting students from different colleges across India was organized by the PGDM-ABM students of the Academy. This two-month-long competitive event is conducted online or offline, with the final round held at the Academy during Oct. 25-26, 2024. The event received an overwhelming response with 10,500+ registrations and 5.27 lakhs of online impressions on the Unstop platform from more than 1,700+ institutions, including IIMs, IITs, several leading B-schools and agricultural colleges. The B-fest included four flagship competitive events: *Adhistatha*



Fig. 3.6. Sankalp 9.0 – Business Festival of NAARM

the best manager competition; *Samadhaan* – the case analysis competition; *Advitiya* – the business plan competition; and *Analytica* – the data analytics competition. There were four sprints or online events: *Flashgun* – the photography competition; *Darpan* – the business quiz competition; *Adjunction* – the ad creation event; and *Sahithya* – the creative writing competition.

Brainstorming Workshop on PGDM-ABM

The Academy organized a brainstorming workshop on “Strengthening PGDM-ABM of ICAR NAARM and making the students industry-ready” on Sept. 28, 2024 during the Reunirse’24, the NPAA Alumni Meet. The session was attended by more than 200 people including faculty, alumni and current students of PGDM-ABM program. Dr. Ranjit Kumar, Principal Scientist, set the context by presenting the landscape of agribusiness management education in India and highlighted the need of the brainstorming session. Dr. T. Srinivas, Head, ABM Division presented the status and

challenges of PGDM-ABM program and flagged a few questions for the discussion: increasing brand value, developing niche area and increasing the batch size. The panelists included two senior faculty members and three senior alumni who made significant progress in their respective fields. From the deliberations and discussions, following major recommendations emerged: (i) imparting practical knowledge and fieldwork-based teaching would be more useful for the students in the agribusiness profession; (ii) keeping in view the fast-changing nature of the industry, a course on digital marketing with more emphasis on rural marketing needs to be added to the curriculum; (iii) in addition to knowledge in agriculture, knowledge in AI, ML, IoT and supply chain management is required; (iv) the faculty are encouraged to write articles in magazines and newspapers which will improve branding of the Academy; (v) in recruiters’ perspective, positive attitude, better communication and negotiation skills are long-term winning attributes, which should be inculcated during the academic program.

PGDM-ABM - Educational Visits

Two educational visits were organized for PGDM-ABM students as part of their study. The first visit was organized on Dec. 7, 2024 to Turkapale, Vanganorepalle, and Kalmalonipalle villages of Nagarkurnool district and Galigudem and Agraharam Potlapalli villages of Mahbubnagar district to collect data related

to the impact of interventions under SCSP of the Academy. The second visit was organized on Dec. 8, 2024 to Gudimalkapur flower market, in which students interacted with various stakeholders viz. traders and wholesalers of flower, vegetable and fish. Students were sensitized about the trading process along with the value chain of these agricultural produces.



Fig. 3.7. PGDM-ABM students' visit to SCSP Villages and Gudimalkapur flower market

Table 3.2. List of projects completed by PGDM-ABM Students of the Academy [14th batch] (2022-24)

Student Name (Reg. No.)	Project Title	Guide Name
Adarsh Kumar (ABM-22101)	Consumption Behaviour of Food Prepared at Home And Away From Home In Hyderabad	Dr. N. Sivaramane
Aishwarya G.S. (ABM-22102)	Sugarcane to Jaggery: Economic Analysis of Costing and Marketing Channels	Dr. S.K. Soam
Akhila (ABM-22103)	Pilot study on Credit behaviour of Farmers Producers Organizations in Kerala	Dr. K. Srinivas
Apoorva Chourasia (ABM-22104)	Purchase Intention Survey of Paddy Hybrid NPH-X4 in Madhya Pradesh and Chattisgarh	Dr. G. Venkateshwarlu
Avinash Kumar Jha (ABM-22106)	Climate Resilient Agriculture by adopting a data driven approach supported by Modern Agriculture Techniques helps to mitigate the effect of a Global Warming.	Dr. M.B. Dastagiri

Student Name (Reg. No.)	Project Title	Guide Name
Baloju Rahul (ABM-22107)	A study on Housing Loan & Mortgage Business: with focus on Market Analysis, Credit Culture, Market Opportunities & Competitive Analysis	Dr. P.C. Meena
Banda Abhinaya (ABM-22108)	Mapping of Operations and development of Standard operating procedures (SOPs)	Dr. M. Balakrishnan
Swapnil Bhor (ABM-22109)	Establishing Dhanuka presence for Chilly Crop and Selected Focus Products in the area of deployment	Dr. Surya Rathore
Chappidi Paul Joseph (ABM-22110)	To assess the impact of Wood Survey on farmers relationship with ITC-PSPD	Dr. P. Supriya
D. Pramod Kumar (ABM-22111)	Oil palm Economics	Dr. Ranjit Kumar
Dilbagh Singh Patel (ABM-22112)	Chilli Farmers Survey- Leveraging Data Insights for Procurement Optimization	Dr. N. Srinivasa Rao
Gunavardhan (ABM-22113)	An overview of Agri-tourism in India : A case study on Agri-tourism destination : Jindam Agro Farms Telangana	Dr. P. Ramesh
Hari Prasad Vasantula (ABM-22114)	Feasibility report on the manufacture of briquettes	Dr. B. Ganesh kumar
Himanshu (ABM-22115)	Market Intention survey on Mustard	Dr. S. Senthil Vinayagam
Indudhar S.M. (ABM-22116)	Analysis of total seed loss focused on packaging and processing loss.	Dr. P.D. Sreekanth
Ishita Chattopadhyay (ABM-22117)	Market Study and Client Mapping of Milk Cooperatives and Milk Unions for the Heat Index Product	Dr. N. Sivaramne
Jeevitha N. (ABM-22118)	Assessment of performance of the product 'Kranti' in Karnataka	Dr. V.V. Sumanth Kumar
Jithin Babu (ABM-22119)	Analysis of Sensor-Based IoT, AI, and ML Startups in Precision Agriculture	Dr. Sanjiv Kumar
Jupally Kavya Sri (ABM-22120)	Market bench marking of IFBL products	Dr. S.K. Soam
Kanchan Dubey (ABM-22121)	Purchase intention survey and consumer behavior of "ASHA-1" variety of NSL	Dr. M. Ramesh Naik
Kartik (ABM-22122)	Navigating ESG Landscapes in India	Dr. Tavva Srinivas
Keerti Raje (ABM-22123)	Evaluating the business operations of Green Delights and suggesting the action plan to enhance the business profitability.	Dr. Surya Rathore
Kharat Mrunal Pandurang (ABM-22124)	Establishing Dhanuka presence for chilly crop and selected focus products in the area of deployment"	Dr. Umesh Hudedamani
Kajal Khobragade (ABM-22125)	Establishing Dhanuka presence for Chilly Crop and Selected Focus Products in the area of deployment	Dr. Ramasubramanian V.
Kritika Subhashini (ABM-22126)	Estimation of seed processing potential of Bihar and suggesting strategies for forward and backward linkages for seed production for FPCs.	Dr. M. Ramesh Naik
M.R. Gourish (ABM-22127)	Market Research & Agri-Tech Startup Leads Generation For Credit analysis to startups.	Dr. Ch. Srinivasa Rao

Student Name (Reg. No.)	Project Title	Guide Name
Madhusudan M. Sakare (ABM-22128)	"Establishing Dhanuka presence for Chilly Crop and Selected Focus Products in the area of deployment."	Dr. M. Balakrishnan
Musale Dhamaraka Kishan Rao (ABM-22129)	Establishing Dhanuka presence for Chilly Crop and Selected Focus Products in the area of deployment	Dr. Tavva Srinivas
Muthina Poojitha (ABM-22130)	Export - import analysis of Coffee	Dr. B.S. Yashavanth
Nalla Pruthwi Teja (ABM-22131)	Optimizing Business Development through Fruit Retailer Onboarding	Dr. N.A. Vijay Avinashilingam
Nitesh Dhakad (ABM-22132)	Analyzing the Kharif Maize variety "Sardar 918" Market Size in Uttar Pradesh.	Dr. P.C. Meena
Saurav Patil (ABM-22133)	Establishing Dhanuka presence for Chilly Crop and Selected Focus Products in the area of deployment	Dr. P. Venkatesan
Kavya Pentakota (ABM-22134)	Market Study on Poultry Industry in India	Dr. N.A. Vijay Avinashilingam
Pooja D.N. (ABM-22135)	Uncovering new market opportunities for Bhoosiri Crop Science Pvt. Ltd.	Dr. P. Ramesh
Poonam (ABM-22136)	Exploratory research on market dynamics, export potential and strategy for profitable trade of agri products	Dr. B.S. Yashavanth
Relangi Gayathri (ABM-22137)	Agriculture insurance in India- A Pmfby perspective	Dr. P.C. Meena
Rokalla Lalitha Bhavani (ABM-22138)	Study of Farmer Market linkages in India	Dr. G.R.K. Murthy
Saineni Abhigna (ABM-22139)	Special projects & Initiatives: Building Asset Class Farmer Producer Organizations	Dr. B. Ganesh Kumar
Samhitha Padhi (ABM-22140)	Comprehensive analysis of Livestock Insurance in India: Status, Business models, Companies, Government Policies and Startups	Dr. Sanjiv Kumar
Satyam Purohit (ABM-22141)	Value chain analysis of ground nut	Dr. D. Thammi Raju
Saurabh Dixit (ABM-22142)	Analyzing the Kharif Maize variety "Sardar 918" Market Size in Uttar Pradesh	Dr. G. Venkateshwarlu
Saurabh Singh Pokhriyal (ABM-22143)	A comprehensive Study Focusing on Indian Scenario & Export Feasibility of Onion and Lead generation	Dr. D. Thammi Raju
Seeram Supriya (ABM-22144)	Mapping of crop monitors (Ag Farm app) and development of SOP	Dr. Sanjiv Kumar
Sk. Karishma Alias Nafish (ABM-22145)	Analyzing the Kharif Maize variety "Sardar 918" Market Size in Uttar Pradesh	Dr. N. Srinivasa Rao
Shikha Arora (ABM-22146)	Purchase Intention survey	Dr. A. Dhandapani
Shivaraj Kumar Chavan (ABM-22148)	Consumer behaviour and value addition of millets and its export potential	Dr. P. Supriya
Sumit Sharma (ABM-22149)	Export-Import Scenario of Onion in India	Dr. A. Dhandapani
Surakanti Nandini (ABM-22150)	Assessment of effectiveness and impact of Agri udaan programme on accelerated start-ups in a-IDEA.	Dr. S. Senthil Vinayagam

Student Name (Reg. No.)	Project Title	Guide Name
Viraj Thombare (ABM-22151)	Feasibility Study on Viability of Cashew Export Business in India	Dr. G.R.K. Murthy
Thukuntla Nithya Sri (ABM-22152)	Scope of Agri-Startups and financial analysis	Dr. P.D. Sreekanth
Tumati Bhardwaj (ABM-22153)	Market Bench Marking of IFBL Products	Dr. Ramasubramanian V.
Umesh Chandra Ravirala (ABM-22154)	Production and Marketing of Lemon in Nalgonda District.	Dr. K. Srinivas
Vaishnavi Bajpai (ABM-22155)	Special projects & Initiatives: Building Asset Class Farmer Producer Organizations	Dr. P. Venkatesan
Vajjala Tejaswini (ABM-22156)	Special projects & Initiatives: Building Asset Class Farmer Producer Organizations	Dr. V.V. Sumanth Kumar
Vinay Kumar (ABM-22157)	Crop Mapping of Non PMFBY crops	Dr. Umesh Hudedamani
Yashwanth Kumar Gusidi (ABM-22158)	To assess the impact of Wood Survey on farmers relationship with ITC-PSPD	Dr. Ch. Srinivasa Rao
Yerrabelli Ramya (ABM-22159)	Food losses and wastage	Dr. Ranjit Kumar

Table 3.3. NAARM Faculty's Contribution to IARI MU-Hyderabad Hub

Faculty	Course No.	Course Title	Whether course Leader/ Associate
Dr. Ramasubramanian V.	STAT522	Data analysis using statistical packages	Leader
	STAT511	Experimental Designs	Associate
Dr. R. Venkattakumar	PGS 505	Agricultural Research, Research ethics and Rural Development programmes	Associate
Dr. G.R. Ramakrishna Murthy	UENGG 102	Farm Machinery and Power	Associate
Dr. D. Damodar Reddy	SOIL 962	PhD Seminar	Associate
	SOIL 502	Soil Fertility and Fertilizer Use	Associate
	GBP 101	Agriculture Heritage	Associate
Dr. Surya Rathore	PGS 505	Agricultural Research, Research ethics and Rural Development programmes	Leader
Dr. M. Ramesh Naik	AGRON 502	Principle and Practices of Soil Fertility and Nutrient Management	Leader
	AGRON 601	Current Trends in Agronomy	Leader
Dr. B.S. Yashavanth	UAS 102	Statistical Methods	Associate

Table 3.4. PG Students Guidance by NAARM Faculty

Major Guide	Student	University	Degree	Award date
Dr. Ramasubramanian V.	Mr. Avinash G.	ICAR-IARI, New Delhi	Ph.D. (Agricultural Statistics)	03.02.24
Dr. Yashavanth B.S.	Mr. Manchina Phaneswar	PJTAU, Hyderabad	M.Sc. (Agricultural Statistics)	08.10.24

4

RESEARCH PROGRAMS

Research programs of the Academy focussed on agribusiness, market reforms, education and technology-driven solutions. Key initiatives included Agri-Business Incubation (ABI) centers and RuTAG 2.0 projects supporting startups, grassroots innovations and rural entrepreneurship. Studies on e-NAM and Farmer Producer Companies (FPCs) identified market challenges, while MOOCs and digital content showed potential to enhance higher education outcomes. Environmental research addressed methane emissions and precision agriculture for sustainability. Gender mainstreaming, agripreneurship and impact assessments promoted inclusivity. Additionally, one project explored NEP-2020 implementation in agricultural universities, emphasizing faculty preparedness and challenges for effective policy rollout.

I. Agribusiness Management (ABM)

Establishment of Agri-Business Incubation (ABI) Centers under XII Plan Scheme for NAIF

Under this externally-funded project, a number of programs and activities were organized during the last one year. During Jan. 22-24, 2024, a mentoring and training program was organized to strengthen start-ups with financial and business management skills. Three immersion programs were organized at ICAR-CTRI, Rajahmundry; VAMNICOM, Pune and ICAR-NAARM, Hyderabad for face-to-face interaction among start-ups and Farmer Producer Organizations (FPOs) that

fostered an environment for knowledge sharing, generating collaborative marketing leads, identifying problems and technology gaps. On Dec. 8, 2024, a Buyer-Seller Meet was organized at BCT-KVK, Visakhapatnam to facilitate marketing and networking among the start-ups and other stakeholders in the industry. On Dec. 19, 2024, a Start-ups and Bankers Interface was organized that created a deal-maker opportunity for the start-ups to scale up their businesses with additional investments from the bankers. One of the incubates supported under the project, Shreel Surya Research Pvt. Ltd., received a patent grant for a Bio-Fertilizer. A testament to the success of these initiatives was the substantial amount of fund generation by the start-ups to the tune of Rs. 1.6 crores over the past year, underscoring the positive impact of mentoring and training enterprises. Three startups viz., Shreel Surya Pvt. Ltd., Maharashtra; Bharat Mandi, Karnataka and Dairy Diary Pvt. Ltd., Karnataka showcased their innovations in the National Conference on Building Sustainable Agricultural Startups in India at the Academy during Nov. 12-13, 2024.

Development of Agripreneurship and Agri-business models for Grassroots Innovations (GRIs)

Under this externally-funded project (RuTAG 2.0), following activities and achievements were made. On Feb. 2, 2024, an Action plan meeting was organised to facilitate market enhancement, upscaling, and registration and certification processes for innovations. Subsequently, a Review-cum-Action plan meeting to shortlist



Fig. 4.1. First Annual Review Meeting of RuTAG 2.0 Project at SKUAST-Kashmir

Grass Root Innovations (GRIs) and finalize work plans for future was organized at IIPM, Bengaluru on April 8, 2024 and at ICAR-CIAE, Coimbatore on May 29, 2024, in which experts evaluated the technologies for feasibility, scalability and social relevance. On Aug. 23, 2024, a review meeting of RuTAG 2.0 projects was convened to assess progress across seven RuTAG 2.0 Centers, including ICAR-NAARM Hyderabad, IIT Bombay, IIT Roorkee, IIT Delhi, IIT Madras, IIT Guwahati, and SKUAST Kashmir. This meeting was chaired by Dr. Monoranjan Mohanty and Dr. Hafsa Ahmad from the Office of the Principal Scientific Advisor to the Government of India. A training workshop on ‘Commercialization Strategies for Rural Innovations through Digital Marketing’ was organized at IIPM, Bangalore during Oct. 21-22, 2024. The first Annual Review Meeting of RuTAG 2.0 Projects was subsequently held during Dec. 5-6, 2024 at SKUAST-Kashmir under the leadership of Prof. Ajay Kumar Sood.

Several innovators were supported to participate in exhibitions and showcase their innovations, such as, Agri-Tech Exhibition in Hyderabad (Feb. 16-18), AGRIBUSINESS Conclave-2024 by NITTE & MANAGE (Mar. 2-3) and Startup MAHAKUMBH in New Delhi (Mar. 18-20). Other events organized under the project

were GI MAHOSTAV in Madurai, CODISSIA Agri Intex Exhibition in Coimbatore (July 11-15), TNAU Farmers Expo in Tamil Nadu (Sept. 26-29) and National Conference on Building Sustainable Agricultural Startups in India at Hyderabad (Nov. 12-13, 2024).



Fig. 4.2. National Conference participation – Farm Sathi

Farmer Producer Companies in India: A Study on their Management Practices and Business Potential

The project identified the following problems faced by the FPCs: (i) lack of adequate capital, (ii) low managerial and technical skills of personnel, (iii) limited market access, (iv) inadequate infrastructure, (v) issues with the regulatory compliance, (vi) poor member

participation, (vii) lack of infusion of modern technologies and (viii) poor networking and policy issues. Strategies for overcoming the

hurdles for the development of Farmer Producer Companies were formulated in the form of do's and don'ts (Table 4.1).

Table 4.1. Do's and Don'ts for Farmer Producer Companies (FPCs)

Problem	Do's	Don'ts
Lack of adequate capital	• Develop strategies for raising capital from diverse sources (i.e., contributions from members, financial institutions, partnerships with private investors, etc.). • Explore microfinance institutions and cooperatives for working capital support. • Build trust among members and supporting institutions through transparent communication on financial management and profit distribution.	• Relying entirely on government schemes and subsidies for growth and expansion. • Improper financial planning and hunting for funds during crisis. • Delayed payment to members due to financial crisis.
Low managerial and technical skills of personnel	• Provide training for executives and members, focusing on business management and technical skills. • Training members to take up the management in due course of time. • Hiring professional managers or consultants with expertise in agribusiness. • Develop incentive schemes for manager with good performance.	• Management by individuals lacking formal education and business experience. • Not seeking external professional assistance or mentorship when necessary.
Limited market access	• Develop innovative marketing models like online platforms, e-commerce and direct-to-consumer channels. • Build partnerships with supply chain players, wholesalers and retailers to expand market reach. • Regularly gather and analyse market information to make informed decisions about pricing and distribution.	• Limiting reach to local areas • Ignoring branding and value-added services • Competing only on price basis and lack of differentiation of products
Inadequate infrastructure	• Access third-party infrastructure (i.e., warehouses, cold storage, processing units, transport) to enhance efficiency. • Investment on infrastructure, wherever necessary like processing, packaging, etc. • Seeking government assistance, private partnerships, or funding to improve infrastructure facilities. • Sharing infrastructure with local FPCs or cooperatives.	• Making huge investment on storage and logistics and locking scare capital • Relying on inefficient, outdated facilities/ technologies that increase operational costs.
Issues with the regulatory compliance	• Staying updated on regulatory requirements (i.e., tax compliance, trademarks, trading licenses, etc.) • Consulting professionals (accountants, legal advisors) to manage compliance-related activities. • Establish efficient systems for record-keeping and filing returns to avoid penalties and legal issues.	• Ignoring compliance regulations • Not maintaining proper records • Lack of audit system
Poor member participation	• Actively engage members in decision-making and governance through regular meetings, communication and incentives. • Develop system and create awareness about the benefits of FPCs to members. • Provision more incentives for members who contribute more to the growth of the FPC.	• Not involving natural leaders in the community in FPC management. • Avoiding meetings and ignoring members who are not active participants.

Problem	Do's	Don'ts
Lack of infusion of modern technologies	• Using digital platforms for marketing, sales and communication with customers. • Adopting e-commerce and digital marketing strategies to reach broader markets. • Using ICTs and social media for communication with farmers and providing advisory support.	• Relying solely on traditional methods of marketing and communication. • Improper use of social media and over-marketing.
Poor networking and policy issues	• Partnership with key value chain players, including suppliers, transporters, warehousing companies and financial institutions, to create a strong support network. • Collaborating with other FPCs to leverage shared resources and solve common problems. • Professional training for management to deal with government agencies and pitch for funds.	• Developing own complete supply chain through heavy investment and efforts. • Considering other FPCs and local retailers as competitors. • Liaising with bankers and micro-finance institutions for meeting working capital requirement. • Exploring new networking opportunities such as collaboration with exporters, processors, etc.

Techno-Economic Feasibility Studies for ICAR Agro-Processing Technologies/ Value Added Products

Among the top ten technologies, techno-economic feasibility of two technologies, viz., (i) preparation of milk-based spray dried nanoencapsulated curcumin formulation and (ii) sour dahi using prolific acidifying lactic cultures were studied during the reported period. Curcumin is the bioactive component of *Curcuma longa* (turmeric) which is reported to have anti-oxidant, anti-inflammatory, anti-parasitic, anti-mutagenic, anticancer, chemoprotective, hepatoprotective, antimicrobial and antiviral properties. Incorporation of curcumin in food poses challenges due to its poor water-solubility, low absorption & bioavailability and rapid degradation at neutral & alkaline pH. Microencapsulation reduces particle size, increases bioavailability, stabilizes and protects bioactive compounds in microscopic particles. Spray-drying was found to be a great encapsulation technique and offered benefits in terms of economics (around ten times cheaper than the lyophilization process) and

versatility (various configurations of air and spray flows exist). The technology to produce nanoencapsulated curcumin formulation maintained physicochemical properties of milk powder except the colour. The formulation was highly dispersible in water and stable under various food-processing conditions like pH, ionic strength and heating. This formulation was prepared with food grade additives and minimal processing and shelf stable at ambient temperature. The process has the potential of up scaling at commercial level.

The type of microbial culture used to make dahi (curd) is an important factor because it determines the flavor and texture of the product. The demand for sour dahi was found to be different in different regions of India. Reportedly, people from southern India preferred sour dahi (made mostly with *Lactobacilli* and yeast cultures), whereas people in northern India preferred mildly sour dahi (made with *Streptococci* culture). The type of culture used also differed with seasons – while *Lactobacilli* dominated the dahi culture during summer, *Streptococci* dominated during winter.

New Market Reforms and E-NAM in India: Effect on Marketing Choices and Participation and Price Realization for Smallholders

e-National Agriculture Market (e-NAM) is an electronic trading platform for agricultural commodities in India. It integrates all the physically dispersed APMC mandis and brings both farmers and different trade functionaries on a single platform, resulting in more transparent and competitive price discovery. As of October 2024, there are 1389 mandis spread over 27 states and UTs on-boarded on e-NAM. These e-NAM mandis trade almost 221 types of agricultural commodities. In the present study, to analyse market arrival and benefits

received by the farmers through eNAM, daily transaction data of all the eNAM mandis were collected from eNAM portal for the period Dec. 2020 to Dec. 2023. For making comparison, transaction details from Agmarknet portal were also collected from non-eNAM mandis for the same period. For detailed analysis, three states having higher number of mandis (i.e., Rajasthan, Madhya Pradesh and Uttar Pradesh) and two commodities (i.e., mustard and wheat) were selected for the present study.

Data showed higher fluctuation in average weekly arrival trend of mustard in Rajasthan mandis on e-NAM platform compared to non-e-NAM. Further, the average modal price was found to be significantly lower at e-NAM

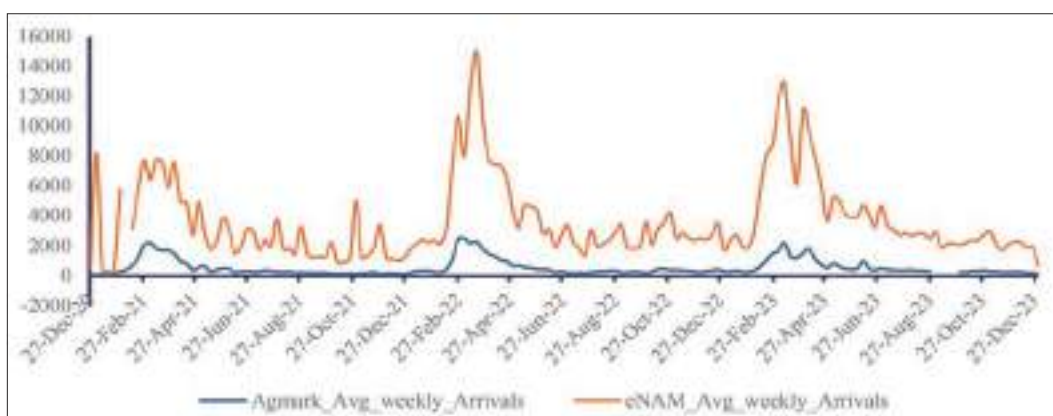


Fig. 4.3. Average Weekly Arrival of Mustard in Rajasthan Mandi (In quintal)

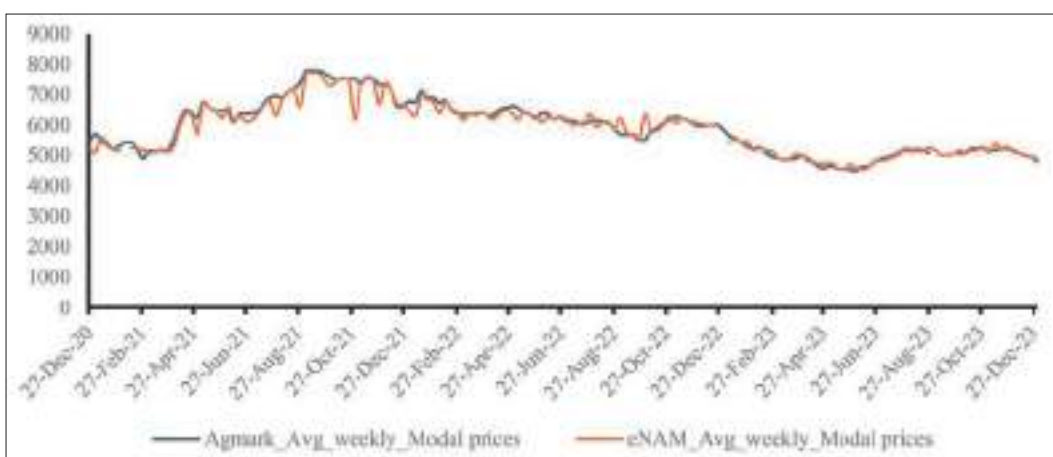


Fig. 4.4. Average Weekly Modal Price of Mustard in Rajasthan Mandi (₹/quintal)

platform during the period [Mean difference: -49.56; $t = 2.9152$, $df = 151$, $p\text{-value} = 0.004096$]. Moreover, there is a cointegrating relationship between eNAM and Non-eNAM average weekly modal prices, suggesting the two prices are related over the long term and tend to move together.

Similarly, the transaction data analyzed for wheat in the mandis of Uttar Pradesh showed higher fluctuation in trade on e-NAM platform. While there is significant difference in average weekly modal price of wheat in the two platforms [Mean difference = -33.46, $t = 4.9772$, $df = 150$, $p\text{-value} = 1.75 \times 10^{-06}$].

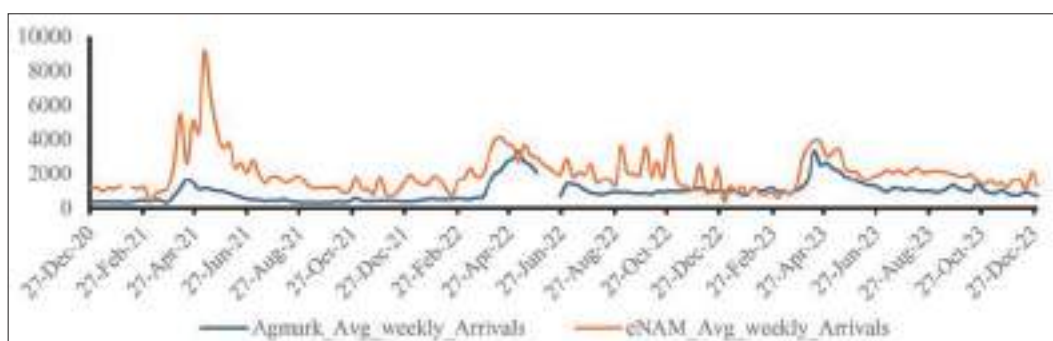


Fig. 4.5. Average Weekly Arrival of Wheat in UP Mandi (In quintal)

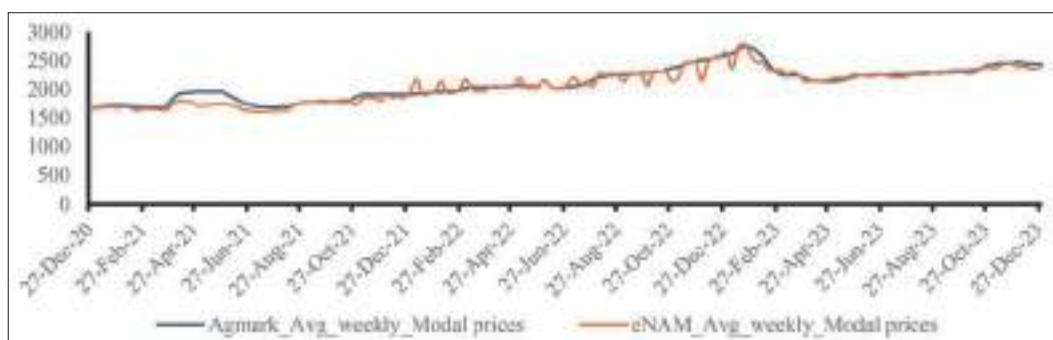


Fig. 4.6. Average Weekly Modal Price of Wheat in UP Mandi (₹/quintal)

From analysis of two large states' data, it appeared that the commodity price on e-NAM platform was lower than non-e-NAM mandi. That might be due to irregular transaction on e-NAM portal. However, in the long-run, the prices on both the mandis are expected to converge.

II. Education Systems Management (ESM)

MOOCs in Digital Agricultural Education - Strategies to Enhance MOOCs

Massive Open Online Courses (MOOCs) have revolutionized education by offering accessible, flexible, and inclusive learning opportunities to global audience. Aligned with

the United Nations Sustainable Development Goal 4 for quality education, MOOCs empower lifelong learning and democratize access to higher education. However, low completion rates remain a critical challenge. Studying the psychological aspects of learners through established theories like the Theory of Planned Behavior, Technology Acceptance Model, Self-Determination Theory and Self-Regulated Learning Theory help uncover reasons for low completion rates. Insights into motivation, self-regulation, and engagement can guide strategies to enhance course design and learner outcomes, maximizing MOOCs' transformative potential. In this project, data were collected

from participants of the Academy's MOOC on "Artificial Intelligence in Agriculture" held in 2024. Out of a total of 4423 registrants, 1197 completed the course. A survey post-completion garnered 458 valid responses. Statistical analysis, including partial least squares (PLS) and confirmatory factor analysis (CFA) was applied to evaluate the measurement model.

Table 4.2 and Figure 4.7 present the partial least squares (PLS) results, highlighting that out of 21 research hypotheses, 18 were supported. The path coefficients ranged from -0.263 to 0.785. The explanatory power of the structural model was assessed using R^2 values, which indicated the variance explained by each endogenous construct. The model showed that Attitudes

(ATT), Subjective Norms (SN) and Perceived Behavioral Control (PBC) explained 63.2% of the variance in Behavioral Intention (BI). ATT was influenced by Perceived Ease of Use (PEOU, 46.8%), Intrinsic Motivation (IM, 40.7%), Self-Efficacy (SE, 22.8%), Perceived Usefulness (PU, 21.1%), Course material (CM, 20.6%), PBC (16%) and SN (14%). SE influenced 61.6% of Goal-Setting (GS), while GS accounted for 48.7% of Task Interest (TI). PBC was influenced by SE (32.5%), IM (26.3%), TI (20.4%), and CM (13.3%). BI influenced 51.4% of performance. Finally, SN was shaped by CM (43.4%) and TI (21.4%). These findings indicated that factors like attitudes, perceived control over behavior, self-efficacy and task interest significantly shaped behavioral intentions and performance.

Table 4.2. The t-value of research hypothesis and path coefficients

Hypothesis	Research Hypothesis	t values	Path coefficient	Result
H1	ATT -> BI	13.401	0.724***	Supported
H2	PBC-> BI	2.517	0.135**	Supported
H3	PBC-> ATT	2.726	0.160***	Supported
H4	SN -> BI	1.265	0.056	Not Supported
H5	SN -> ATT	2.565	0.140**	Supported
H6	PU -> ATT	3.536	0.468***	Supported
H7	PEOU -> ATT	6.87	0.211***	Supported
H8	IM-> ATT	6.309	0.407***	Supported
H9	IM-> PBC	3.779	0.263***	Supported
H10	CM -> ATT	3.908	-0.206***	Supported
H11	CM -> PBC	2.643	0.133***	Supported
H12	IM-> SN	1.649	0.115	Not Supported
H13	CM -> SN	7.908	0.434***	Supported
H14	SE -> GS	12.134	0.785***	Supported
H15	SE -> ATT	3.379	0.228***	Supported
H16	SE -> PBC	4.771	0.352***	Supported
H17	GS -> TI	12.076	0.698***	Supported
H18	GS -> SN	0.082	-0.006	Not Supported
H19	TI -> PBC	2.908	0.204***	Supported
H20	TI -> SN	2.636	0.214***	Supported
H21	BI -> Performance	13.987	0.514***	Supported

Note. ** $p < .05$, *** $p < .01$

This study explored psychological factors influencing MOOC performance by integrating multiple theoretical models. Data from a diverse demographic sample highlighted the role of behavioral intentions, shaped by variables like attitude, self-efficacy and perceived control

in determining performance. The findings reinforce the utility of merged models for understanding learner behavior in MOOCs and propose a unified structural framework for future testing in varied e-learning contexts.

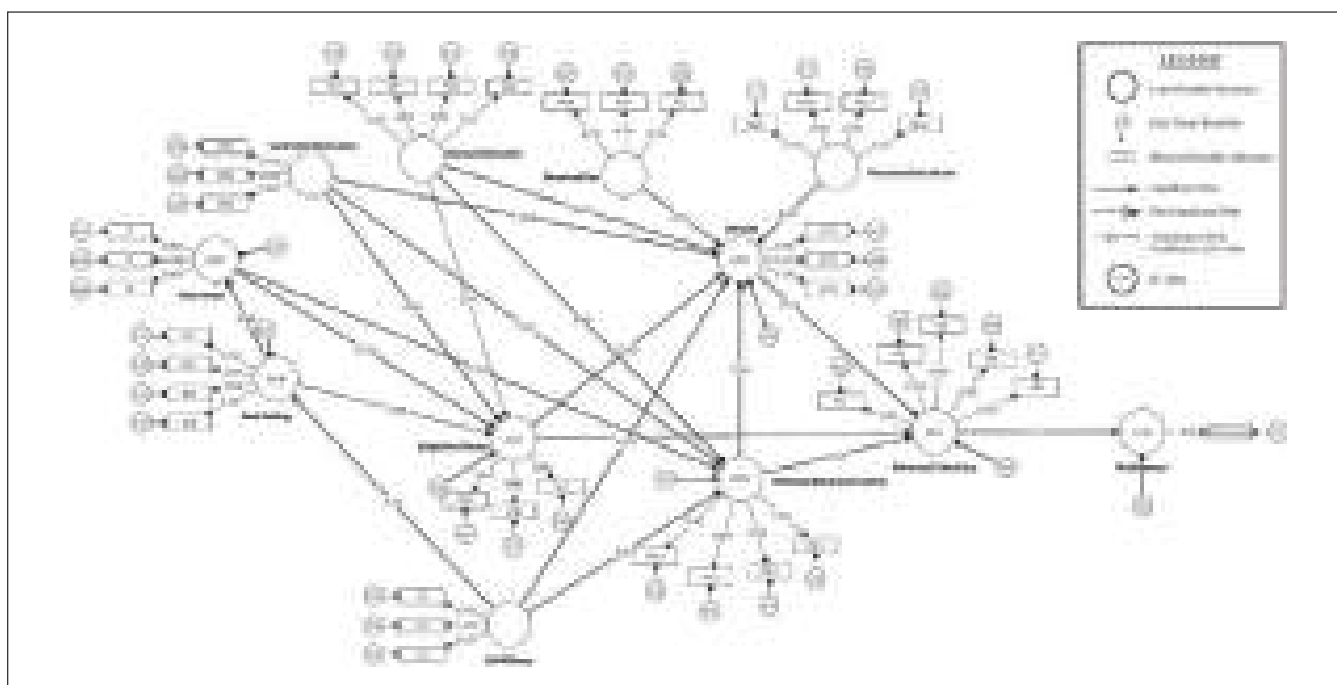


Fig. 4.7. Proposed Theoretical framework for understanding Learner Behaviour in MOOCs

Operationalising Digital Content for MOOCs in Dairy Education

Under this consultancy project, a MOOC on Fat Rich Dairy Products consisting of 17 modules was developed for ICAR-National Dairy Research Institute, Karnal. By leveraging the power of Generative AI tools, virtual sets were created that added a visually appealing and dynamic feel to each video module, enhancing the learning experience. By integrating AI-driven tools into the production and editing processes, high-quality educational content was produced and delivered optimizing

time and resources, at the same time making it more accessible and impactful for learners.



Fig. 4.8. Operationalising Digital Content for MOOCs in Dairy Education

Development of implementation framework for National Education Policy - 2020 in Agricultural Education

During the reported period, the project investigated awareness about National Education Policy (NEP) 2020 among different actors of Agricultural Universities (AUs), preparedness and challenges in its effective implementation. Data were collected from 305 respondents comprising of 273 faculty and 32 administrators from 65 AUs, whose average age was 46.06 years with 23.30% females and 76.70% males; majority of the respondents (88.9%) held doctorates. The majority (34.80%) of respondents had more than 20 years of

experience. Awareness about NEP varied from 8.04 to 8.91 with an average of 8.36 out of 10. Among the different categories of respondents (Assistant Professors, Associate Professors, Professors, HoDs and Administrators (Deans / Directors), the Associate Professors and HoDs had higher awareness on NEP 2020. The mean attitude scores varied from 50.63 to 53.87; Assistant Professors, Associate Professors and HoDs were reported to have more favourable attitude than administrators. The preparedness for implementation of NEP 2020 indicated that majority of administrators (53.13%) were in the medium category while majority of the faculty (41.39%) were in the high category (Table 4.3).

Table 4.3. Preparedness for implementation of NEP

S. No	Category	Administrators (n = 32)		Faculty (n = 273)	
		Fre	%	Fre	%
i.	Low	5	15.63	61	22.34
ii.	Medium	17	53.13	99	36.26
iii.	High	10	31.25	113	41.39
	Mean (SD)	70.06 (17.27)		51.94 (9.97)	

The challenges for implementation of NEP 2020 reported were (i) shortage of faculty, (ii) lack of sufficient infrastructure and budget, (iii) many report responsibilities of the academicians, (iv) lack of competitive peer-reviewed research funding across disciplines, (v) many legislative

changes are needed for implementation, (vi) limited teacher and institutional autonomy, (vii) a severely fragmented higher educational ecosystem and (viii) suboptimal governance and leadership of HEIs (Table 4.4).

Table 4.4. Major issues/challenges perceived by faculty/administrators in the implementation of NEP 2020

S.No	Issue/challenge	Mean	Rank
i.	Shortage of faculty	4.47	I
ii.	Lack of sufficient infrastructure and budget	4.43	II
iii.	Many report responsibilities of the academicians	4.29	III
iv.	Lack of competitive peer-reviewed research funding across disciplines	4.22	IV
v.	Many legislative changes are needed for implementation	4.16	VI
vi.	Limited teacher and institutional autonomy	4.16	V
vii.	A severely fragmented higher educational ecosystem	4.07	VII
viii.	Suboptimal governance and leadership of HEIs	4.05	VIII

For effective implementation of NEP 2020, following points were identified to be important: (i) evolution into large multidisciplinary universities with at least 3000 students, (ii) enhancing gross enrolment ratio, (iii) education management – competencies, pedagogy, (iv) governance and academic leadership – change management, (v) integration of vocational and professional education and (vi) digital education.

III. Human Resource management (HRM)

Motivational Factors Driving Scientific Personnel: Identification of Strategies for Scientific Excellence

The objectives of the project were to (i) identify the motivational factors driving the scientific personnel; (ii) assess the stress levels among scientific personnel; (iii) find out the relationship between motivational factors and stress levels and (iv) develop strategies for improved performance of scientific personnel. The tools used were Big-Five Inventory (BFI-10) and Short Index of Self-Actualization (SISA), Meaning in Life Questionnaire (MLQ) and Depression, Anxiety and Stress Scale (DASS 21). The data collected through an online survey using Google form from research scientists working in National Agricultural Research and Education System (NARES) in India, who were participants in different training programs organized by the Academy. A total of 443 participants (137 females, 30.9%) responded to the study. The participants' age varied from 24 to 63 years with an average of 37.2 years and a standard deviation of 7.23. Their educational levels varied from post-graduation to Ph.D. in agriculture and allied subjects. The participants had professional experience ranging from 1 to 34 years. The study concluded that males had

higher extraversion and neuroticism among personality traits; meaning in life and similar levels of psychological distress, whereas females scored high in self-actualization. Extraversion, conscientiousness and presence of meaning were improved with age, whereas anxiety and depression were decreased with age and this relationship was stronger in males. Agreeableness and openness as personality traits positively contributed towards self-actualization. Higher scores on neuroticism and lower scores on agreeableness, conscientiousness and openness were associated with increased level of psychological distress.

IV. Extension Systems Management (XSM)

Development of a framework for gender mainstreaming in the development schemes of Agriculture sector in India

The project aimed at developing a framework for assessment of gender attribution in developmental schemes of agriculture across Ministries and to develop a status report on mainstreaming gender. Data collection and analysis showed that four development schemes of agriculture had 100% budget allocation (Part A) for women, namely Namu Drone Didi (Ministry of Agriculture & Farmers' Welfare), National Rural Livelihood Mission – Aajeevika (Ministry of Rural Development), Mahila Coir Yojana (Ministry of MSME) and Biogas Programme (Ministry of New & Renewable Energy). There were only two schemes in Part B having 30% budget allocation for women called Krishonnati Yojana formed because of amalgamation of 16 developmental schemes of the Ministry of Agriculture. Another scheme in this category was Pradhan Mantri Matsya Sampada Yojana under the Ministry of Fisheries, Animal Husbandry and Dairying. Analysis of last

five year's data showed that 62 Ministries and 33 States/Union territories had gender budget cells in the year 2024, whereas 57 ministries and 23 States/Union territories had these cells during 2020-22. In 2023, the scenario was like 2020-22 but it increased to 27 for States// Union territories. Maharashtra (85%), Manipur (75), Uttarakhand (6.10), Andhra Pradesh (5.16) and Gujarat (4.78%) were the top five states in agriculture gender budget allocation.

Management and Impact Assessment of Farmer FIRST Project

The Academy, as a key partner institution in the consortium, tasked with supporting the management of the Farmer FIRST Programme (FFP) through capacity building, documentation, information dissemination and impact assessment under this project. During the reported period, field visits were conducted in FFP villages associated with ICAR-Indian Grassland and Fodder Research Institute, Jhansi, Uttar Pradesh and ICAR-Indian Institute of Soil Science, Bhopal, Madhya Pradesh. These visits facilitated focused group discussions with farmers and FFP teams, leading to the documentation of best practices for upscaling and wider adoption. Recognizing exemplary contributions, three farmers from FFP villages were honoured with the "Innovative Farmers Award" during the 49th Foundation Day celebrations of ICAR-NAARM on Sept. 1, 2024.

Agripreneurship for Sustainable Agricultural Development: Technological and Institutional Innovations and Strategies

Under the project, several workshops and training programs were conducted during 2024 to advance agripreneurship and sustainable agriculture. The Lead and Collaborating Institutes conducted the following activities: (i) reviewing

project activities that focussed on identifying agricultural challenges, technological innovations, and an agripreneurship (ICAR-NAARM). ICAR-IISWC trained Women Self-Help Groups in fruit and vegetable processing and held a workshop on medicinal crops and cluster farming. A Farmer Field School at Patanjali Yoga Peeth engaged 50 farmers in value-added products and marketing. ICAR-CSSRI hosted a multi-stakeholder workshop in Karnal on agribusiness challenges, including visits to five Farmer Producer Organizations. ICAR-NIBSM trained participants in mushroom and poultry farming, emphasizing Kadaknath poultry and oyster mushrooms. ICAR-CIFA organized a workshop on agripreneurship in eastern India and a consultation on aquaculture technology commercialization. Surveys on agribusiness ecosystems across multiple regions further promoted innovation and sustainability.

Impact of SCSP on Socio-economic Status of Beneficiaries

Several interventions were implemented under the Scheduled Caste Sub-Plan (SCSP) to support SC farmers, including youth and women in the districts like Nandyal and Kurnool, Andhra Pradesh, which were selected for their high SC population. During the reported period, one village from each district was adopted to provide training and key agricultural inputs, aiming to improve income and livelihoods. A total of 22 training programs were conducted addressing agriculture and allied activities benefiting 1034 farmers (362 male, 672 female) from Telangana, Andhra Pradesh, Karnataka, Kerala, Maharashtra, Uttar Pradesh and Tamil Nadu. Some of these were Integrated Farming, Poultry, Crop Management, Value Addition to Millets, Milk and Meat and Mushroom Cultivation. Primary data on socio-economic status and

feedback on training programs were collected through interviews and questionnaires. Critical inputs were distributed to beneficiaries and two custom hiring centres (having 71 farm implements each) were established in two villages.

V. Information and Communication Management (ICM)

Comprehensive Geo-spatial Analysis of Environmental Change in Farming

The objective of this project was to assess GHGs emission (CO_2 , NO_2 , CH_4) and to develop a suitable methodology for estimation methane (CH_4) emission and environmental pollution distribution pattern. A significant emphasis was placed on CH_4 emission from paddy cultivation,

a major contributor to agricultural GHG emission. Estimating paddy area accurately is crucial for effective agricultural management and addressing environmental challenges. Traditional ground-based survey methods, although widely used, are time-consuming, and labour-intensive. To address these limitations, Sentinel-1 Ground Range Detected (GRD) Synthetic Aperture Radar (SAR) data was utilized to estimate the paddy area. Data provides C-band at 5.405 GHz data, which is ideal for detecting vegetation characteristics under varying weather conditions. The study leveraged dual-polarization configurations, specifically VH (Vertical Transmit, Horizontal Receive), to monitor and classify paddy fields. The process of estimating paddy areas was conducted using the Google Earth Engine (GEE)



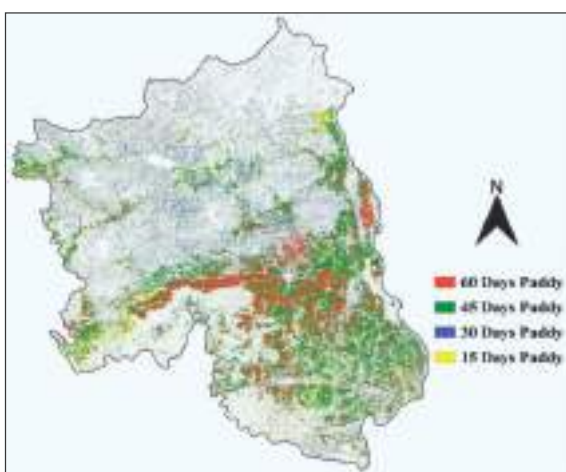
Aerial View of Paddy Fields



Polarization Signature Analysis for Paddy



Classified Paddy Fields

Fig. 4.9. Polarization Signature Analysis

Fig. 4.10. Paddy fields classification with 15 days interval

cloud-based platform. Temporal dynamics of paddy crops are analysed bi-monthly intervals. it allowed tracking of growth stages, including water inundation, vegetative growth and harvesting. A Random Forest classifier, trained on labelled data from various land cover classes (Fig. 4.9). The results of SAR image derived from paddy fields classified into four distinct growth stages (a detailed view of crop maturity depicted in Fig. 4.10) (West Godavari district of Andhra Pradesh). The estimated paddy area and literature data (225.84, 227 in '000' ha, respectively) with error variation of 0.51%.

Identification of similar research studies using machine learning techniques

The project aimed at developing a methodology for identifying similar research studies. Initially, the model was developed and tested for rice-related studies. In the study, a total of 17,744 abstracts containing the word “paddy” or “rice” published in scientific journals during 2013-23 were collected from the Scopus database. From this data, a simple random sample of 5,000 abstracts were utilized for finding out similarity. After pre-processing the text data, each pair of abstracts were tested for similarity using cosine similarity measure. About 85% of the pairs were found to have a similarity of <0.1 . In future, the large language models will be used for finding similarity.

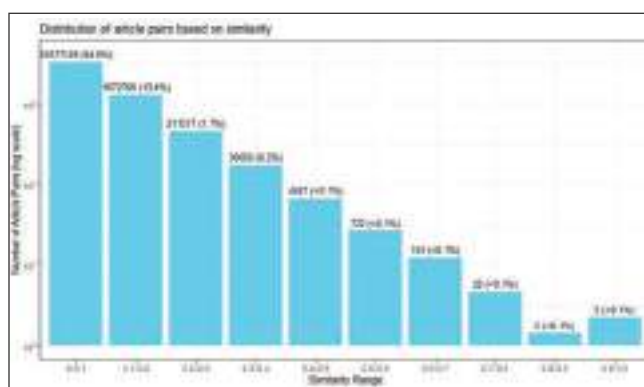


Fig. 4.11. Similarity of research on ‘paddy’ or ‘rice’ published in scientific journals during 2013-23

Mega Environment Analysis Tools for AICRP Trials

The project is aimed at incorporating weather variables in AICRP data to develop prediction models for genotype models using statistical and other approaches. The prediction models are expected to be useful in predicting the performance of genotypes in untested locations using the information available on tested location performances and weather variables in both tested and untested locations. Different types of models were tried to predict the yield

based utilizing the information on genotypes and locations. A growing degree day model was utilized to convert weather information at different locations into a summary measure. The two phenological stages of maize, namely, days to 50% pollen shed and days to 50% silking. These GDD values were calculated from the date of sowing at each location and included in Random effect and fixed effects models. The results showed that (Table 4.4), only in 15 to 24% of models, there is a significant reduction in $G \times E$ component using GDD as covariate in the model as compared to the usual $G \times E$ models.

Table 4.5. No. of Experiments where significant observed over usual model by incorporating GDD as a covariate in Maize

GDD Type	Fixed Location		Random Location*	
	Sig.	Non-Sig.	Sig.	Non-Sig.
Silking	5	28	5	28
Pollen	8	25	7	26

* 10 models had singular fit.

A prediction model was developed using random forest method. Using 33 experiments with at least 5 locations per experiment, cob weight in maize was estimated. The total dataset was divided into 70:30 as Training and Test data with GDD as quantitative and locations and genotypes as categorical predictors. The average mean absolute percent error was 13 % and in the range of 9.8 to 19.5%. The proposed methodology with more fine tuning is expected to be useful in AICRP in understanding the performance of the entries which were not promoted to AVT-I or AVT-II. As a part of this project, a SAS Macro was created for calculation of GDD from weather data and date of sowing for maize crop and the same has been proposed to be included as a part of AICRP Automation system for maize.

GDD based on Days to 50% Silking			
expID=1018			
Entry	Barapani_MG Silking_GDD	Kangra_HP Silking_GDD	Srinagar_JK Silking_GDD
AH 4158	9033.37	9810.43	8689.77
AH 4271	10145.90	9862.83	8588.80
AH 4668	9774.62	9916.47	8184.80
AH 8089	9637.32	9591.90	8132.70
AH 8178	9591.80	9754.43	8790.63
AH 8662	8883.45	9810.43	8699.93
Bio 9544 (Check)	9914.68	9861.73	8523.80
CP 509	10052.98	10025.10	8238.57
DH 353	10004.82	9916.03	8115.50
DH 356	10100.52	9969.70	8359.40
DH 357	9171.27	9756.77	8343.70
DH352	9682.83	9753.77	8689.33
DHM 121 (Check)	10145.90	9861.73	8686.97
IQ 8393	9820.88	9758.37	8625.47
IQ 8624	9837.32	9754.80	8499.87
IW 8477	9913.80	10025.10	8376.53
JH 20096	10055.13	9169.10	8374.70
JH20088	9682.83	9647.73	8696.60
KDMH 132	8834.50	10024.43	8337.53
KMH-21-67	8834.67	8583.80	8549.30
KMH-21-71	9200.27	8635.60	8574.93
PH 42105	10145.90	9861.73	8494.40
PH 42114	10100.52	9755.70	8476.63
PH 42122	9365.68	9275.07	8450.47

Fig. 4.12. Output of GDD Calculation from SAS Macro

KRISHI-ICAR Research Data Repository for Knowledge Management

Under ICAR-Research Data Repository for Knowledge Management, the major activities involve maintaining the AICRP Information System for Sorghum, Small Millets, Maize and support for AICRP on Castor, Sunflower, Safflower, Vegetable Crops. During last year, the AICRP Maize Automation System was used to create, collect and analyze 82 experiments in 811 locations. The analyzed output contains nearly 2.41 lakhs of rows in the database. For AICRP Sorghum, 16 experiments in 188 locations and for AICRP Small Millets, 9 experiments in 118 locations were conducted and data were analyzed using the Automation System producing 25,000 rows and 45,000 rows of output, respectively.

A SAS Macro for analyzing data from factorial experiments was developed and tested which

The SAS System

The GLM Procedure

Class Level Information		
Class	Levels	Values
Irrigation	4	1 2 3 4
Fertilizer	3	1 2 3
Blk	2	1 2

ANOVA Table for yield

Source	DF	SS	MS	FValue	ProbF
Blk	1	287.733750	287.733750	1.28	0.2815
Irrigation	3	957.911250	319.303750	1.42	0.2885
Fertilizer	2	1510.225833	755.112917	3.37	0.0724
Irrigatio*Fertilizer	6	907.097500	151.182917	0.67	0.6739
Error	11	2468.131250	224.375568	—	—

ANOVA Table for yield

CV	DepMean	RootMSE
21.69192	69.05417	14.97917

LSMean Table for yield

Irrigation	Fertilizer			Mean
	1	2	3	
1	82.70	84.65	67.75	78.37
2	74.40	54.05	62.90	63.78
3	78.30	83.00	52.70	71.33
4	69.55	70.00	48.65	62.73
Mean	76.24	72.93	58.00	69.05

Source	DF	SS	MS	FValue	ProbF
Blk	1	3447.507273	3447.507273	8.91	0.0137
Irrigation	3	694.220130	231.406710	0.60	0.6306
Fertilizer	2	190.002593	95.001297	0.25	0.7868
Irrigatio*Fertilizer	6	922.621832	153.770305	0.40	0.8645
Error	10	3868.352727	386.835273	—	—

ANOVA Table for s_yield

CV	DepMean	RootMSE
38.42737	51.18261	19.66813

LSMean Table for s_yield

Irrigation	Fertilizer			Mean
	1	2	3	
1	42.80	58.70	56.30	52.60
2	60.95	40.32	44.55	48.61
3	65.55	61.90	50.30	59.25
4	51.45	38.90	43.30	44.55
Mean	55.19	49.95	48.61	51.25

Warning: Use CD Values or Group Letters, only when the corresponding effect is significant

Dependent Variable: s_yield

Warning: Use CD Values or Group Letters, only when the corresponding effect is significant

Output created by the Macro developed for SSCNARS

Fig. 4.13. SAS Macro sample output

can be used either as a stand-alone macro or can be added to AICRP Automation system. The inputs to the macro are name of the SAS Dataset; variables in the dataset representing the factors, separated by space; names of the variables to be analyzed (for multiple variables using space as delimiter); an optional parameter for replication variable name and options for multiple comparison test, alpha level and group letters (yes/no). The macro can analyze multiple dependent variables with any number of factors in either RBD or CRD setup. The output contains the ANOVA table, CV, Group Letters, adjusted mean (2-way or 3-way tables). A sample output is shown in Fig. 4.13.

Identification and characterization of superior haplotypes for agronomically important traits in Rice

Major genes related to heat tolerance and low phosphorus tolerance in rice (*Oryza sativa*) were

collected from public databases (Heat tolerance genes - *OsSIZ1*, *OsHTAS*, *OsRab7*, *OsBHT*, *HTH5*, *OsHsfB2b*, *OsHSFA2d*, *OsHSF7*, *HTG3*. Low phosphorus tolerance - *PSTOL1*, *Pup1*, *OsPht1;8*, *OsPT8*, *OsMYB2P-1*, *OsSPX1*, *OsSPX2*). Single nucleotide polymorphism (SNP) variations for selected target genes were retrieved from the IRIC SNPSeek database. The haplotype blocks were identified by using the haploview software. Nucleotide-based haplotypes for representative genes were visualized using flapjack. For example, *OsHTAS* (*LOC_Os09g15430*), a key gene associated with heat tolerance was about 20 haplotypes in the 3K Rice Genome panel with variant frequency ranging from 1-2854. The haplotype group 13 was having the highest number of genotypes (2854). A single haplo block was formed for the non-synonymous SNPs of *OsHTAS* gene using haploview.

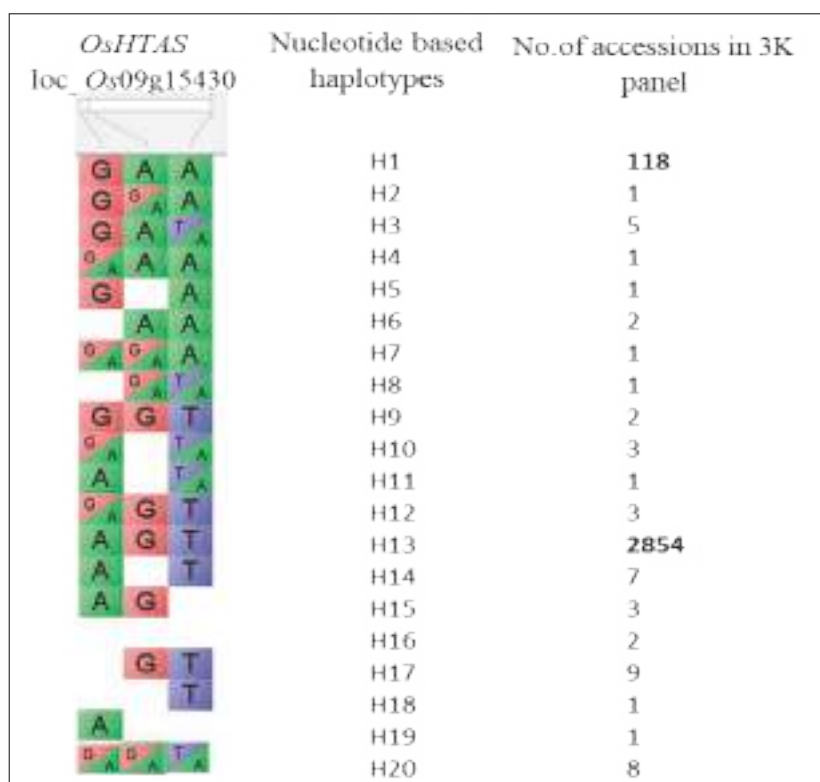


Fig. 4.14. Haplotype analysis of *OsHTAS* across the 3K rice genome panel

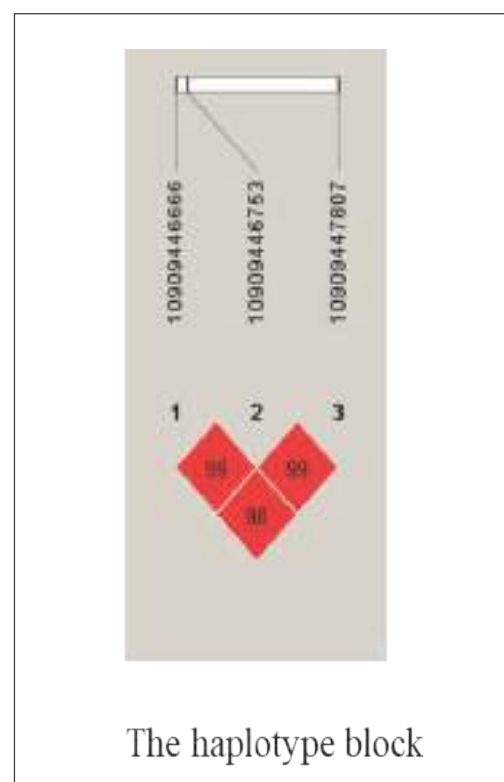


Fig. 4.15. Haplotype block of 3K Rice Genome panel

VI. Research Systems Management (RSM)

IP Management and Commercialization of Agricultural Technology (under NAIF)

Prior art patent searches were successfully conducted by ZTMU for ICAR-IIOR and ICAR-DCR aiding these Institutes in refining their patent filings and strengthening their intellectual property portfolios. Four copyright registrations were secured for intellectual contributions developed at the Academy, namely, (i) Promotion of Interdisciplinary Research in Indian Agriculture: Strategies and Opportunities, (ii) Fostering Entrepreneurial Ecosystem in India: The Agriversity Way, (iii) Green Campus @ ICAR-NAARM: A Dossier and (iv) Organic Farming in India: Production Constraints and Policy Recommendations. As a key enabler of agri-business innovation, the Academy's ZTMU served as a help desk for agri-business incubators, startups and SAUs, providing capacity-building programs and training on technology transfer and IP management. In recognition of its commitment to IP awareness, the World Intellectual Property Day was celebrated on April 26, 2024, with the theme "IP and the SDGs: Constructing Our Common Future Through Innovation and Creativity," reinforcing its role in aligning agricultural research with SDGs.

Dynamics of Agriculture and Food Systems in Different Agroecosystem of India

An assessment of factors driving India's future agricultural food systems and the dietary diversification was made using NSSO data from the 68th (2011–12) and 78th (2019) rounds. The findings revealed a shift towards high-value foods like fruits, meat, and processed items, while traditional staples remain significant. The observed dietary diversification could mainly be attributed to urbanization, income growth, and lifestyle changes. The dietary diversification was conspicuous in urban areas than in rural India.

Emerging picture of food surpluses and deficits was assessed by considering the Telangana state as an example. In the state of Telangana, there has been production surplus of various magnitude for commodities like rice, groundnut, tomato, mango, and papaya, with production exceeding consumption. For instance, the surplus was to tune of 2.85 million tonnes for rice and 1.17 million tonnes for mango. On the other hand, the commodities such as milk, banana, and brinjal recorded huge deficits, with consumption surpassing production. For instance, the deficit was of the order of 3.20 million tonnes for bananas and 2.03 million tonnes for milk. This indicates a need for increased local production or improved supply management for these high-deficit commodities to meet consumption demands effectively.

Indian Food Grain Production towards 2050: Implications for Self Sufficiency, Food Security, SDG's achievements

The study examined the rates of growth, price responsiveness and future forecasts of major crops on a worldwide scale, with a specific focus on India. Its objective was to offer policymakers valuable information to develop sustainable plans for food supply. Cereals and millets play a crucial role in world nutrition, but they encounter substantial obstacles as a result of climatic fluctuations and environmental stresses. Between 1990 and 2022, various crops such as wheat, rice, maize, barley, sorghum and bajra showed different rates of increase on global scale and in India. Crops such as chickpea, lentil, green pea and pigeon pea showed significant growth trends in the pulse sector. Oilseed crops such as soybean, sesame, peanut and safflower exhibited fluctuating production patterns that impact global market dynamics and consumer behavior. Price elasticity revealed changes in consumer behavior in India. These observations are crucial for formulating efficient agricultural policies that promote the well-being of farmers and consumers.

5

THINK TANK/ POLICY RESEARCH ACTIVITIES

The Academy initiated several policy-related studies during 2024 primarily focused on agriculture, food security, education and sustainable development in India with particular emphasis on climate adaptation and women empowerment. The eight policy papers published by ICAR-NAARM addressed critical issues in Indian agriculture and allied sectors, offering strategic recommendations for sustainable development, self-reliance and climate resilience. Viksit Goa 2047 focused on revitalizing Goa's agriculture and allied sectors to achieve self-reliance by 2047 through technological innovation and sustainable practices. The Agro-Eco-Tourism paper proposed strategies to promote and regulate agro-eco-tourism in Goa, balancing economic growth with ecological conservation. The Sustainability of vegetable street vendors paper examined challenges faced by street vendors and recommended policy interventions to improve their livelihoods and ensure food security. The Sheep and goat meat sector paper highlighted the potential of organizing this sector to contribute to India's self-reliance goals, emphasizing breeding, processing and marketing improvements. The Net zero emission paper discussed agriculture's role in achieving India's 2070 net-zero target, advocating for climate-smart practices and renewable energy adoption. The Public-private partnership paper identified challenges in agricultural extension services and recommended strategies for effective collaboration between public and private entities. The Farmers' innovations paper

emphasized integrating traditional knowledge into mainstream agriculture for sustainability, while the Farmer-centric extension system paper called for reforming extension services to better serve farmers' needs through technology and stakeholder collaboration. Collectively, these papers provided a roadmap for sustainable agriculture, farmer empowerment and achieving national development goals like Atmanirbhar Bharat and Net zero emissions.

Brainstorming Workshop on 'Sustainable Rice Production: Strategies and Policies'

A brainstorming workshop on 'Sustainable Rice Production: Strategies and Policies' was organized on Oct. 16, 2024 in which 26 distinguished scientists participated. Dr. B. Venkateswarlu, Former Vice Chancellor, Vasant Rao Naik Marathwada Agricultural University, Parbhani, Maharashtra emphasized on the need for rationalization of rice areas and suggested to focus on improving the paddy yields in low productive North Eastern states of India. Dr. R.M. Sundaram, Director, ICAR-Indian Institute of Rice Research, Hyderabad apprised about the important developments in rice research since Green Revolution and elaborated upon the modern tools and approaches that can be deployed for achieving the goal of sustainable rice production. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM highlighted the importance of sustainable rice production in view of demand for rice from growing population, changing consumer preferences, depleting natural resources, adverse impacts of climate change and the

need for policy interventions for achieving the sustainable development goals linked to sustainable rice production. Five invited talks

by the rice experts followed by two panel discussions were held during the workshop. [Program Director: Dr. B. Nirmala]



Fig. 5.1. Participants of Brainstorming Workshop on 'Sustainable Rice Production: Strategies and Policies' (Oct. 16, 2024)

6

AGRI-INNOVATION DEVELOPMENT AND NURTURING

Centre for Agri-Innovation (CAI), a premier Centre of Excellence of the Academy successfully launched numerous initiatives in agricultural research and innovation management. It is hosting a Technology Business Incubator (named a-IDEA - Association for Innovation Development of Entrepreneurship in Agriculture), with support from Department of Science & Technology (DST) and Agri Business Incubator (NAARM-ABI) for encouraging budding start-ups. The a-IDEA, which was funded by ICAR-NAIF (National Agriculture Innovation Fund) in 2011 received support from DST and the National Bank for Agriculture and Rural Development (NABARD). Additionally,

it has BioNest, an Agri-Bio-incubator for Agri-Biotechnology Startups funded by BIRAC, DBT, Gol.

The a-IDEA nurtures innovations and entrepreneurship in the field of agriculture and allied sectors. The ABI's primary focus is to sensitize, guide, ideate, incubate and accelerate the innovative early-stage startups/entrepreneurs to help them become competitive food and agri-business ventures. Besides supporting start-ups, it has been dealing with issues that affect farming community. Since its establishment in 2014, a-IDEA supported startups across 14 domains of agriculture and allied sectors.



Fig. 6.1. Focus areas of Agri-Innovation Development and Nurturing in the Academy

Agri Udaan 6.0

The Demo Day of Agri Udaan 6.0 was organized on Mar. 22, 2024 at Mumbai by a-IDEA, the TBI of ICAR-NAARM and ICAR-CIFE in collaboration with NABARD and the State Bank of India. The event aimed to support food and agribusiness startups by providing a platform to pitch their ideas before investors, enabling them to scale up and secure funding while fostering networking opportunities. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM emphasized the role of agri-startups in addressing critical challenges in agriculture, including water scarcity and climate change to ensure sustainable livelihoods for farmers. Dr. Ajay Kumar Sood, Deputy Managing

Director, NABARD highlighted the need for startups to work collaboratively, while Shri Shantanu Pendsey, Chief General Manager, SBI stressed the importance of cohesive efforts to solve farmers' problems through institutional interventions.

Dr. C.N. Ravi Shankar, Director & VC, ICAR-CIFE encouraged startups to explore opportunities in the fisheries sector, and Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM commended a-IDEA's mentorship in making startups market- and investment-ready. Out of 156 applicants, 10 startups were selected to participate, and 120 stakeholders and 30 investors attended the event.



Fig. 6.2. Agri Udaan 6.0 at Mumbai (Mar. 22, 2024)



Agri Udaan 7.0

The launch of Agri Udaan 7.0, the Food & Agribusiness Accelerator Program of a-IDEA, TBI, ICAR-NAARM was held at Novotel, HICC, Hyderabad on Aug. 22, 2024. The program aimed to invite applications from food and agribusiness startups nationwide, offering mentorship, industry networking, investor meetups, and funding opportunities to selected startups. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, highlighted the role of startups and Farmer Producer Organizations (FPOs) in addressing key agricultural challenges such as erratic monsoons, market access and

post-harvest losses, while reflecting on the achievements of earlier Agri Udaan editions.

Dr. C. Tara Satyavathi, Director, ICAR-IIMR stressed upon the need for TBIs to support startups in scaling and managing risks. Shri Shantanu Pendsey, CGM, SBI underscored startups' role in improving resource use efficiency and bridging agri-supply chain gaps, and Shri Partho Saha, CGM, NABARD advocated for more startups in the agri-ecosystem to enhance farmers' livelihoods through value addition and reduced drudgery. The event drew 110 participants, including startups, investors, incubators, mentors, and industry professionals.



Fig. 6.3. Udaan 7.0 at Hyderabad organized (Aug. 22, 2024)

To raise awareness, five roadshows were held at key locations viz., PUSA KRISHI in New Delhi, ICAR-CIFE in Mumbai, Anand Agricultural University in Gujarat, TANUVAS in Tamil Nadu, and ICAR-CTCRI in Kerala, each were attended by about 100 participants. As a result of this, the program received an overwhelming response and received 257 applications.

Aggnite 4.0

It was organized during Nov. 23 to Jan. 18, 2024 as a national-level ideation competition to inspire students towards agricultural entrepreneurship and educate them on startup ecosystems. The competition received 133 applications. Following a three-tier virtual evaluation process assessing innovation, problem-solving, team competency, and presentation skills, 3 winners were selected viz., Mr. Bejawada Chanikya Naidu (1st, ₹25,000),



Mr. N. Asif Khan (2nd, ₹15,000), and Ms. Pola Snehaja (3rd, ₹10,000). Awards were presented during the launch of Agri Udaan 7.0 by Dr. Ajay Kumar Sood, Deputy Managing Director, NABARD on Mar. 22, 2024.



Fig. 6.4. Aggnite 4.0 organized during Nov. 23 to Jan. 18, 2024

Immersion Programs

During the past year, two immersion programs were organised to connect FPOs/FPCs, Farmers and Startups. The details are given below.

Location	Date	Startups attended	No. of FPOs attended
VAMNICOM, Pune	Mar. 2, 2024	15	47
ICAR-NAARM in Collaboration with Dhan Academy, Madurai	Jun. 21-22, 2024	14	300 (farmers)



Fig. 6.5. Immersion program organised at Hyderabad (Jun. 21-22, 2024)

Sensitization Programs (SPs) on Entrepreneurship Development in Agriculture and Allied Sectors

During 2024, the Academy organized 20 Online Sensitization Programs (SPs) focused on 'Entrepreneurship Development in Agriculture and Allied Sectors', that recorded participation of 12,713 students from across the country (Table 2.2). Basically, these SPs were supported by the National Bank for Agriculture and

Rural Development (NABARD) to rejuvenate student's minds for taking entrepreneurship as a profession with the objectives: (i) to orient the students about entrepreneurship, (ii) to make students understand the scope for startups in agriculture and (iii) to spread awareness about the startup ecosystem in agriculture and the process of incubation. [Program Director: Dr. N.A. Vijay Avinashilingam]

Table 6.1. Sensitization programs on 'Entrepreneurship Development in Agriculture and Allied Sectors' organized (online) by the Academy during the year 2024.

Sl. No.	Name of the University	No. of students	Date
1	Sri Konda Lakshman Telangana State Horticultural University (SKLTSHU), Hyderabad	629	09.01.24
2	School of Agricultural Sciences & Technology, NMIMS, Pune, Maharashtra	152	18.01.24
3	Acharya N G Ranga Agricultural University (ANGRAU), Guntur, AP	1124	22.01.24
4	School of Agricultural Sciences, IES, Bhopal, MP	105	29.01.24
5	Tamil Nadu Dr. J. Jayalalithaa Fisheries University (TNJFU), Nagapattinam, TN	520	07.02.24
6	Faculty of Agricultural Sciences & Tech., Integral University, Lucknow, UP	206	09.02.24
7	Shobhit University, Meerut, UP	105	24.02.24
8	Assam Agricultural University (AAU), Jorhat, Assam	71	05.06.24
9	Tamil Nadu Agricultural University (TNAU), Coimbatore, TN	6000	14.06.24

Sl. No.	Name of the University	No. of students	Date
10	Veterinary College & Research Institute, TANUVAS, Udumalpet, Tamil Nadu	141	23.07.24
11	Shri Guru Ram Rai University (SGRRU), Dehradun, Uttarakhand	235	30.07.24
12	Chaudhary Sarwan Kumar Himachal Pradesh Krishi Vishwavidyalaya, Palampur, HP	1122	27.08.24
13	Lala Lajpatrai College of Veterinary and Animal Sciences, Hisar, Haryana	290	28.08.24
14	SRS Institute of Agriculture and Technology, Vedsandhur, TN	275	12.09.24
15	Junagadh Agricultural University, JAU, Gujarat	350	17.09.24
16	Kerala Agricultural University, Kerala	137	23.10.24
17	Rajiv Gandhi Institute of Veterinary Education & Research (RIVER), Puducherry	285	26.11.24
18	PGP College of Agricultural Sciences, Namakkal, TN	385	28.11.24
19	Swami Keshawanand Rajasthan Agricultural University, Bikaner	260	03.12.24
20	Anand Agricultural University, Anand	321	13.12.24
Total number of student beneficiaries		12,713	

Accomplishments of Startups incubated/ supported by NAARM

- Garudalytics, a-IDEA's Incubatee was selected for receiving funds from the IITTNiF startup fund.
- AgriVijay's Co-founder & Director Ms. Shobha Chanchlani was a part of the Top 25 Women Entrepreneurs in Green Business organized by WEP (Women Entrepreneurship) - NITI Aayog Unnati Award to Reward Program in association with MicroSave Consulting & SIDBI at NITI Aayog, Delhi.
- AHB Suraksha's, LIVSURAKSHA - herbal dominant liver tonic has received government supply approval through Aanchal Dairy Cooperative located in Uttarakhand.
- Cowbit Technologies Pvt. Ltd., Nidhi Prayas Grantee from the first Call based in Cuttack successfully operating in the

B2B space featured in DST Compendium of Impactful Women-Led Startups Under NIDHI Program, compendium released during 'DST Startup Utsav@ Celebrating 8 years of DST-NIDHI'.

- Fermentech Labs Pvt. Ltd., a Nidhi Prayas Grantee from the third Call based in Uttarakhand successfully operating in the B2B space featured in DST compendium of 50 successful startups Under NIDHI Program, compendium released during 'DST Startup Utsav@ Celebrating 8 years of DST-NIDHI'.
- Renewagri Om Ecommerce Pvt. Ltd., officially signed an MoU with German Companies Next2Sun / Wattkraft represented by Mr. Sascha Krause-Tuenkerto (Investor & CEO Next2Sun) and Mr. Yogendra (Wattkraft) to scale AgriVoltaics projects in India.
- Aqua Doctor Solutions, founded by Dr. Debtanu Barman and one of the leading

innovators in Aquaculture & Fisheries, was awarded the Seed Grant by NABARD and IIT Kharagpur during Startup Expo. 2024 organized by ABIF at IIT Kharagpur during Oct. 24-25, 2024.

- Bhojpatta Agripreneur Pvt. Ltd., Bihar, founded by Mr. Nitish Kumar, participated in the National Agri-Startup Expo (NASE-2024), Empowering Agri-Innovation for a Sustainable Farming organized by RKVY KRISHIK-Agri Business Incubator, UASD and University of Agricultural Sciences, Dharwad during Dec. 27-28, 2024.

Awards/ recognitions of Startups incubated/ supported by NAARM

- Nyasta Gramojvala Solutions Pvt. Ltd., founded by Ms. Bala Bhargavi Narisetty, received the Best Early Stage Start-up award on the occasion of 49th Foundation Day of NAARM on Sept. 1, 2024.
- Chimertech Pvt. Ltd., founded by Dr. Ragul Paramasivam and Dr. Karthik Deekonda, was awarded the Best Growth Stage Start-up award on 49th Foundation Day of NAARM on Sept. 1, 2024.
- Shree Kamdhenu Panchgavya Udyog was conferred with "Best Startup Award 2024" which was presented by Aasha Kiran Sansthan, APC Academy, CoE.
- Chimertech Private Limited won the Aavishkaar Kisan Samarthan Award along with a cash prize of Rs. 1 lakh at the AgriTech Pavilion in the Startup Mahakumbh 2024 held in New Delhi during Mar. 18-20, 2024.
- Dr. Sumit Saxena, Managing Director of Healventure Biosciences became a distinguished member of the International Trade Council's Business Council on Sustainability and Environment.
- Mr. Raman Kumar, founder of Agrifeeder Agricultural Services Pvt. Ltd. was awarded as "The Upcoming Agritech Startup of 2024" at the prestigious Wolfden Investors Summit in Mumbai.
- Khadyam Speciality Foods Pvt. Ltd. founded by Mr. Sitaram, an innovative Agri startup specializing in millets, was honored as the 'Best Agri Startup in Millets' by FICCI and Govt. of Rajasthan during the Rajasthan Millet Conclave 2024.
- Renewagri Om Ecommerce Pvt. Ltd. earned 1st place under the 'Best Agtech in Sustainable Agriculture' at the 6th edition of FICCI Agri Startup Awards held in New Delhi.
- Garudalytics founded by Dr. V.S.S. Kiran was recognized as an HDFC Tech Innovator 2024 out of 2,100+ startups.

7

FARMER OUTREACH PROGRAMS

In the year 2024, the Academy organized a multitude of Farmer Outreach Programs particularly under the Scheduled Caste Sub-Plan (SCSP) to create awareness about the emerging profitable farm enterprises/ sustainable practices, skill development and capacity building for scheduled caste farmers across India. A total of 19 programs were conducted under SCSP with a budget allocation of Rs. 91.75 lakhs and benefitted 1,452 farmers. Key initiatives included training on integrated farming systems, scientific beekeeping, poultry production, vermicomposting and eco-friendly

bagmaking. Programs also emphasized women's empowerment through training in fruit/ vegetable processing, fabric printing and tailoring. Additionally, demonstrations and distribution of agricultural inputs were held to support farmers. These programs aimed to enhance rural livelihoods, promote sustainable agricultural practices and improve productivity, aligning with the Academy's mission to empower farmers and strengthen Indian agricultural sector. A detailed note on the programs organized is furnished in Table 7.1.

Table 7.1. Farmer Outreach Programs organized under SCSP during 2024

Sl. No.	Program Name	Duration	Collaborator/ Venue	No. of participants	Program Director
1.	Skill Development Training Program on Propagation and Nursery Management in Horticulture	Jan. 21-27, 2024	ICAR-KVK, Kalaburagi	35	Dr. M. Balakrishnan
2.	Skill Development Training Program on Integrated Farming System	Jan. 30 to Feb. 3, 2024	ICAR-K.H. Patil KVK, Gadag, Karnataka	40	Dr. M. Balakrishnan
3.	Skill Development Training Program on Integrated Farming System	Feb. 12-16, 2024	ICAR-KVK, KSNUAHS, Shivamogga	37	Dr. M. Balakrishnan
4.	Skill Development Training Program on Scientific Beekeeping	Feb. 19-26, 2024	ICAR-KVK (DK), Mangaluru	25	Dr. M. Balakrishnan
5.	Skill Development of Women on Fruits/ Vegetable Processing and IFS (2 programs)	Feb. 19-24, 2024	KVK II, Sitapur, Lucknow	130	Dr. M. Ramesh Naik
6.	Skill Development Program on Vermicompost Production Technology	Feb. 21-22, 2024	ICAR-KVK (DK), Mangaluru	50	Dr. M. Balakrishnan
7.	Training Program on 'Self-employment through Poultry Production'	Feb. 26-28, 2024	KVK, Ashokenagar, West Bengal	75	Dr. M. Ramesh Naik

Sl. No.	Program Name	Duration	Collaborator/ Venue	No. of participants	Program Director
8.	Skill Development Training on Profitable Sheep and Goat farming	May 21-25, 2024	KVK, Yagantipalle, AP	30	Dr. M. Balakrishnan
9.	Training Program on Vegetable crop production	Jun. 5-7, 2024	KVK, Pune, Maharashtra	60	Dr. M. Ramesh Naik
10.	Empowering Rural Women Through Fabric Printing and Dyeing Techniques	Jul. 10-12, 2024	KVK, Darsi, AP	30	Dr. M. Ramesh Naik
11.	Skill Development Training Program on Backyard Poultry Management	Jul. 24-26, 2024	SVVU, Tirupati, AP	30	Dr. M. Balakrishnan
12.	Skill Development Training Program on Tailoring Trade	Aug. 21-30, 2024	ICAR-K.H. Patil KVK, Gadag, Karnataka	20	Dr. M. Balakrishnan
13.	Skill development training program on Eco-friendly Bag making	Aug. 27-31, 2024	Dr. K.L. Rao KVK, Garikapadu, AP	30	Dr. P. Supriya
14.	Skill Development Training Program on Dairy Farming	Aug. 27 to Sept. 2, 2024	CVSc, Rampura Phul, Bathinda, Punjab	50	Dr. M. Balakrishnan
15.	Skill Training program on Backyard Poultry Farming	Aug. 29 to Sept. 2, 2024	Dr. K.L. Rao KVK, Garikapadu, AP	30	Dr. P. Supriya
16.	Demonstration-cum-distribution of Agricultural Inputs and Farm Tool Kits	Sept. 5, 2024	Midthurat Nandhyal, AP	350	Dr. M. Ramesh Naik
17.	Demonstration-cum-distribution of Agricultural Inputs and Farm Tool Kits	Sept. 6, 2024	Kurnool, AP	350	Dr. M. Ramesh Naik
18.	Skill Development Training Program on Organic-cum-Natural Farming	Nov. 25-29, 2024	ICAR-KVK, Kallakurichi, TN	40	Dr. M. Balakrishnan
19.	Skill Development Training on Backyard Poultry Management	Dec. 17-21, 2024	ICAR-KVK, Mamnoor, TG	40	Dr. M. Balakrishnan
		Total		1452	



Fig. 7.1. SD-TP on Propagation and Nursery Management in Horticulture (Jan. 21-27, 2024)



Fig. 7.2. SD-TP on Integrated Farming System (Jan. 30 to Feb. 3, 2024)



Fig. 7.3. SD-TP on Scientific Beekeeping at ICAR-KVK (Dakshina Kannada), KVAFSU, Mangaluru (Feb. 19-26, 2024)



Fig. 7.4. Skill Development Training on Profitable Sheep and Goat farming (May 21-25, 2024)



Fig. 7.5. SD-TP on Backyard Poultry Management (Jul. 24-26, 2024)



Fig. 7.6. SD-TP on Tailoring Trade (Aug. 21-30, 2024)



Fig. 7.7. SD-TP on eco-friendly bag making (Aug. 27-31, 2024)



Fig. 7.8. SD-TP on Dairy Farming (Aug. 27 to Sept. 2, 2024)

8

LINKAGES AND COLLABORATIONS

- (i) Memorandum of Understanding (MoU) signed between ICAR-NAARM, Hyderabad and the School of Agriculture at S.R. University, Warangal, Telangana, on Feb. 12, 2024.



Fig. 8.1. Signing MoU with School of Agriculture, S.R. University, Warangal

- (ii) Memorandum of Understanding (MoU) between ICAR-NAARM, Hyderabad and Acharya N.G. Ranga Agricultural University (ANGRAU), Guntur, Andhra Pradesh for facilitating Students' Training/Postgraduate Research on Oct. 4, 2024.



Fig. 8.2. Signing MoU with Acharya N.G. Ranga Agricultural University, Guntur

- (iii) Memorandum of Understanding (MoU) between ICAR-NAARM, Hyderabad and Vignan University, Guntur, Andhra Pradesh for facilitating Students' Training/Postgraduate Research on Oct. 22, 2024.

9

REVIEW AND ADVISORY COMMITTEES

VIII Quinquennial Review Team (QRT) Meeting

The VIII QRT reviewed Academy's activities and achievements for the last seven years (01.04.17 to 31.03.24) by organizing three separate meetings (April 15-16, April 29-30 and May 24) in 2024. The QRT looked at the evolving processes and practices, interacting with the Faculty & Staff, as well as a wide range of stakeholders comprising of senior Management of the Council, Deputy Directors General &

Assistant Directors General and Directors of ICAR Institutes, Vice Chancellors and Deans & Directors of Agricultural Universities, ARS scientists and trainees, students, farmer producer organizations, agri-startups and farmers. The team opined that various programs were elegantly carried out by the Academy including capacity building, research and academic programs, which were wide ranging and addressed a very large number of functionaries in the NARES.



Fig. 9.1a. Dr. S. Ayyappan, Chairman addressing at the first meeting of VIII QRT



Fig. 9.1b. VIII Quinquennial Review Team of the Academy

As part of the meeting, an interaction of the QRT was organized on April 29, 2024 with ATARI Directors, DEEs of SAUs, Heads of KVKs and Extension Scientists/ HoDs of ICAR Institutes.

This program was attended by 41 scientists and Coordinated by Dr. R. Venkattakumar, Head, XSM Division and Dr. P. Venkatesan, Principal Scientist.



Fig. 9.1c. Members of the VIII QRT with the participants of the program

Second Meeting of IX Research Advisory Committee (RAC)

The second meeting of IX Research Advisory Committee (RAC) was held at the Academy during Sept. 3-4, 2024. Chairman of the Committee Dr. Chintala Govindarajulu, Former Chairman, NABARD and the members complimented the Institute for its contribution since last meeting held during Oct. 2023. The Committee opined that the Academy should

focus on skill development programs as per the National Education Policy and to take up projects related to meta-analysis and MSME for providing policy suggestions to stakeholders. Dr. Ch. Srinivasa Rao, Director presented Institute's activities for the past year and Dr. Tavva Srinivas, Head, ABM Division & Member Secretary presented the action taken report on the recommendations given during the first meeting.



Fig. 9.2a. Second Meeting of IX-Research Advisory Committee of the Academy (Sept. 3-4, 2024)



Fig. 9.2b. IX-Research Advisory Committee of the Academy

29th Institute Research Council (IRC) Meeting

The 29th Institute Research Council (IRC) meeting was conducted during Nov. 5-6, 2024. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, in his opening remarks urged that scientists could translate the outcomes of their projects into capacity building modules and policies. Dr. Seema Jaggi, ADG (HRD), ICAR who attended the meeting virtually as an external expert acknowledged and complimented NAARM for

expanding its span of activities in NARES. She suggested to take up more externally funded projects by the Academy. During the meeting, a total of 14 in-house, 10 extra-mural and 17 new in-house projects were presented and discussed in seven technical sessions – one each extra-mural and in-house project were completed, 17 new in-house projects were approved, and 10 projects were extended to accomplish the objectives.



Fig. 9.3a. Chairman and Members of 29th Institute Research Committee (IRC) (Nov. 5-6, 2024)



Fig. 9.3b. 29th Institute Research Committee (IRC) meeting (Nov. 5-6, 2024)

10

INFRASTRUCTURE DEVELOPMENT

Name of the infrastructure	Details	Budget (Rs. in lakhs)	Date of completion/ inauguration
Farmers' Hostel	The Farmers' Hostel has 72 beds, one dining hall, kitchen and classroom facilities for skill development training of farming community.	3,99,50,000/-	19.04.2024
Multi-input Digital Content Production Unit	This IP-based system, facilitates the capture, production, and streaming of high-quality multimedia content with ease and efficiency and provides format flexibility up to Full HD (1080p) resolution. It features in-built streaming with multi-destination delivery, enabling live streaming to multiple platforms. The system includes over 20 live virtual templates, inbuilt graphics and effects, live title overlay, and the Record-and-Play option.	35,67,257/-	19.04.2024
Center for Agri Innovation (Technology Bhawan)	Extended the incubation space for startups and created a structure for organizing exhibitions.	27,08,510/-	30.04.2024
Custom Hiring Centres	Custom Hiring Centre having 71 farm implements in Midthur village, Nandyal District, Andhra Pradesh (under SCSP).	35,56,393/-	30.09.2024
	Custom Hiring Centre having 71 farm implements in H. Muravani village, Peddakadubur Mandal, Kurnool District, Andhra Pradesh		01.10.2024
Elsevier-Scopus	Subscription of Abstract-cum-Citation Database of Elsevier-Scopus	21,64,324/-	01.02.2024



Fig. 10.1. Inauguration of Multi-input Digital Content Production Unit at Centre of Lifelong Learning in Agricultural Education (COLAgE)



Fig. 10.2. Inauguration of Farmers Hostel by Dr. Himanshu Pathak, Secretary DARE & DG, ICAR at the Academy (Apr. 19, 2024)



Fig. 10.3. Establishment of Custom Hiring Centre in Midthur village, Nandyal District, Andhra Pradesh



Fig. 10.4. Establishment of Custom Hiring Centre in H. Muravani village, Peddakadubur Mandal, Kurnool District, Andhra Pradesh



Fig. 10.5. Inauguration of renovated 'Technology Bhawan' on Apr. 30, 2024

Table 11.1. Training Programs/Summer schools/Winter schools attended in India

Staff Name	Program	Duration (From-To)	Organizers/ Place
Dr. Ramasubramanian V.	Interpersonal Effectiveness and Team Building	Jan. 22-25, 2024	IIM, Ahmedabad
	MDP on Leadership (a Pre-RMP Program)	Jun. 3-12, 2024	ICAR-NAARM, Hyderabad
Dr. Tavva Srinivas	Training on Managerial Effectiveness	Jan. 29 to Feb. 3, 2024	IIM, Ahmedabad
	MDP on Leadership Development (a Pre-RMP Program)	Feb. 2-13, 2024	ICAR-NAARM, Hyderabad
Dr. R. Venkattakumar	MDP on Leadership (a Pre-RMP Program)	Jun. 3-12, 2024	ICAR-NAARM, Hyderabad
Dr. B. Ganesh Kumar	MDP on Leadership Development	Jun. 3-7, 2024	ICAR-NAARM, Hyderabad
Dr. K. Srinivas	Advanced MDP "Branding practices and strategies in Digital Age for Startups"	Dec. 23-28, 2024	IBPR, SKUAST-K, Shalimar, Srinagar
Dr. M. Balakrishnan	Executive Education Program on 'Design Thinking'	Jul. 15-17, 2024	IIM, Bangalore
Dr. D. Thammi Raju	Design Thinking	Aug. 23-24, 2024	IIM, Bangalore
Dr. Dipesh Debnath	Training program on Enhancement of Teaching Competencies of Agri. Faculty	Mar. 11-15, 2024	ICAR-NAARM, Hyderabad
Dr. Sona Yengkokpam	Training Program on Multivariate analysis using R	Aug. 26-30, 2024	ICAR-NAARM, Hyderabad
Dr. K. N. Yamini	Training Program on Enhancement of Teaching Competencies of Agri. Faculty	Mar. 11-15, 2024	ICAR-NAARM, Hyderabad
Dr. Umesh Hudedamani	Training on Commercialization Strategies for Rural Innovations through Digital Marketing	Oct. 21-22, 2024	IIPM, Bangalore
Dr. P. Supriya	Online course management through MOOCs	Jan. 1-12, 2024	ICAR-NAARM, Hyderabad
	MOOCs on Artificial Intelligence in Agriculture	Mar. 1-31, 2024	ICAR-NAARM, Hyderabad
Dr. B.S. Yashavanth	Winter School on "Use of Big Data Analysis in Agriculture"	Jan. 18-Feb. 7, 2024	UAS, Dharwad
	Data Visualization and Storytelling using Data	Aug. 26-28, 2024	IIM, Bangalore

Table 11.2. Visits Abroad

Staff Name	Program	Date (from – to)	Organized / Sponsored by	Place	Purpose
Dr. Ch. Srinivasa Rao	5 th INSA-Royal Society, UK Yusuf Hamied Bilateral Workshop and part of delegation to UK.	Nov. 26-27, 2024	The Royal Society, London, UK and INSA, New Delhi	London, UK	Distinguished Speaker in Session 7: Moving to a system-levels approach: Biodiversity-water-energy-food production

Table 11.3. Participation in Workshop/Conferences/Symposia/ Meetings (outside NAARM)/Farmers' meet etc.

Staff Name	Program	Organized by/ Venue	Date
Dr. Ch. Srinivasa Rao	National Seminar on Agricultural Education and Research Empowerment for Climate Resilient Production Systems to Enhance Bio-Economy	University of Agricultural Sciences, Dharwad	Jan. 3, 2024
	National Workshop on Ecosystem Services in Agricultural Landscape of Himalayan Region of India	ICAR-ATARI, Zone VI, Ziro, Arunachal Pradesh	Jan. 11, 2024
	National Seminar on Smart Technologies for Sustainable Agriculture and Environment	ICAR-CRIDA, Hyderabad	Feb. 22, 2024
	Vice Chancellors and Directors Conference of ICAR	NASC Complex, New Delhi	Feb. 26- 27, 2024
	Golden Jubilee program organised by the National Academy of Sciences, India	IIT, New Delhi	May 18, 2024
	Skill Development Training program on Profitable Goat and Sheep Farming under SCSP	KVK, Yagantipalle, Nandyal, AP	May 21, 2024
	Training program on "Cultivation Technology on Vegetables" under SCSP	KVK, Narayangav, Pune, Maharashtra	Jun. 5, 2024
	'Training-cum-Workshop on Entrepreneurship and IPR' for students of SKUAST	SKUAST, Kashmir at Srinagar	Jun. 25, 2024
	Interaction of the SMDs and the Directors of ICAR Institutes for 100 days Action Plan, Five Year Plan, MoUs etc.	NASC Complex, New Delhi	Jul. 15, 2024
	Industry-Institute Interaction	NASC Complex, New Delhi	Jul. 15, 2024
	96 th ICAR Foundation and Technology Day Ceremony	NASC Complex, New Delhi	Jul. 16, 2024
	Secretary, DARE and DG, ICAR Nominee as Expert in ASRB Selection Committee Meetings	ASRB, New Delhi	Jul. 26, 2024
	4th National Postgraduate Research Conference	UAS, Raichur	Aug. 13, 2024
	ORWs of Soil Science Students of IARI Hub	ICAR-CRIDA, Hyderabad	Aug. 19, 2024
	Launch of Agri Udaan 7.0	Novotel, Hitex, Hyderabad	Aug. 22, 2024
	Regional Advisory Group (RAG) - Stakeholders meeting on "Climate Resilient Models in Agriculture for Telangana State"	NABARD, Hyderabad	Sept. 2, 2024
	Agri Udaan Roadshow	CIFE, Mumbai	Sept. 24, 2024
	Interactive Session for Scientists	CTCRI, Thiruvananthapuram	Sept. 30, 2024
	International Nutricereal Convention 6.0 of IIMR, Hyderabad	ICAR-NAARM	Oct. 17, 2024

Staff Name	Program	Organized by/ Venue	Date
Dr. Ch. Srinivasa Rao	National Learning Week inaugurated by Hon'ble Prime Minister, Shri Narendra Modi	Ambedkar Int. Centre, New Delhi	Oct. 19, 2024
	Orientation to UG, PG and PhD Students of IARI Mega University-Hyderabad	ICAR-CRIDA, Hyderabad	Oct. 23, 2024
	31 st Swadeshi Science Congress under Swadeshi Science Movement-Kerala (SSMK)	ICAR-CIFE, Mumbai	Nov. 7, 2024
	National Conference on Building Sustainable Agricultural Startups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
	Chairperson of Sub-theme on Impact of Climate Change on Carbon, Water, Nitrogen and Energy Foot Prints, Global Science Conference 2024	NASC, New Delhi	Nov. 19, 2024
	94th Annual Session of NASI	IISER, Bhopal	Dec. 1, 2024
	Fertilizer Association of India Annual Seminar on Sustainable Fertilizer and Agriculture	New Delhi	Dec. 4-5, 2024
	Smart India Hackathon conducted by Ministry of Education	IIT, Hyderabad	Dec. 11, 2024
	Foundation Day Celebrations of ICAR-IIRR, Hyderabad	ICAR-IIRR, Hyderabad	Dec. 16, 2024
	PJTAU Diamond Jubilee Celebrations	PJTAU, Hyderabad	Dec. 20, 2024
	Natural & Organic Farmers Summit	KVK, Tuniki, Medak	Dec. 25, 2024
Dr. Ramasubramanian V.	International Conference on Emerging Trends of Statistical Sciences in AI and its Applications	Society of Statistics, Computer and Applications, Banasthali Vidyapith, Rajasthan	Feb. 26-28, 2024
	National Symposium: Transforming IARI into a Global Leader in Agricultural Research and Education	NASC, New Delhi	Jun. 22, 2024
	AGRI-UDAAN 7.0 Launch	Hyderabad	Aug. 22, 2024
	Brainstorming Workshop on "Sustainable Rice Production in India: Strategies and Policies"	ICAR-NAARM, Hyderabad	Oct. 16, 2024
	National conference on Building sustainable agricultural start-ups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
	International Indian Statistical Association (IISA 2024) Conference	CUSAT, Kochi	Dec. 27-31, 2024
Dr. Tavva Srinivas	Brainstorming Workshop on "Sustainable Rice Production in India: Strategies and Policies"	ICAR-NAARM, Hyderabad	Oct. 16, 2024
	National conference on building sustainable agricultural start-ups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024

Staff Name	Program	Organized by/ Venue	Date
Dr. Ranjit Kumar	32 nd International Conference of Agricultural Economists (ICAE)	NASC, New Delhi	Aug. 2-7, 2024
	IMC of ICAR- NIAP, New Delhi	ICAR-NIAP, New Delhi	Sept. 9, 2024
	32 nd Annual Conference of Agricultural Economics Research Association (AERA)	IGKV, Raipur	Dec. 11, 2024
Dr. N. Sivaramane	84th Annual Conference of Indian Society of Agricultural Economics (ISAE)	PAJANCOA&RI, Karaikal	Nov. 11-13, 2024
	National workshop on Market intelligence & Digital Innovations	State Agricultural Prices Board, Thiruvananthapuram, Kerala	Oct. 3-4, 2024
Dr. B. Nirmala	National Conference on Building Sustainable Agricultural Start-ups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
	94 th Annual Session of NASI	IISER, Bhopal	Dec. 1-3, 2024
Dr. Sanjiv Kumar	Conference on Digital Agriculture & New Technologies	CII, Hyderabad	Nov. 27, 2024
	National Conference on Building Sustainable Agricultural Start-ups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
Dr. S.M. Feroze	National Workshop on "Natural Farming and its scope under Changing Climate Scenario"	RPCAU, Pusa	Apr. 20-21, 2024
	82 nd Annual conference of ISAE held during	PJANCOA&RI, Karaikal	Nov. 11-13, 2024
	32 nd Annual conference of AERA, India	IGKV, Raipur	Dec. 11-13, 2024
Dr. B. Ganesh Kumar	Workshop on 'Mainstreaming Biodiversity into Food Systems and Micro enterprises: What are the Possibilities under RD & PR Programmes?'	NIRD&PR, Hyderabad	Apr. 18-19, 2024
	Workshop on Mission Karmayogi	DoPT, New Delhi	Oct. 3, 2024
	National Conference on Building Sustainable Agricultural Startups in India	ICAR-NAARM Hyderabad,	Nov. 12-13, 2024
	Poultry Expo	Indian Poultry Equip. Man. Asso. (IPEMA), Hyderabad	Nov. 27-29, 2024
Dr. Sona Yengkokpam	Workshop on Mission Karmayogi	DoPT, New Delhi	Oct. 3, 2024
	Symposium on Accelerated R & D towards a Viksit Bharat	IISER, Bhopal	Dec. 1-3, 2024
Dr. V.V. Sumanth Kumar	ITU/FAO Workshop on "Cultivating Tomorrow: Advancing Digital Agriculture through IoT and AI" and in "Precision Agriculture Conclave for Public Private Partnership"	ITU/FAO & ICAR	Mar. 18-20, 2024
	Digital Agriculture: Empowering Indian Farming	ICT Unit of ICAR	Dec. 16-18, 2024

Staff Name	Program	Organized by/ Venue	Date
Dr. S. Senthil Vinayagam	Harnessing Opportunities: NABARD transforming Agri-Landscape	NABARD HQs, Mumbai	Mar. 6, 2024
	Flagship start-up promotion event – ‘Aarambh’	Adm. Staff Coll. of India, Hyderabad	Mar. 15, 2024
	1st Institution’s Innovation Council (IIC) meeting/ workshop	DEIS, JNTUH, Hyderabad	Mar. 23, 2004
	43rd Foundation day of NABARD	NABARD, Hyderabad	Jul. 19, 2024
	National Conference on “Stepping up Agricultural Research and Education for Next Generation: Researchers Perspectives” organized by ARS Scientists’ Forum, ICAR, New Delhi	ICAR-IIRR, Hyderabad	Jul. 29, 2024
	Workshop on “Initiatives based on Innovative Agricultural Technologies for Landless and Resource Poor Households in Rural Areas”	NIRD&PR, Hyderabad	Aug. 28, 2024
	National Conclave on Incubation of Rural Women Entrepreneurs organized by Rural Livelihood Division, MoRD, New Delhi	Karnataka SRLM, Bangalore	Aug. 29, 2024
	National Seminar on Sustainable Rural Livelihoods Organized by Rural Development Vertical, NITI Aayog	NITI Aayog, New Delhi	Sept. 10, 2024
	Industry Meet-cum-Workshop on Emerging Technologies and Business Prospects of Biofortified Sweet Potato in India	ICAR-CTCRI, Regional Station, Bhubaneswar	Dec. 17, 2024
Dr. D. Thammi Raju	National Conference on Building Sustainable Agricultural Startups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
Dr. Navneet Kumar	7 th Education Leaders Conclave & Awards	ObserveNow Media, Hyderabad	Jul. 9, 2024
	National Conference on Building Sustainable Agricultural Startups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
Dr. Dipesh Debnath	7 th Education Leaders Conclave & Awards	ObserveNow Media, Hyderabad	Jul. 9, 2024
	National Conference on Building Sustainable Agricultural Startups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
	94 th Annual Session of NASI and Symposium on ‘Accelerated R & D towards a Viksit Bharat’	IISER, Bhopal	Dec. 1-3, 2024
Dr. K. N. Yamini	Two-day National Conference on “Shaping the Future of Education @2047”	MANUU, Hyderabad	Feb. 29 & Mar. 1, 2024
	7 th Education Leaders Conclave & Awards	ObserveNow Media, Hyderabad	Jul. 9, 2024
	4 th National Post Graduate Research Conference-2024	UAS, Raichur	Aug. 12-13, 2024
	National Conference on Building Sustainable Agricultural Startups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
	94 th Annual Session of NASI and Symposium on ‘Accelerated R & D towards a Viksit Bharat’	IISER, Bhopal	Dec. 1-3, 2024

Staff Name	Program	Organized by/ Venue	Date
Dr. R. Venkattakumar	Agripreneurship for Sustainable Agricultural Development: Technological and Institutional Innovations and Strategies (workshop)	ICAR-NAARM, Hyderabad	Mar. 20-21, 2024
	Seminar on "IP Management" on the theme "IPs and SDGs"	ICAR-NAARM, Hyderabad	Apr. 26, 2024
	Preparing for Success: A Collaborative Workshop for PGDM-ABM Educators	ICAR-NAARM, Hyderabad	May 22, 2024
	Editor's Workshop-Enabling Research Ecosystem	Elsevier and Education Division of ICAR	Sept. 9, 2024
	Launching event of FPO Gativardhan programme	Vijayawada, Andhra Pradesh	Oct. 7, 2024
	94 th Annual Session of NASI and Symposium on 'Accelerated R & D towards a Viksit Bharat'	IISER, Bhopal	Dec. 1-3, 2024
	Buyer-Seller meet between FPOs and star-ups	Vijayawada, Andhra Pradesh	Dec. 19, 2024
Dr. Surya Rathore	National Seminar on Transformative Agriculture: and Sustainable Development: Rethinking Agriculture for a Changing World	MPUAT, Udaipur, Rajasthan	Mar. 5-7, 2024
	Workshop on Community Science Education: Challenges and Opportunities	University of Agricultural Sciences, Dharwad	Oct. 27, 2024
Dr. P. Venkatesan	94 th Annual Session of NASI and Symposium on 'Accelerated R & D towards a Viksit Bharat'	IISER, Bhopal	Dec. 1-3, 2024
	Natural & Organic Farmer's Summit - 2024 at Dr Ramanaidu Ekalavya Grameena Vikas Foundation	ICAR-KVK, Tuniki, Medak, Telangana	Dec. 25, 2024
Dr. N.A. Vijay Avinashilingam	Training need assessment of clientele states to prepare annual training calendar for 2024-25	EEL, Hyderabad.	Feb. 6, 2024
	RuTAG 2.0 Project Team Meeting & Field Visits Report	ICAR-CIAE, Coimbatore	May 29, 2024
	Equine Conference on Health and Nutrition	VCRI, Udumalpet TANUVAS	Aug. 14, 2024
	Grass Roots Innovators - Stakeholders Interaction	Udumalpet, TANUVAS	Aug. 14, 2024
	Attended Networking meeting hosted for visiting Kenyan Delegation and Indian MPs	IIT, Hyderabad	Oct. 14, 2024
Dr. Mahantesh Shirur	94 th Annual Session of NASI and Symposium on 'Accelerated R & D towards a Viksit Bharat'	IISER, Bhopal	Dec. 1-3, 2024
	National Conference on Digital Agriculture	NASC, New Delhi	Dec. 16-18, 2024
Dr. P. Supriya	National Conference on Building Sustainable Agricultural Startups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
	India Genomics Conclave 2024 by Amazon Web Services	CSIR-CCMB, Hyderabad	Sept. 13, 2024
	Editor's Workshop-Enabling A Research Ecosystem	Elsevier and Education Division of ICAR	Sept. 24, 2024

Staff Name	Program	Organized by/ Venue	Date
Dr. M. Balakrishnan	AAHP-2024 5th Biennial Poultry Health Conference and National Symposium	DPR, Hyderabad	Feb. 23-24, 2024
	Assessment Committee Meeting of CAS promotion of ARS scientist	ICAR-CIFT Cochin, Kerala.	Jul. 10, 2024
	94 th Annual Session of NASI and Symposium on 'Accelerated R & D towards a Viksit Bharat'	IISER, Bhopal	Dec. 1-3, 2024
	Natural and Organic Farmers Summit-2024	Tuniki, Medak	Dec. 26, 2024
Dr. D. Damodar Reddy	Pedagogical Development Programme on Enhancing Pedagogical Competencies	NAAS, New Delhi	Jan. 29-Feb. 2, 2024
	Regional Consultation on Science of Natural Farming	MANAGE, Hyderabad	Jun. 14, 2024
	Brainstorming Workshop on Sustainable Rice Production: Strategies and Policies	ICAR-NAARM, Hyderabad	Oct. 16, 2024
	Caring Soils Beyond Food Security: Climate Change Mitigation & Ecosystem Services	NAAS, New Delhi	Nov. 19-22, 2024
Dr. Umesh Hudedamani	Online Course Production and Management for MOOCs	ICAR-NAARM, Hyderabad	Jan. 1-12, 2024
	AgriIP-2024 (Online short course on patents)	IPTM Unit, CIFT, Kochin	Jan. 15-Feb. 14, 2024
	SRIJAN: Empowering ZTMCs/ITMCs of ICAR Institutes	NASC, New Delhi	Feb. 13-15, 2024
Dr. M. Ramesh Naik	Workshop on Creating awareness for Rainwater Harvesting and its Efficient Use	ICAR-CRIDA, Hyderabad	May 7, 2024
	Conference on Stepping Up Agricultural Research and Education for Next Generation Research Perspectives	ICAR-IIRR, Hyderabad	Jul. 29-30, 2024
	Workshop on Sustainable Rice Production in India: Strategies and Policies	ICAR-NAARM, Hyderabad	Oct. 16, 2024
	Workshop on Mainstreaming of Biodiversity in Agriculture	TS Biodiversity Board, Hyderabad	Nov. 11, 2024
	National Conference on Building Sustainable Agricultural Startups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
	Conference on Caring Soils Beyond Food Security: Climate Change Mitigation & Ecosystem Services	NASC, New Delhi	Nov. 19-22, 2024
	94 th Annual Session of NASI and Symposium on 'Accelerated R & D towards a Viksit Bharat'	IISER, Bhopal	Dec. 1-3, 2024
	Workshop on Geospatial Analytical tools for Natural Resource Management	ICAR-NAARM, Hyderabad	Dec. 17-19, 2024
Dr. Sivakumar S.	International Conference on "Innovative education, research and extension approaches for transmitting scientific know-how to augment livestock production in the contemporary scenario"	VCRI, Thanjavur, Tamil Nadu	Jul. 10-12, 2024

Staff Name	Program	Organized by/ Venue	Date
Dr. Sivakumar S.	National Seminar on “Emerging Trends in Food Processing Technologies for Food security and Safety”	CSIR-IICT, Hyderabad	Jul. 20, 2024
	National Conference on Building Sustainable Agricultural Startups in India	ICAR-NAARM, Hyderabad	Nov. 12-13, 2024
	National Seminar on Policy and Strategies for Indian Goat Sector in ‘Amrit Kaal’	ICAR-CIRG & DAHD	Nov. 19, 2024

Table 11.4. Lectures delivered in other Universities/Institutes etc. by the Faculty

Staff Name	Topic	Program/ Organized by/ at	Date
Dr. Ch. Srinivasa Rao	Climate Change and Ecosystem-based Solutions for improving Resilience and Good Quality of Life	Training Program (TP) on “International Instrument on Biodiversity and Wetlands” for IFS Officers organised by NIRD&PR at Hyderabad	Jan. 8, 2024
	Climate Change: Coping Strategies, Policy and Programmes	International Conference on Climate Change and Agroecosystem: Threats, Opportunities and Solutions organised by Asso. of Agro-meteorologists at Varanasi	Feb. 8, 2024
	National Education Policy 2020: Changing Role of Teachers	TP on Enhancing Pedagogical Competencies for Agricultural Education organised by NAAS, New Delhi	Jan. 29, 2024
	Climate Smart Agriculture for Development Planning and Action: Technologies, Programs and Policies	TP on “Building Local Resilience in a Changing Climate” for Women Scientists & Technologists working in Govt. Sector organised by LBS National Acad. of Admin., Mussoorie	Mar. 1, 2024
	Input Management – Government Policies and Delivery by Public and Private Sectors in Agriculture	IAS Phase-I Professional Course for 2023 batch organised by LBS National Academy of Administration, Mussoorie	Mar. 4, 2024
	National Education Policy 2020: Changing Role of Teachers	TP on Enhancing Pedagogical Competencies for Agricultural Education organised by NAAS, New Delhi	Mar. 5, 2024
	National Educational Policy 2020: Changing Role of Teachers	TP on Enhancing Pedagogical Competencies for Agricultural Education organised by NAAS, New Delhi	Apr. 1, 2024
	National Education Policy 2020: Changing Role of Teachers	TP on Enhancing Pedagogical Competencies for Agricultural Education organised by NAAS, New Delhi	May 3, 2024
	C-farming in dryland agriculture critically dissecting the potential, inherent difficulties and opportunities for adoption	Strategy Workshop on Carbon Farming organized by NAAS, New Delhi	May 4, 2024

Staff Name	Topic	Program/ Organized by/ at	Date
Dr. Ch. Srinivasa Rao	Aribandi Laxminarayana 6 th Memorial Lecture on “Climate resilient agriculture”	Seminar on Rainfed Agriculture – 2024 organised by PJTSAU in collaboration with Aribandi Foundation & Telangana Farmers Society at Hyderabad	Jun. 9, 2024
	NAARM initiatives in agricultural entrepreneurship: few takeaways for students	‘Training-cum-Workshop on Entrepreneurship and IPR’ organised by NAHEP at SKUAST, Kashmir, Srinagar	Jun. 25, 2024
	NAARM Initiatives in agri-business management	‘Training-cum-Workshop on Entrepreneurship and IPR’ organised by NAHEP at SKUAST, Kashmir, Srinagar	Jun. 25, 2024
	Agriculture & Economy: Challenges and Prospects	“Refresher Course in Commerce & Management” organized by UGC-MMTTC, Hyderabad	Aug. 29, 2024
	Farmer Participatory Climate Adaptive Agriculture	Webinar on “PRE-COP29” entitled “Impact of Climate Change on the Dry Areas” organized by Int. Dryland Dev. Commission hosted by Alexandria University	Sept. 10, 2024
	Climate Change in Agriculture and Indian Economy	Webinar during Karmayogi Saptah organized by ICAR-NAARM, Hyderabad	Oct. 25, 2024
	Participatory Soil Care Alienating Local Governance	Global Science Conference 2024 organised by Indian Society of Soil Science at NASC, New Delhi	Nov. 21, 2024
	World Soil Day Lecture Eminent Alumni Lecture Series	World Soil Day Lecture Organized by Division of Soil Science & Agricultural Chemistry & IARI Alumni in Asso. with Delhi Chapter of Indian Society of Soil Science	Dec. 5, 2024
	Developing business model for FPOs	Training on ‘Business opportunities for FPOs in value addition in ICAR-IIOR, Hyderabad	Jan. 4, 2024
	Business Plan Development in Agri-horti enterprises for creating viable agri-preneurship ventures	Training on ‘Gender- Sensitive Agri-Horti based Dimensions for addressing Livelihood, Nutrition, and Entrepreneurship by ICAR-CIWA, Bhubaneswar	Jan. 18, 2024.
	Natural Farming Practices and Farmers’ Income	National Workshop on ‘Natural Farming and its Scope under Changing Climate Scenario’ by RP-CAU, Samastipur	Apr. 20-21, 2024
	Analytical tools applied in value chain management	Key Speaker in webinar by ICAR-NINFET, Kolkata	Nov. 25, 2024
	Making FPO Business Scalable and Sustainable	Key Speaker in webinar by ICAR RCER, Patna	Dec. 19, 2024

Staff Name	Topic	Program/ Organized by/ at	Date
Dr. Ch. Srinivasa Rao	Lead lecture on IP & Tech transfer in Agricultural Research: Perspective and policies	National Seminar on “Medicinal and Aromatic Plants in Perspective of Health and Wealth” by MAPAI and ICAR-DMAPR at Anand, Gujarat	Oct. 22-23, 2024.
	‘Expert Speaker’ in Online training programme on “Value Chain Management in Natural Fibres”	ICAR-National Institute of Natural Fibre Engineering and Technology, Kolkata	Nov. 29, 2024
	Climate Resilient Interventions Through Livestock and Poultry Farming	National training program on “Mission LiFE (Lifestyle For Environment) and its Implementation for Rural Development” at NIRD&PR, Hyderabad	Mar. 19, 2024
	Livestock Management and Mitigation Strategies during Disasters	Training Program on “ECHO Disaster Management and Mitigation Strategies in Agriculture and Allied Sectors” at EEI, Hyderabad	Jul. 31, 2024
	Livestock, Management and Methane Mitigation	Training Program on “Carbon Credits in Food and Agriculture, Carbon Finance and Carbon Farming” at EEI, Hyderabad	Aug. 23, 2024
	PPP in Animal Husbandry Sector: Challenges/Issues, Working Models and Experiences	Training Program on “EHCO Building Networks and Partnerships for Profitability in Agri and Allied Sectors” at EEI, Hyderabad	Sept. 9, 2024
	Concepts of Value Chain Management	Training Program on “Value Chain Management in Natural Fibres” at ICAR-NINFET, Kolkata	Nov. 25, 2024
	Integrating animal husbandry with watershed management to enhance sustainable income	Training Program on “Participatory Watershed Management for Enhancing Sustainable Income under WDC-PMKSY 2.0” at NIRD&PR, Hyderabad	Nov. 26, 2024
Dr. Ramasubramanian V.	Two lectures, viz., (i) CART - Classification and regression trees & (ii) Kohonen self-organization maps	Winter School on Use of Big Data Analysis in Agriculture organized by UAS, Dharwad	Feb. 3, 2024
	A statistical standpoint on scientometrics in horticultural domain with special reference to fruits and vegetables	International Conference on Emerging Trends of Statistical Sciences in AI and its Applications organized by Society of Statistics, Computer and Applications at Tonk, Rajasthan	Feb. 26-28, 2024
	Two lectures, viz., (i) PCA - Principal Component Analysis and (ii) CART - Classification And Regression Trees	CBP on ‘Applications of Machine Learning in Agricultural Meteorology’ organized by AICRP on Agrometeorology at ICAR-CRIDA, Hyderabad	Mar. 1, 2024

Staff Name	Topic	Program/ Organized by/ at	Date
Dr. Ramasubramanian V.	Three lectures, viz., (i) Classification and regression trees, (ii) Multi-dimensional scaling and (iii) Principal component analysis	Research methodology training on Data Analysis for Decision Making organized by UAS, Dharwad	Aug. 28-29, 2024
	Cross impact analysis based models for technology forecasting in agriculture	International Indian Statistical Association (IISA 2024) Conference organized by CUSAT, Kochi	Dec. 27, 2024
Dr. R. Venkattakumar	Networking in Agribusiness	Guest Lecture for RAWI Students of Kerala Agricultural University, Thrissur	Feb. 2, 2024
	Communicating Science through Mainstream Media	For newly-recruited scientists of ICFRE-Forest Research Institute, Dehradun	Feb. 14, 2024
	Role of FPOs in Tapping the Value Addition Opportunities in Horticulture	Training program jointly organized by ICAR-IIHR and MANAGE, Hyderabad	Aug. 7, 2024
Dr. Surya Rathore	Gender Issues in Agricultural Research and Development	ICAR Winter School on Gender Sensitization: A Step Towards Agricultural Development during Jan. 18-Feb. 7, 2024 organized by AAU, Anand	Jan. 19, 2024
	Advocacy and Capacity building for NextGen Extension practices	Training program on NextGen Extension: Transforming Roles and Sparking Innovations organized by UBKV, Coochbehar and MANAGE, Hyderabad	Aug. 29, 2024
	Vocational Scope of Home Science Extension and Communication Management	Workshop on Community Science Education: Challenges and Opportunities organized by UAS, Dharwad	Oct. 27, 2024
Dr. P. Venkatesan	Presentation skill for Extension Management	CBP on Training Management for Effective Extension organized by CAU, Imphal	Feb. 28, 2024
	Farm-based innovations for agripreneurship a farmer's first perspective	Advanced Entrepreneurship Development Program for Women in Agriculture organized by ICAR-CIFA, Bhubaneswar	Mar. 7, 2024
Dr. N.A. Vijay Avinashilingam	Innovative Digital Technologies in Extension Deliveries	State Nodal Officers under VISTAAR Program organized by MANAGE, Hyderabad	Oct. 24, 2024
	Harnessing Digital Innovations for Productivity	National Facilitators of MANAGE, Hyderabad	Nov. 19, 2024
Dr. Mahantesh Shirur	Personal Management Strategies for becoming Effective Extension Personnel	Induction training to agri-extension professionals organized by MANAGE, Hyderabad	Jun. 19, 2024
	Unlocking Potential: Advancing Women as Agri-Input Dealers	For the officials of SAMETIs organized by MANAGE, Hyderabad	Aug. 30, 2024

Staff Name	Topic	Program/ Organized by/ at	Date
Dr. Mahantesh Shirur	Market led extension (Online)	Training Program on Agricultural Marketing for Extension Professionals organized by PAMETI, Ludhiana and MANAGE, Hyderabad	Sept. 27, 2024
	Utilising DAESI trained input dealers as para-extension professionals	National Workshop on DAESI organized by MANAGE, Hyderabad	Dec. 10, 2024
	Training mind and body to make effective performance	Faculty Development Program for EEIs and SAMETIs organized by MANAGE, Hyderabad	Dec. 20, 2024
Dr. M. Balakrishnan	AI tools for Agriculture	Training Program on Climate Change Adaptation with ICT Applications organized by MANAGE, Hyderabad	Jul. 22, 2024
	Transforming Livestock Management with AI: Innovations, Applications, and Future Prospects	National Conference on “Biotechnological Innovations to Augment Health and Productivity of Livestock and Poultry for Sustainable Livelihood” organized by CVSc, SVVU at Proddatur	Oct. 24, 2024
	AI Models and its applications in Biological Sciences	National Seminar on AI innovations for advanced biological science and research organized by NGP College, Coimbatore	Dec. 27, 2024
Dr. M. Ramesh Naik	Organic farming and its benefits for NRM	Training program on ‘Natural Resource Management - Soil and water Management’ organized by EEI, Hyderabad	Feb. 29, 2024
Dr. Sivakumar S.	Challenges faced by the small and medium scale Dairy Entrepreneurs - A way forward for success	Inter. Con. On ‘Innovative education, research and extension approaches for transmitting scientific know-how to augment livestock production’ organized by VCRI, Tamil Nadu	Jul. 10-12, 2024
Dr. Laxman M. Ahire	Agro Tourism-A Way Forward for Biodiversity Conservation	Training Program on ECHO-Agro Tourism: A Way Forward for profitability organized by EEI, PJTAU, Hyderabad	Sept. 11, 2024
	Agro Tourism: A Forward for Biodiversity Conservation	Training Program on Agro Tourism and Entrepreneurship Development organized by RARS, KAU and MANAGE at Pattambi	Oct. 11, 2024

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1. Ahire, L.M., Rao, C.S. Baral, K. and Rani, V., 2024. Agri-tourism for Rural Development. Kheti 8: 4-7.
2. Ahire, L.M., Venkatesan, P. and Balakrishnan, M., 2024. Agri Tourism-An Innovation to Earn an Extra Income from Farm Activities: A Case of Tourism Centers in Maharashtra, India, AgriGate- An International Multidisciplinary e-Magazine 4: 1-12.

13

HONORS, AWARDS AND RECOGNITIONS

Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM got selected as the Director-cum-Vice Chancellor of ICAR-IARI, New Delhi

Dr. D. Damodar Reddy, Principal Scientist, got selected as the Fellow of the Indian Society of Soil Science, conferred at the 88th Annual Convention of the ISSS on 19th November 2024 at New Delhi.

Table 13.1. NAARM Foundation Day Awards 2024

Award	Awardee
Annual Performance Awards (2023-24)	
Best Faculty	Dr. Ranjit Kumar, Principal Scientist, ABM Division
Best Trainer	Dr. M. Ramesh Naik, Scientist, RSM Division
Best Scientist	Dr. B.S. Yashavanth, Scientist, ICM, Division
Best Promising Young Scientist	Dr. Umesh Hudedamani and Dr. P. Supriya
Best Policy Paper	Organic Farming in India: Production Constraints and Policy Recommendations (by M. Ramesh Naik, Umesh, H., Gopinath, KA., Ramanjeneyulu, A.V., Ravisankar, N., Tavva Srinivas, Ansari, M.A. and Singh, V.K.)
Best Research Paper	Computational Tools and Scientometric for CRISPR based Genome Editing. Journal of Plant Biochemistry and Biotechnology (by Balakrishnan, M., Kotla, A., Agarwal, S., Krishnan, P., Supriya, P.)
Best Administrative Staff (Officer)	Smt. N. Vijaya Lakshmi, AF&AO and Smt. K.K. Rukmani Ammal, AAO
Best Technical Staff (T-6 and above)	Sri P. Mohan Singh, ACTO and Dr. Kirttiranjan Baral, STO
Best Skilled Support Staff	Sri G. Venkatesham, SSS, Audit and Accounts
Best Young Professional (YP I/II/Project Staff)	V. Veeresham, YP-II, PGS Unit; Ms. Ch. Lalitha Navya, YP-II, XSM Division and Mr. P. Sai Kiran, YP-II, Cash and Bills
Best PG Student	Mr. Monish Gowda T.P. (2023-25 Batch) and Ms. Jog Anuradha Vijay (2023-25 Batch)
Institution Building Award	Dr. B. Ganesh Kumar, Head, HRM Division and Mr. Vivek Purwar, CAO (SG)
Decadal Performance Awards (2013-24)	
Best Faculty	Dr. G.R.K. Murthy, Principal Scientist, ESM Division
Best Administrative Staff	Mr. M. Narsing Rao, UDC
Best Technical Staff	Mr. P. Namdev, CTO and Mr. M. Shekhar Reddy, CTO
Best Skilled Support Staff	Mr. B. Raju, SSS

Table 13.2. Honors/ Awards/ Recognitions received by the staff of the Academy during 2024

Name of the Staff	Honors/ Awards/ Recognitions
Dr. Ch. Srinivasa Rao	Best Policy Paper Award conferred by ICAR for 'Climate Change and Indian Agriculture: Impacts, coping strategies, programmes and policy' (by Rao, C.S., Prasad, R.S. and Mohapatra, T.)
Dr. Tavva Srinivas	Editorial Board Member for the Journal of Research ANGRAU PG Faculty in Agricultural Economics in Hyderabad hubs of ICAR-IARI, New Delhi

Name of the Staff	Honors/ Awards/ Recognitions
Dr. Tavva Srinivas	Research Guide for P.G Program in Agricultural Economics in Hyderabad hubs of ICAR-IARI, New Delhi
	Co-author of Best Policy paper award for 'Organic Farming in India: Production Constraints and Policy Recommendations' conferred during 49 th Foundation Day of ICAR-NAARM
	Reviewer of Biotechnology Ignition Grant (BIG) Scheme applications (B RAC BIG call 24)
	Evaluator of NIDHI PRAYAS 5th Call proposals
	Alternate Member, Food and Agriculture Division (FADC) Council of BIS for 2024
	Evaluator for MBA Agribusiness Management thesis of PJTAU
Dr. Ranjit Kumar	Fellow of Agricultural Economics Research Association
	Fellow of Bihar Agricultural Science Academy (BASA)
	Member, Institute Management Committee (IMC) of ICAR-Central Institute for Women in Agriculture (CIWA), Bhubaneswar (2023-2026)
	Member, Institute Management Committee (IMC) of ICAR-National Institute of Agricultural Economics & Policy Research (NIAP), New Delhi (2023-2026)
	Member, Academic Council, Prof. Jayashankar Telangana State Agricultural University (PJ TSAU), Hyderabad
	Member, Board of Studies in the faculty of Agri-Business Management at UAS, Bangalore
	External Member, Institute Technology Management Committee (ITMC) of ICAR-Indian Institute of Oilseeds Research (IIOR), Hyderabad.
	Member, Editorial Board, Indian Journal of Agricultural Economics, Mumbai
	Section Editor, Indian Journal of Agricultural Sciences, New Delhi.
	Best Faculty Award of ICAR-NAARM
Dr. N. Sivaramane	Member of Executive Committee of Indian Society of Agricultural Economics
	Member of Niti Aayog Working group on Crop Husbandry, Agriculture inputs, Demand & Supply
Dr. B. Nirmala	Fellow, Telangana Academy of Sciences, Hyderabad
Dr. Sanjiv Kumar	External Examiner for PhD thesis evaluation and Viva-Voce examination of Agribusiness Business Management of IGKV, Raipur
	Expert in Selection Committee for Guest Faculty in College of Agribusiness Management, Sabour, Bihar
Dr. Sanjiv Kumar	Moderator in Session on Harnessing Agricultural Data for Higher Productivity and Cost Reduction organized by Confederation of Indian Industry, Telangana on Nov. 27, 2024
	Question Paper Setter for ICAR-SRF for Agricultural Economics and Agribusiness Management
Dr. B. Ganesh Kumar	Member, RAC for ICAR-Central Institute of Fisheries Technology (CIFT), Cochin
	Member, RAC for ICAR-National Institute of Foot and Mouth Disease (NIFMD), Bhubaneswar
	Member, Training Quality Improvement Measures Committee (TQIMC), NIRD&PR, Hyderabad
	ICAR-NAARM Best Institution Building Award
Dr. K. Srinivas	Editor of Indian Journal of Agricultural Economics
	Member, QRT for ICAR-National Research Center on Equines, Hisar, Haryana
Dr. Navneet Kumar	Member, Selection Committee for Direct selection and CAS promotion of Teachers at Junagadh Agricultural University, Junagadh
	Guest Editor for Frontiers in Nutrition Journal
Dr. P. Venkatesan	External Expert in IRC of ICAR-IIRR Hyderabad, for Transfer of Technology and Training Section
	Distinguished Alumni Award from Gandhigram Rural Institute, Gandhigram, Dindigul

Name of the Staff	Honors/ Awards/ Recognitions
Dr. Surya Rathore	Best Paper Award for Oral presentation in the National Seminar on Transformative Agriculture and Sustainable Development: Rethinking Agriculture for a Changing World held during Mar. 5-7, 2024 at MPUAT, Udaipur
	Expert in Selection committee for SMS (Home Science) in Bihar Agricultural University, Sabour (Bihar)
	Expert for review of research proposals (2 PhD and 3 Masters students) of Dept. of Extension Education & Communication Management, PJTAU, Hyderabad
	External Examiner for PhD (Agricultural Extension Education) thesis of Kerala Agricultural University, Thrissur
	Evaluator of M.Sc. thesis 'Career preferences and Career preparedness of Under Graduate Girl students of PJTAU- An analytical study', Dept. of Agri. Ext., CoA, PJTAU, Hyderabad
Dr. Yashavanth B.S.	Member of Technical Advisory Committee of the project on 'Development of DUS guidelines for Curry Leaf' constituted by the PPVFRA, MoAFW, New Delhi.
	Member of Advisory Committee of Master's students of Dept. of Agricultural Statistics, Agriculture College, UAS, Dharwad
Dr. M. Balakrishnan	Member, IMC, ICAR-Directorate of Poultry Research, Hyderabad
	Academic Council Member, Mall Reddy University, Hyderabad, Telangana
	External Examiner in Anna University, TN; NIT Trichy, TN and ICAR-IARI, New Delhi for PhD thesis evaluation. External Examiner in IVRI Hyderabad Hub for Information Technology in Agriculture course for both MVSc and PhD students
	Member, IMC, ICAR-Indian Institute of Rice Research, Hyderabad
Dr. Ramasubramanian V.	External Examiner, M.Sc. (Agricultural Statistics) thesis of two students of Dr. Rajendra Prasad Central Agricultural University, Bihar
Dr. Dipesh Debnath	Editor, Journal of the Inland Fisheries Society of India, ICAR-CIFRI, Barrackpore, Kolkata during Apr. 2023 to Mar. 2025
	Chairman, Application Screening Committee in NFDB, Hyderabad for recruitment of Executive Assistants (F&A and Technical)
	Recognized by ICAR-IARI, New Delhi as Research Guide in ICAR-Hyderabad Hub
	Recognized by ICAR-IARI, New Delhi as Faculty Member in ICAR-Hyderabad Hub
	Member, Selection Committee in NFDB, Hyderabad for recruitment of Executive Director (Tech.)
Dr. M. Ramesh Naik	Member in DPC of Technical Category, ICAR-IIRR, Hyderabad
	External Member for selection of SRF, YP-I and YP-II, ICAR-CRIDA, Hyderabad
	Joint Secretary for the Indian Society of Oilseeds Research (ISOR) for the period of 2024-2026
	Second Best Research Article Award for the article on Organo-Mineral Fertilizers for Sustainable Agriculture by the Fertilizer Association of India
	Best Poster Award for the paper 'Redefining Food Systems in India: Managing Dietary Changes and Environmental Impacts for Sustainable Development' in 94 th Annual Session of NASI & Symposium held during Dec. 1-3, 2024 in IISER Bhopal
	Member, Editorial Board of International Journal of Economic Plants
Dr. Sivakumar S.	Associate Fellow Award of National Academy of Dairy Science, Karnal
	Executive Committee Member of Society for Veterinary and Animal Husbandry Extension Education, Chennai
Dr. Umesh Hudedamani	Member, Institute Technology Management Committee of ICAR-National Meat Research Institute, Hyderabad

Table 13.3. Technologies/ Products recognized by ICAR during 2024

Sl. No.	Name of the Technology/product	Developers
(i)	Model of Entrepreneurial Ecosystem for Agricultural Universities	Rathore, S., Sonatakki, B.S., Venkateshwarlu, G. and Rao, C.S.
(ii)	Placement linked skill Development for CEOs of FPOs (Model)	Vinayagam, S.S., Kumar, S.A.D., Venkateshwarlu, G. and Rao, C.S.
(iii)	Impactful model for Farmer First Programme Implementation	Singh, A.K., Gautam, U.S., Prakash, V.C., Burman, R.R., Venkatesan, P., Sivaramane, N., Rao, C.S. et al.
(iv)	Organic Farming in India: Production Constraints and Policy Recommendations	Naik, M.R., Hudedamani, U., Gopinath, K.A., Ramanjaneyulu, A.V., Ravisankar, N., Srinivas, T., Ansari, M.A., Singh, V.K., Venkateshwarlu, G. and Rao, C.S.
(v)	A Rapid method to estimate crop loss due to Hailstorm damage in farmers fields using remote sensing	Prabhakar, M., Gopinath, K.A., Reddy, A.G.K., Thirupathi, M., Rao, C.S., Rao, M.S., Singh, R. and Singh, V.K.
(vi)	Low cost Portable Kiln for Biochar Production	Venkatesh, G., Venkateshwarlu, B., Gopinath, K.A., Rao, C.S., Korwar, G.R., Reddy, B.S., Prasad, J.V.N.S., Rao, K.V. and Singh, V.K.
(vii)	Development of Biochar production technology from Pigeon Pea stalk	Venkatesh, G., Gopinath, K.A., Reddy, K.S., Reddy, B.S., Prabhkar, M., Rao, C.S., Kumari, V.V., Singh, V.K., Prasad, J.V.N.S. and Rao, K.V.
(viii)	Non-Methanol Based Extraction Method for Estimation of Nicotine, Reducing Sugars and Chlorides in FCV Tobacco Leaf	Prasad, L.K., Rao, C.C., Reddy, D.D., Padmaja, K. and Jhonson, N.
(ix)	Dense Planting Technique – A Strategy for Minimization of Yield Loss under Delayed Monsoon Conditions in FCV Tobacco Grown on Rainfed Alfisols of Andhra Pradesh	Anuradha, M., Reddy, D.D., Chenchiah, K.C., Prasad, L.K. and Bindu, J.P.
(x)	Integrated Agro-techniques for Orobanche Management in Chewing Tobacco	Kumaresan, M., Reddy, D.D., Krishna, S.K., Manivel, P. and Annadurai, V.
(xi)	Maize-Tobacco: Profitable Cropping System for Higher System Productivity and Profitability in Tobacco Growing Vertisols of Andhra Pradesh	Kumar, T.K., Reddy, D.D., Rao, C.C., Krishna, S.K. and Madhav, M.S.
(xii)	Soil Fertility Maps of Southern Light Soils (SLS) Region of FCV Tobacco in Andhra Pradesh	Prasad, L.K., Rao, C.C., Reddy, D.D., Ravishankar, H. and Bindu, J.P.
(xiii)	Strengthening Farmer-Centric Extension System in India	Venkattakumar, R., Venkatesan, P., Vinayagam, S.S., Venkateshwarlu, G. and Rao, C.S.
(xiv)	Android-based dynamic mobile application on “Kadakhnath Info”	Mooventhan, P., Ghosh, P.K., Dixit, A., Kaushal, P., Sharma, G.L., Verma, P.K., Verma, L.K., Khan, M.A., Singh, U., Venkatesan, P. and Singh, S.R.K.

CONCLUDED PROJECTS

In-house

S. No.	PME Code	Project Title	Team	Duration	
				From	To
1	AGED/NAARM/SIL/2020/07/ 00173	Mega Environment Analysis Tools for AICRP Trials	A. Dhandapani, B.S. Yashavanth, S.K. Bal	06/2020	05/2024

Extra-mural

S. No.	PME Code	Project Title	Team	Budget (In Lakhs)	Sponsored by	Duration	
						From	To
1.	AGED/NAARM/COP/2019/001/ 00155	Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education	S.K. Soam, D. Thammi Raju, N. Srinivasa Rao, Alok Kumar, V.V. Sumanth Kumar, Surya Rathore, Sanjiv Kumar, M. Balakrishnan, S. Senthil Vinayagam, P. Krishnan, B.S. Yashavanth	1704.17	ICAR NAHEP	2/2019	10/2024

ON-GOING PROJECTS

In-house

S. No.	PME Code	Project Title	Team	Duration	
				From	To
1.	AGED/NAARM/SIL/ 2022/ 004/00184	Motivational Factors Driving Scientific Personnel: Identification of Strategies for Scientific Excellence	B. Ganesh Kumar, Sona Yengkokpam, B.S. Yashavanth, R.V. S. Rao	11/24	10/27
2.	AGED/NAARM/SIL/ 2024/ 002/00195	Women leadership in NARES: Challenges and policies	Sona Yengkokpam, B. Ganesh Kumar, B. Nirmala, D. Debnath, S. Sivakumar	11/24	10/26
3.	AGED/NAARM/SIL/ 2020/ 001/00167	New Market Reforms and E-NAM in India: Effect on Marketing Choices and Participation and Price Realization for Small holders	Ranjit Kumar, Sanjiv Kumar, A. Dhandapani P.C. Meena	06/20	03/25

S. No.	PME Code	Project Title	Team	Duration	
				From	To
4.	AGED/NAARM/ SIL/2020/002/00168	Techno-Economic Feasibility Studies for ICAR Agro-Processing Technologies/Value Added Products	Srinivas Tavva, B. Ganesh Kumar, G.R.K. Murthy, S. Sivakumar	06/20	10/25
5.	AGED/NAARM/ SIL/2018/011/00150	Farmer Producer Companies in India: A Study on their Management Practices and Business Potential	N. Sivaramane, Ranjit Kumar, P.C. Meena, N.A.V. Avinashilingam, P. Venkatesan, Tavva Srinivas, S. Senthil Vinayagam, P.D. Sreekanth, B.S. Yashavanth	10/18	04/25
6.	AGED/NAARM/ SIL/2023/002/00190	Disaggregated food product demand: Insights and predictive analysis	N. Sivaramane, P. Venkatesan, Tavva Srinivas, S.M. Feroze, B. Nirmala, Mahantesh Shirur, P.C. Meena, Shivendra Shrivastava	09/23	10/26
7.	AGED/NAARM/ SIL/2024/003/00196	Exploring the Financing Landscape for Agri Startups in India: Key Challenges and Strategic Solutions	Sanjiv Kumar, K. Srinivas, P.C. Meena, S. Senthil Vinayagam, P.D. Sreekanth, N.A.V. Avinashilingam	11/24	10/27
8.	AGED/NAARM/ SIL/2024/004/00197	Efficiency of dairy farms and yield gap in milk production: Unlocking the dairy business potential in rural India	S.M. Feroze, Ranjit Kumar, P.C. Meena, Sanjiv Kumar, Sivakumar. S	12/24	11/27
9.	AGED/NAARM/ SIL/2024/005/00198	Analysing India's Competitiveness in Global Agricultural Trade	B. Nirmala, S.M. Feroze, B.S. Yashavanth, Tavva Srinivas	11/24	10/27
10.	AGED/NAARM/ SIL/2024/006/00199	Assessing Crop Insurance in India: An Analysis of Accessibility, Awareness, and Operational Gaps	P.C. Meena, S.M. Feroze, Sanjiv Kumar, B. Nirmala	11/24	10/25
11.	AGED/NAARM/ SIL/2020/ 010/00176	Development of a framework for gender mainstreaming in the development schemes of Agriculture sector in India	Surya Rathore, P. Krishnan, Lipi Das, Veenita Kumari, B.S. Yashavanth	01/21	03/25
12.	AGED/NAARM/ SIL/2024/007/00200	Competency Mapping and Skill Gap Assessment of KVK Professionals in India	R. Venkattakumar, P. Venkatesan, Mahantesh Shirur, V. Ramasubramanian, N. Sivaramane, V. Venkatasubramanian, R.R. Burman, R.K. Padaria	11/24	03/27
13.	AGED/NAARM/ SIL/2024/008/00201	Impact of Pradhan Mantri Samman Nidhi (PM-KISAN) Yojana on Farmers	N.A.V. Avinashilingam, N. Sivaramane, S. Senthil Vinayagam, R. Venkattakumar, D. Debnath, P.C. Meena, B.S. Yashavanth	12/24	11/26

S. No.	PME Code	Project Title	Team	Duration	
				From	To
14.	AGED/NAARM/ SIL/2024/009/00202	FPO Typologies and Adoption of Sustainable Agriculture Practices	Mahantesh Shirur, R. Venkattakumar, P. Venkatesan, S.M. Feroze, B.S. Yashavanth, N. Sivaramane	11/24	10/26
15.	AGED/NAARM/ SIL/2022/001/00181	MOOCs in Digital Agricultural Education-Strategies to Enhance Effectiveness, Inclusivity and Reach	G.R.K. Murthy, V.V. Sumanth Kumar, S. Senthil Vinayagam, P. Supriya, Navneet Kumar	04/22	03/26
16.	AGED/NAARM/ SIL/2022/005/00185	Development of implementation framework for National Education Policy -2020 in Agricultural Education	D. Thammi Raju, G.R.K. Murthy, P. Ramesh, P. Venkatesan, N.A.V. Avinashilingam, N. Sivaramane, S. Senthil Vinayagam, Ch. Srinivasa Rao, K.N. Yamini, Navneet Kumar	10/22	09/25
17.	AGED/NAARM/ SIL/2024/010/00203	Assessment of AI Tools on National Agricultural Research and Education System	V.V. Sumanth Kumar, Navneet Kumar, S. Senthil Vinayagam	11/24	10/26
18.	AGED/NAARM/ SIL/2024/011/00204	Good Educational Practices and Innovations for Quality Agricultural Higher Education	K.N. Yamini, S. Senthil Vinayagam, D. Thammi Raju, D. Debnath, Mahantesh Shirur, Navneet Kumar, S. Siva Kumar	11/24	03/27
19.	AGED/NAARM/ SIL/2024/012/00205	Educational interventions for enhancing employability of fishery graduates in accordance with NEP 2020	D. Debnath, D. Thammi Raju, Sona Yengkokpam, K.N. Yamini	11/24	10/26
20.	AGED/NAARM/ SIL/2024/013/00206	Assessing the effectiveness of Virtual Reality (VR) resources in Agricultural Higher Education	Navneet Kumar, G.R.K. Murthy, V.V. Sumanth Kumar, D. Debnath, S. Senthil Vinayagam, Siva Kumar S.	11/24	10/27
21.	AGED/NAARM/ SIL/2021/003/ 00179	Impact Assessment of SC Sub Plan (SCSP) implemented by Agricultural R & D Institutes on Socio-economic Status of Beneficiaries	M. Balakrishnan, B. Ganesh Kumar, P. Venkatesan, M. Ramesh Naik, P. Supriya, B.S. Yashavanth, D. Thammi Raju, N. Sivaramane, P.C. Meena, N.A.V. Avinashilingam, R. Venkattakumar, B. Nirmala, Sona Yengkokpam	07/21	03/26
22.	AGED/NAARM/ SIL/2022/002/ 00182	Comprehensive Geo-spatial Analysis of Environmental Change in Farming	P.D. Sreekanth, D. Thammi Raju, P. Venkatesan, V.V. Sumanth Kumar, M. Ramesh Naik, Navneet Kumar, D. Damodar Reddy	02/22	03/25
23.	AGED/NAARM/ SIL/2022/007/00187	Identification of similarity in research studies using machine learning	B.S. Yashavanth, A. Dhandapani, V. Ramasubramanian	02/22	10/28

S. No.	PME Code	Project Title	Team	Duration	
				From	To
24.	AGED/NAARM/ SIL/2024/014/00207	Development and Validation of Integrated workflow for Next Generation Sequencing Data Analysis in Agriculture	M. Balakrishnan, S. Jayakumar, P. Supriya, S.P. Yadav, S.K. Mangrauthia, Vignesh	11/24	10/27
25.	AGED/NAARM/ SIL/2024/015/00208	NAARM eLibrary: Mobile Access to Library Resources and Digital Collections	N. Srinivasa Rao, P.D. Sreekanth, A. Dhandapani	11/24	10/26
26.	AGED/NAARM/ SIL/2024/016/00209	Development of Information System for Managing Frontline Extension Activities Using Geo-spatial Tools	P.D. Sreekanth, R. Venkattakumar, A. Dhandapani, P. Venkatesan, Manoranjan Kumar, V. Subramanian	11/24	03/27
27.	AGED/NAARM/ SIL/2023/003/00191	Indian Food Grain Production towards 2050: Implications for Self Sufficiency, Food Security & SDGs	M.B. Dastagiri, V. Ramasubramanian, D. Damodar Reddy, M. Ramesh Naik, H. Umesh	10/23	05/25
28.	AGED/NAARM/ SIL/2020/009/00175	Dynamics of Agriculture and Food Systems in Different Agroecosystem of India	M. Ramesh Naik, H. Umesh, N. Sivaramane, B.S. Yashavanth, N. Subash, M. Mohanty, Nishant K. Sinha, V. Ramasubramanian, D. Damodar Reddy, Ch. Srinivasa Rao	01/22	12/25
29.	AGED/NAARM/ SIL/2024/017/00210	Profiling of Agricultural Technology Commercialization Processes in National and International R&D Organizations	Umesh Hudedamani, D. Damodar Reddy, K. Srinivas, Vikram Singh, S. Siva Kumar, V. Ramasubramanian	11/24	09/27
30.	AGED/NAARM/ SIL/2024/018/00211	Evaluation of Research-for-Development (R4D) in Indian Dairy Sector	S. Siva Kumar, H. Umesh, Navneet Kumar, Sona Yengkokpam, S.M. Feroze, M. Ramesh Naik, V. Ramasubramanian	11/24	10/27

Extra-mural

S. No.	PME Code	Project Title	Team	Budget (In Lakhs)	Sponsored by	Duration	
						From	To
1.	AGED/NAARM/ COL/2024/ 001/00194	National S&T survey on resources devoted to R&D institutions in Agriculture	B. Ganesh Kumar, V. Ramasubramanian, Kanyini, M. Balakrishnan, Sona Yengkokpam	21.43	NSTMIS, DST	07/24	06/25
2.	AGED/NAARM/ COP/2017/ 001/00121	Management and Impact Assessment of Farmer FIRST Project	P. Venkatesan, N. Sivaramane, Mahantesh Shirur	114.05	ICAR	03/17	03/26

S. No.	PME Code	Project Title	Team	Budget (In Lakhs)	Sponsored by	Duration	
						From	To
3.	AGED/NAARM/ COL/2023/ 001/00189	Agripreneurship for Sustainable Agricultural Development: Technological and Institutional Innovations and Strategies	P. Venkatesan, R. Venkattakumar, N. Sivaramane, S. Senthil Vinayagam	150.63	NASF, ICAR	06/23	05/26
4.	AGED/NAARM/ SOL/2022/ 008/00188	Distance Education Courses on Extension Management for Agricultural Professionals	Sandhya Shenoy		ICAR	01/22	01/25
5.	AGED/NAARM/ SOL/2015/ 007/00110	KRISHI- ICAR Research Data Repository for Knowledge Management	A. Dhandapani, S.K. Soam, H. Umesh, P. Supriya, Navneet Kumar	76.00	ICAR	11/15	03/26
6.	AGED/NAARM/ COL/2023/ 005/00193	Identification and characterization of superior haplotypes for agronomically important traits in Rice (<i>Oryza sativa</i> L.)	P. Supriya, S.K. Mangrauthia, M. Balakrishnan, R.M. Sundaram	20.52	DST	09/23	08/26
7.	AGED/NAARM/ SOL/2016/ 002/00112	Establishment of Agri-Business Incubation (ABI) Centers under XII Plan Scheme for NAIF	S. Senthil Vinayagam, H. Umesh, N.A.V. Avinashilingam, B.S. Yashavanth	79.15	NAIF Scheme	01/16	12/23
8.	AGED/NAARM/ COL/2023/ 004/00192	Development of Agripreneurship and Agri-business models for Grassroots Innovations (GRIs)	S. Senthil Vinayagam, N.A.V. Avinashilingam, H. Umesh, Sanjiv Kumar	166.72	O/o PSA to GOI, New Delhi	10/23	09/26
9.	AGED/NAARM/ SOL/2008/ 003/00067	IP Management and Transfer/ Commercialization of Agricultural Technologies Scheme	H. Umesh, N. Sivaramane, M. Ramesh Naik, Sanjiv Kumar, S. Sivakumar	19.15	ICAR, New Delhi	04/14	03/25

BUDGET UTILIZATION DURING JAN-DEC 2024

Budget Allocation and Expenditure (Rs. in Lakhs)

Sl. No.	Head of Account	Budget	Expenditure
A. Grant-in-Aid Capital			
I.	Works	1389.11	1167.65
II.	Equipments	150.80	58.83
III.	Information Technology	65.17	25.20
IV.	Library Books & Journals	26.43	31.22
V.	Vehicles & Vessels	3.11	3.11
VI.	Furniture & Fixtures	78.02	4.61
Total Capital		1712.64	1290.62
Grant-in-Aid Salaries (REVENUE)			
I.	Establishment Charges	2915.12	2872.71
II.	Wages	0.00	0.00
III.	OTA	0.01	0.01
Total Salaries		2915.13	2872.72
B. Grant-in-Aid General (REVENUE)			
I.	Pension & Other Retirement Benefits	6087.87	6087.87
II.	Domestic TA / Transfer TA	73.54	64.32
III.	Research & Operational Expenses	307.99	307.99
IV.	Administrative Expenses	1060.00	1154.22
V.	HRD	22.74	14.37
VI.	Miscellaneous Expenses	206.22	122.47
Total Grant-in-Aid General		7758.36	7751.24
GRAND TOTAL (Capital + Revenue)		12386.13	11914.58

REVENUE GENERATION DURING JAN-DEC 2024

Sponsored Projects Budget (Rs. in Lakhs)

Sl. No.	Name of the Project	Funds received	Expenditure
I.	GIZ Consultancy Project	29.17	4.74
II.	Development of MOOC Course, NDRI Karnal	23.70	6.27
III.	Identification & Characterization of superior haplotypes for agronomically important traits in Rice under DST-SERB	3.50	2.29
IV.	National S&T Survey on Resources devoted to R&D Institutions in Agriculture under DST-NSTMIS	17.14	1.95
	Total	73.51	15.25

Off-campus and Sponsored Programs (Rs. in Lakhs)

Sl. No.	Name of the Program	Budget received
I.	FLAME for the staff of M/s. Coromandel International Ltd. (Dec. 4-8, 2023)	7.06
II.	Induction training on Project Management and Research Methodology for the newly recruited Scientists of ICFRE, Dehradun	13.12
III.	Training on Entrepreneurial Skill Development for the III yr B.Tech (Dairy Technology) students of ICAR-NDRI, Karnal (Jun. 5 to Jul. 2, 2024)	15.00
IV.	Training new Scientists of Central Agricultural University, Jhansi (Sept. 8-11, 2024)	4.46
V.	MORE Batch 1 for M/s. Coromandel International Ltd. (Jun. 3-7, 2024)	5.74
VI.	MORE Batch 2 for M/s. Coromandel International Ltd. (Jun. 10-14, 2024)	8.40
VII.	NEAT Batch 1 for M/s. Coromandel International Ltd. (Jun. 17-21, 2024)	9.61
	Total	63.39

Educational Programs (Rs. in Lakhs)

Sl.No.	Agribusiness Management Batches	Receipts
I.	PGDM (ABM) 2022-24	0.50
II.	PGDM (ABM) 2023-25	251.20
III.	PGDM (ABM) 2024-26	223.85
	Total	475.55

Other Activities (Rs. in Lakhs)

Sl. No.	Name of Other Activities	Receipts
I.	Receipts from Farm Produce	4.08
II.	Receipts from Guest House and Quarters	132.95
III.	Receipts from Institute Training Programmes	75.67
IV.	Miscellaneous Receipts	32.15
	Total	244.85

Net receipts from Sponsored Research Projects (Rs. in Lakhs)

Sl. No.	Name of the Sponsored Research Project	Receipts
I.	NASF, New Delhi - Entrepreneurship Development through Farmer Led Innovations	1.43
II.	Identification & Characterization of superior haplotypes for agronomically important traits in Rice under DST-SERB	0.51
III.	RuTAG 2.0 Project	1.39
IV.	Net Receipts from Educational Programs	304.13
V.	Net Receipts from IRGS Programs/Schemes	17.91
	Total	325.37

Plan Schemes (Rs. in Lakhs)

Sl. No.	Name of the Scheme	Fund/Budget	Expenditure
I.	National Agricultural Innovation Fund (Component - I) ZTMC	9.50	6.19
II.	National Agricultural Innovation Fund (Component - II) ABI	10.15	5.99
III.	XII Plan EFC Main Scheme KRISHI Project	13.29	10.79
IV.	Management & Impact Assessment of Farmer First Project	12.27	5.65
V.	NASF, New Delhi - Entrepreneurship Development through Farmer Led Innovations	16.41	5.77
VI.	Emeritus Scientist Scheme	6.35	4.4
VII.	NAHEP	90.80	41.74
	Total	158.77	80.53

- Shri P. Prasad, Hon'ble Minister of Agriculture, Govt. of Kerala
- Dr. S. Ayyappan, Former Secretary, DARE & DG, ICAR, New Delhi
- Dr. Arvind Kumar, Former-VC, RLBCAU, Jhansi
- Dr Sanjay Kumar, Chairman, Agricultural Scientists Recruitment Board, New Delhi
- Dr. A.K. Singh, Former VC, Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior
- Shri Mohan Kumar Mishra, Former Secretary, National Council for Cooperative Training, New Delhi
- Dr. Chintala Govindarajulu, Former Chairman, NABARD, Mumbai
- Dr. Adeela Abdulla, IAS, Director of Agriculture, Kerala
- Dr. P. Sateesh Kumar, Executive Director, M/s. Nuziveedu Seeds Limited (NSL), Hyderabad
- Dr. V.K. Unni, Professor, Public Policy Management, IIM, Calcutta
- Dr. P. Rajasekharan, Chairman, Kerala State Agricultural Prices Board
- Dr. Umakanth Dash, Director, Institute of Rural Management Anand, Gujarat
- Dr. Gopala Krishna, Vice President, Nagarjuna Fertilizers Pvt. Ltd.
- Dr. Ajay Kumar Sood, Deputy Managing Director, NABARD, Mumbai
- Smt. Susheela Chintala, CGM, TSRO, NABARD, Telangana
- Shri M.R. Gopal, CGM, APRO, NABARD, Andhra Pradesh
- Mr. K.C. Choudhury, Assistant Director Gr.I and Nodal Officer (BI) at MSME-DFO Hyderabad

National Startup Day

The National Startup Day was celebrated by the Academy on Jan. 16, 2024 in hybrid mode to honour the Indian Startup Ecosystem, as announced by the Government of India. Mr. N.V. Ramana, an Agribusiness Expert, was the keynote speaker who delivered a talk on 'Emerging Trends & Innovations in Agri & Allied Sectors' emphasizing futuristic agriculture and the vital role of youth in embracing startups to benefit farmers. Sixty participants from India and abroad including Lesotho, Kenya, South Africa, Zambia and Nigeria joined the program,



Fig. 17.1. Participants of National Startup Day (Jan. 16, 2024)

sharing innovative ideas and advancements in agriculture. The program was coordinated by Dr. N.A. Vijay Avinashilingam, Principal Scientist.

Second Graduation Ceremony of Agri-Startups

The 2nd Graduation Ceremony of Agri-Startups was organized on Jan. 23, 2024 celebrating the achievements of 43 startups from 13 states across 12 domains, including AgTech and Precision Farming. These startups participated in programs like FPO Immersion and Cohort Networking, which saw six startups advancing to the acceleration phase. In this program, Shri Govinda Rajulu Chintala, Former Chairman, NABARD, emphasized cultivating independence over reliance on external funding and encouraged them for contributing towards national growth. Dr. Ch. Srinivasa Rao, President of a-IDEA and Director of ICAR-NAARM, thanked NABARD for its support in fostering the agricultural startup ecosystem. Shri Raymond B. D'Souza, CGM of OFDD, NABARD, highlighted the importance of market-ready products. Dr. Ravi Shankar, Co-Founder of Gram Paathshala, praised a-IDEA's PPP model and urged youth to focus on job creation while keeping small farmers in mind. The event was attended by 200 participants, including bankers, investors, incubation heads, students, and officials.



Fig. 17.2. 2nd Graduation Ceremony of Agri-Startups (Jan. 23, 2024)

National Science Day

The Academy celebrated 'National Science Day (NSD)' in collaboration with National Academy of Agricultural Sciences (NAAS)-Hyderabad Chapter on Feb. 28, 2024. The theme of this year NSD was 'Indigenous Technology for Viksit Bharat'. On the occasion of NSD 2024, the Academy, organized essay and quiz competitions for the middle grade students of Pallavi International School, Attapur, Hyderabad on the theme 'Science for Sustainable Future'. Smt. Annapurna Koduru, Principal, Pallavi International School, in her speech appreciated and thanked the efforts of ICAR-NAARM for selecting their school for NSD celebrations and motivating the students through competitions. As part of the celebration, the students were taken around for a campus tour, which



Fig. 17.3. One participant receiving award during National Science Day Celebration (Feb. 28, 2024)

included a visit to nursery, herbal garden and bio-diversity park of the Academy, as it was an occasion to nurture and seed scientific attitude among them. Dr. R. Venkattakumar, Head, XSM explained the significance of NSD. Smt. Aneeja, Chief Technical Officer explained the students about activities of ICAR-NAARM.



Fig. 17.4. Participants of National Science Day celebration at NAARM, Hyderabad (Feb. 28, 2024)

Seminar on 'Intellectual Property Management'

A seminar on 'Intellectual Property Management' was organized by the Academy as a part of World Intellectual Property Day celebration on April 26, 2024 under the theme 'IP and the SDGs: Constructing our shared future through innovation and creativity'. The keynote address was delivered by Dr.

P. Lakshmi Santhi, Senior Manager, SciTech Patent Art Services Pvt. Ltd., Hyderabad, who highlighted the importance of intellectual property management. She also discussed on various aspects pertaining to IP and the SDGs, focusing on fostering innovation and creativity for a collective future. Dr. G. Venkateshwarlu, Joint Director, highlighted the pivotal role of innovation in burgeoning economies like India and stressed the significance of IP filings in

catalysing innovations. The seminar garnered over 60 participants from the Academy. Dr. Umesh Hudedamani, Scientist & Director

of the program explained the significance of observing World IP Day.



Fig. 17.5. World Intellectual Property Day (Apr. 26, 2024)

Workshop on Official Language Implementation

The Academy organized a workshop on “Practical Compliance of Official Language Implementation” mainly involving participants from Southern Region of India during May 28-29, 2024. The workshop discussed the importance of languages in transfer of technology and approaches for practical implementation of



Fig. 17.6. Workshop on Practical Compliance of Official Language Implementation (May 28-29, 2024)

Hindi in scientific institutions. The revised proforma of the parliamentary questionnaire was also discussed for its proper compliance. The workshop emphasized and recommended that the abstracts of scientific papers be presented/ published in Hindi as well. A total of 28 Hindi Officers, Technical Officers and Scientists participated in the workshop, which was coordinated by Dr. J. Renuka, Joint Director (Official Language).

Review Meeting of ITMUs of Agricultural Division

A review meeting of ZTMUs/ITMUs in the agricultural education division under the NAIF Scheme was held online on Jul. 30, 2024. This meeting was organized by ZTMC, ICAR-NAARM, Hyderabad, and witnessed participation of ITMUs from IASRI, New Delhi; NIAP, New Delhi; CIWA, Bhubaneswar and ITMU, NAARM. The meeting reviewed activities of the innovation and incubation fund of NAIF in the concerned institutes, including budget utilization and outcomes and to sensitize the participants about IPR and related issues. The program was organized by Dr. Shiv Datt, Principal Scientist, Dr. Vikram Singh, Sr. Scientist from the IP&TM Unit and Dr. Umesh Hudedamani, Scientist from ICAR-NAARM, Hyderabad.

Hindi Pakhwada Celebration

The Academy celebrated Hindi Pakhwada (fortnight) during Sept. 14-27, 2024. This year, various competitions were organized such as noting and drafting, essay writing, just-a-minute, elocution and singing for Officers, Staff and PGDM-ABM students. A competition on presentation of scientific papers in Hindi was organized for Scientists. The celebration culminated with the distribution of certificates and cash awards for the winners of various



Fig. 17.7. Celebration of Hindi Pakhwada (Sept. 14-27, 2024)

competitions. The Hindi version of the Annual Report 2023 was released by Dr. Ch. Srinivasa Rao, Director during the valedictory session of the program. The program was coordinated by Dr. J. Renuka, Joint Director (Official Language).

Swachhata Hi Seva Campaign

The Academy organized Swachhata Hi Seva Campaign during second fortnight of Sept., 2024 with an array of awareness and shramdaan activities covering focal points of *Swabhav Swachhata - Sanskaar Swachhata*

and *Zimmedari - Bhaagidari*. The action plan of Swachhata Hi Seva-2024 entailed a series of daily programs including display of banners in the campus, administering Swachhta pledge to staff, involvement of public representatives, *ek ped maa ke naam*, zero-waste orchard management, swachhata walkathon, engagement with school children by arranging drawing competition, *shram daan* for in-house cleaning, creative transformation of waste to art (on-farm farmers' lounge), human chains to solicit commitment for cleanliness at places of worship/ schools, identification and transformation of cleanliness target units or CTUs (black spots), *safai mitra suraksha shivir* (health camp, distribution of protective gears, social welfare schemes for sanitation workers) and *safai mitra samman*. The fortnight-long campaign culminated with the observance and celebration of *Swachh Bharat Diwas* on 2nd Oct., 2024 and felicitation of *safai mitras*.



Display of Banners on SHS



Swachhata Pledge



Ek Ped Maa Ke Naam



Swachhata Run (Walkathon)



Creative Transformation of Waste



Farm Lounge Made of Farm Waste



Drawing Competition on SHS



Human Chain on Reduce-Reuse-Recycle



CTU (Farm Pond)



Shramdaan



Safai Mitra Suraksha Sivr



Felicitation of Safai Mitras on SBD

Swachhta Special Campaign 4.0

A Special Campaign 4.0 was organized for Institutionalizing Swachhta and Minimizing Pendency in Government Offices during Oct. 2-31, 2024. As part of the Campaign, cleanliness activities were undertaken across the campus covering 27 sites. A successful attempt was made to identify and collect scrap and junk materials from around the campus, which were sold through online bidding process generating a revenue of ₹ 5,25,272/- and cleared an approx. area of 5,000 sq.ft. The program was coordinated by Dr. M. Balakrishnan.



Fig. 17.8. Swachhta Special Campaign 4.0 (Oct. 2-31, 2024)

Deeksharambh program

Under this program, 24 undergraduate and postgraduate students of IARI Mega Hub Hyderabad visited the Academy on Oct. 24, 2024 to gather knowledge on agricultural education and research in India and discussed the transformative impact of modern farming

techniques, including precision farming and digital tools. As part of the program, students visited key facilities such as a-IDEA and got exposed to innovation and entrepreneurship opportunities in agriculture. They also toured MOOC lab and the Agribusiness Management division. The program aimed to familiarize students with the Academy's operations, research and the evolving agricultural landscape. Dr. D. Damodar Reddy, Principal Scientist and Dr. M. Ramesh Naik, Scientist, were the Coordinators of the Program.

Communal Harmony Week

The Academy observed Communal Harmony Campaign Week 2024 during Nov. 19-25, 2024. As part of the same, Display of Banners, Cash Collection Boxes placed at Canteen Building and Halls of Residence was done. As part of the same, an Essay Writing Competition in English for Staff members on "Strategies for Inculcating Communal Harmony and National Integration among Youth", A Debate Competition was also conducted for PGDM-ABM students on the topic 'Communal Harmony in 21st Century: A myth or reality'? In addition, banners on Communal Harmony were placed at prime locations of the Institute during the week. Flag Day was observed Nov. 25, 2024 and the monetary contribution made by the staff of the Institute towards the distribution of Flags was sent to National Foundation for Communal Harmony, New Delhi.

Awareness program on 'Prevention of Sexual Harassment of Women at Workplace'

As a part of the 'Sexual Harassment of Women at Workplace Prevention Week', an awareness program was organized on Dec. 9, 2024 on 'Sexual Harassment of Women at Workplace' (Prevention, Prohibition and Redressal) Act, 2013. Dr. B. Nirmala, Chairperson, Internal Complaints Committee (ICC) and Nodal Officer, Prevention of Sexual Harassment of Women at Workplace (POSH) at the Academy sensitized the women employees as to what needs to be done for registering complaints regarding any acts of sexual harassment that they encountered and elaborated upon the provisions of POSH Act, 2013. This Act is applicable not only in government offices but also in private/

unorganized and organized establishments as well. Also, the participants were appraised about the SHe-Box portal of Government of India for registering the complaints. Dr. Ramasubramanian V., Joint Director (In-charge) emphasized on the need to provide a safe and dignified workplace for women free from all forms of harassment and intimidation. He suggested that an awareness lecture, preferably in local language Telugu, can be organized on forthcoming Women's Day (March 08), among other events, so that the employees are made aware of their rights. Shri Vivek Purwar, Head of Administration & Registrar elucidated the punishments for the offenders and mentioned that such offences are treated as misconduct as per service rules. Dr. B. Nirmala, Principal Scientist, was the Director/ Coordinator of the Program.



Fig. 17.9. Awareness program on 'Prevention of Sexual Harassment of Women at Workplace' (Dec. 9, 2024)

Swachhta Pakhwada Celebration

Swachhta Pakhwada was observed during Dec. 16-31, 2024 at the Academy. A series of activities were organized to create awareness, promote cleanliness, and encourage sustainable practices among staff, students, and the community. The event witnessed enthusiastic participation from various stakeholders, including faculty members, outsourcing workers, and residents. The activities included

the Swachhta Pledge taken by all staff, a Swachhta Awareness Day featuring training and demonstrations on converting leaf litter into vermicompost, and cleanliness maintenance through eOffice promotion and digitization. A Swachhta-themed selfie booth was set up at the Faculty Center lawns, while a Green Drive at the Farmers' Hostel saw the planting of 1,267 ornamental and foliage plants. Community outreach activities such as Swachhta pledge



Fig. 17.10. Observation of Swachhta Pakhwada (Dec. 16-31, 2024)

by residents at Bandlaguda Jagir, and Kisan Diwas celebrations at Agraharam Potlapally village, where seven innovative farmers were felicitated and herbal plants were distributed, were also conducted. Signature campaigns on cleanliness were held in Rajendranagar, and a Swachhta-themed drawing competition was organized at Adarsha Vidyalaya High School. Posters on wastewater recycling were distributed, and public cleaning was undertaken at the area opposite Venkateswara Swamy Temple. Organic waste management was promoted using lawn grass clippings as mulch and vermicompost in the floriculture block. Publicity efforts ensured that many activities were covered in print media. A Swachhta Run (walkathon) was conducted at the Academy, culminating in the concluding program. It successfully engaged employees, students, and the community in promoting cleanliness and environmental sustainability. The activities emphasized hygiene, waste management, and sustainable practices, reinforcing the commitment towards a cleaner and greener India. The initiative received positive feedback and extensive participation, contributing to the overall success of the Swachhta movement.

Kisan Diwas

The Academy celebrated 'Kisan Diwas', as part of the Swachhata Pakhwada on Dec. 23, 2024 at Agraharam Potlapalli village of Mahbubnagar district, Telangana. The theme of the celebration was "Empowering Farmers for Sustainable Agriculture". About 100 farmers and farm-women benefitted through this programme. Seven progressive farmers/ farm women were felicitated on the occasion, for their innovative methods of practicing agriculture and allied activities and being model for fellow farmers of the village. Dr. P. Venkatesan, Principal Scientist, explained the farmers about the significance of "Kisan Diwas", while Dr. Sandhya Shenoy, Emeritus Scientist, ICAR-NAARM explained the farm women about strategies for empowerment through agriculture and allied activities. Dr. R. Venkattakumar, Head, XSM suggested the farmers to diversity the farm activities for achieving sustainability. Planting materials of medicinal plants (200 no.), such as multivitamin plant (*Sauropus androgynous*) and insulin plant (*Costus igneus*) multiplied by at our own nursery were distributed to the farmers. The program was coordinated by Dr. Laxman M. Ahire and Mr. Shekhar Reddy.



Fig. 17.11. Kisan Diwas Celebration at Agraharam Potlapalli (Dec. 23, 2024)



Fig. 17.12. Farmers attending Kisan Diwas Celebration at Agraharam Potlapalli (Dec. 23, 2024)

Human resources (as on 31.12.2024)

Category	Sanctioned strength	In position	Abolished	Vacant
Scientific – RMP	02	00	00	02
Scientific – Faculty	60	36	00	24
Technical	42	22	00	20
Administrative (including legal)	44	29	00	15
Multi-tasking Staff (regular)	33	16	00	17
Multi-tasking Staff	18	14	04*	00
Total	199	117	04	78

*These posts got abolished as and when vacated, as these are additionally sanctioned posts as a one-time measure for regularization of CLTS

Research Management Positions

Dr. Ch. Srinivasa Rao, Director (till 26.12.24)

Dr. G. Venkateswarlu, Joint Director (till 31.05.2024)

Dr. Ramasubramanian, V., Joint Director (Acting Charge) (since 09.08.2024)

Scientific Staff

S.No.	Name & Designation
Agribusiness Management Division	
1.	Dr. Tavva Srinivas, Principal Scientist & Head
2.	Dr. Ranjit Kumar, Principal Scientist
3.	Dr. Prem Chand Meena, Principal Scientist
4.	Dr. N. Sivaramane, Principal Scientist
5.	Dr. B. Nirmala, Principal Scientist
6.	Dr. Sheikh Mohammad Feroze, Principal Scientist
7.	Dr. Sanjiv Kumar, Senior Scientist
Education Systems Management Division	
8.	Dr. S. Senthil Vinayagam, Principal Scientist & Head
9.	Dr. Dhumantarao Thammi Raju, Principal Scientist
10.	Dr. G. R. Ramakrishna Murthy, Principal Scientist
11.	Dr. V. V. Sumanth Kumar, Principal Scientist
12.	Dr. K. N. Yamini, Principal Scientist
13.	Dr. Dipesh Debnath, Principal Scientist
14.	Dr. Navneet Kumar, Principal Scientist

S.No.	Name & Designation
Human Resource Management Division	
15.	Dr. B. Ganesh Kumar, Principal Scientist & Head
16.	Dr. K. Srinivas, Principal Scientist
17.	Dr. Sona Yengkokpam, Principal Scientist
Extension Systems Management Division	
18.	Dr. R. Venkattakumar, Principal Scientist & Head
19.	Dr. Surya Rathode, Principal Scientist
20.	Dr. P. Venkatesan, Principal Scientist
21.	Dr. N.A. Vijay Avinashilingam, Principal Scientist
22.	Dr. Mahantesh Shirur, Principal Scientist
Information and Communication Management Division	
23.	Dr. A. Dhandapani, Principal Scientist & Head
24.	Dr. S.K. Soam, Principal Scientist
25.	Dr. M. Balakrishnan, Principal Scientist
26.	Dr. P.D. Sreekanth, Principal Scientist
27.	Dr. N. Srinivasa Rao, Principal Scientist
28.	Dr. B.S. Yashavanth, Scientist
29.	Dr. P. Supriya, Scientist

S.No.	Name & Designation
Research Systems Management Division	
30.	Dr. Ramasubramanian, V., Principal Scientist & Head
31.	Dr. D. Damodar Reddy, Principal Scientist
32.	Dr. M.B. Dastagiri, Principal Scientist
33.	Dr. P. Krishnan, Principal Scientist (on deputation since 29.12.2021)
34.	Dr. Siva Kumar, S., Principal Scientist
35.	Dr. Umesh Hudedamani, Scientist
36.	Dr. M. Ramesh Naik, Scientist

Administrative Staff

S.No.	Name & Designation
1.	Shri Vivek Purwar, Chief Administrative Office (Senior Grade)/ HoAR
2.	Shri Z.H. Khilji, Comptroller
3.	Dr. J. Renuka, Joint Director (Official Language)
4.	Shri Mukul Rajsingh, Senior Administrative Officer
5.	Shri B.P.R. Vittal, Principal Private Secretary
6.	Smt. A. Mercy, Principal Private Secretary
7.	Shri M. Naga Venkateswara Rao, Finance & Accounts Officer
8.	Smt. K.K. Rukmani Ammal, Assistant Administrative Officer
9.	Dr. K.R. Ghanshyam, Assistant Administrative Officer
10.	Shri T. Srinivas, Assistant Administrative Officer
11.	Shri G. Raj Reddy, Assistant Administrative Officer
12.	Smt. N. Vijayalakshmi, Assistant Finance & Accounts Officer
13.	Smt. T. Vanisri, Private Secretary
14.	Smt. S. Shanthi, Private Secretary
15.	Smt. Y. Anuradha, Private Secretary
16.	Smt. S. Sessa Sai, Private Secretary
17.	Smt. B. Padma Saroja, Assistant
18.	Shri R. Chandra Babu, Assistant
19.	Shri M. Sridhar, Assistant
20.	Smt. Y. Gayathri, Assistant
21.	Shri K. Suryanarayana, Assistant
22.	Shri M. K. Samson, Assistant
23.	Shri Krishna Chaitanya Vangara, Assistant (since 18.09.2024)

S.No.	Name & Designation
24.	Ms. Nabanita Sarkar, Assistant (since 10.09.2024)
25.	Shri Kapil Gaur, Assistant (since 02.09.2024)
26.	Ms. Shivangi Singh, Assistant (since 02.09.2024)
27.	Shri Dhikollu Upali, Assistant (since 11.10.2024)
28.	Smt. V. Shailaja, Personal Assistant
29.	Smt. Rajashri S. Bokde, Upper Division Clerk

Technical Staff

S.No.	Name & Designation
1.	Shri P. Namdev, Chief Technical Officer
2.	Dr. P. Vijender Reddy, Chief Technical Officer
3.	Shri Sohail Ahmad Khan, Chief Technical Officer
4.	Smt. G. Aneja, Chief Technical Officer
5.	Shri M. Shekhar Reddy, Chief Technical Officer
6.	Dr. Laxman M. Ahire, Chief Technical Officer (since 23.09.2022)
7.	Shri M. Ravi, Chief Technical Officer (since 11.12.2022)
8.	Shri P. Mohan Singh, Assistant Chief Technical Officer
9.	Smt. Savithri Murali, Assistant Chief Technical Officer
10.	Dr. Kirttiranjan Baral, Senior Technical Officer
11.	Shri K. Shivaiah, Technical Officer
12.	Shri S. Rajukumar, Technical Officer
13.	Shri N. Prabhakar Rao, Technical Officer
14.	Shri M. Srinivasa Rao, Technical Officer
15.	Shri P. Srinivas, Sr. Technical Assistant
16.	Shri B. K. Venkatram, Sr. Technical Assistant
17.	Shri R. Siva Prasad, Sr. Technical Assistant
18.	Shri Danam Murahari, Sr. Technical Assistant (since 21.06.2023)
19.	Shri Dharmendra Pratap Singh Gautam, Technician (T-1) (since 25.04.2024)
20.	Shri Sabbavarapu V.S. Narayana, Technician (T-1) (since 15.05.2024)
21.	Shri Venkateshwar Pandey, Technician (T-1) (since 24.05.2024)
22.	Shri Ankush Kumar, Technician (T-1) (since 05.06.2024)

Multi-tasking Staff

S.No.	Name
1.	Shri K. Pentaiah
2.	Shri P. Yadaiah
3.	Shri L. Satyanarayana
4.	Smt. C. Kausalya
5.	Shri Chilumula Venkatesham
6.	Shri Kyasam Satyanarayana
7.	Shri B. Premdas
8.	Smt. V. Saroja
9.	Shri S. Nayab Rasool
10.	Shri B. Ashok
11.	Shri K. Daniel
12.	Shri D. Srisailam
13.	Shri B. Ratnaiah
14.	Shri D. Babaiah
15.	Shri K. Samson
16.	Shri K. Narasimha
17.	Smt. K. Lakshmi
18.	Smt. L. Chandramma
19.	Shri B. Raju
20.	Shri MD Sadath
21.	Shri T. Srinivas
22.	Smt. E. Yadamma
23.	Smt. S. Lakshamma
24.	Shri P. Venkatesh
25.	Shri G. Ramulu S/o. G. Laxmaiah
26.	Shri P. Yadaiah/K
27.	Smt. R. Pramila
28.	Shri G. Ramulu S/o. G. Danaiah
29.	Smt. M. Sharada
30.	Shri N. Yadagiri

Retirements

S.No.	Name, Designation & Date
1.	Dr. G. Venkateshwarlu, Joint Director (31.05.2024)
2.	Dr. P. Ramesh, Principal Scientist (30.04.2024)
3.	Dr. S. K. Soam, Principal Scientist (31.12.2024)
4.	Shri P. Swamy, Upper Division Clerk (31.03.2024)
5.	Shri M. Narsing Rao, Upper Division Clerk (30.09.2024)
6.	Smt. S. Shakuntala, Multi-tasking Staff (30.11.2024)
7.	Shri G. Pentaiah Multi-tasking Staff (30.09.2024)
8.	Shri R. Sattaiah Multi-tasking Staff (31.05.2024)
9.	Shri M. Ganesh Kumar Multi-tasking Staff (30.06.2024)
10.	Shri Kukkala Satnarayana Multi-tasking Staff (31.03.2024)
11.	Smt. K. Andalu Multi-tasking Staff (31.08.2024)
12.	Shri N. Krishna Multi-tasking Staff (30.06.2024)
13.	Shri Sham Bahadur, Asst. Chief Technical Officer (31.01.2024)
14.	Shri C.J. Samuel, Asst. Admin. Officer (06.02.2024 (FN))
15.	Smt. M. Laxamma, Multi-tasking Staff (29.02.2024)

Transfers from ICAR-NAARM:

- Dr. Ch. Srinivasa Rao, Director relieved from ICAR-NAARM on 26.12.2024(AN) to join as Director-cum-Vice Chancellor in ICAR-IARI, New Delhi

New appointments/ joining

S. No.	Name	Post	Date of joining
1.	Dr. Mahantesh Shirur	Principal Scientist in Extension Policy and Management	10.01.2024
2.	Dr. B. Nirmala	Principal Scientist in Agricultural Business Policy	22.01.2024 (AN)
3.	Dr. K.N. Yamini	Principal Scientist in Education Management	26.02.2024
4.	Dr. Dipesh Debnath	Principal Scientist in Education Management	01.03.2024
5.	Dr. Sheikh Momammad Feroze	Principal Scientist in Agricultural Economics	14.03.2024

S. No.	Name	Post	Date of joining
6.	Dr. Sona Yengkokpam	Principal Scientist in Human Resource Management	18.03.2024
7.	Dr. Siva Kumar S.	Principal Scientist in Research Policy and Governance	27.03.2024
8.	Dr. Navneet Kumar	Principal Scientist in Education Technology	16.04.2024
9.	Shri Kapil Gaur	Assistant	02.09.2024
10.	Ms. Shivangi Singh	Assistant	02.09.2024
11.	Ms. Nabanita Sarkar	Assistant	10.09.2024
12.	Shri Krishna Chaitanya Vangara	Assistant	18.09.2024
13.	Shri Dhikollu Upali	Assistant	11.10.2024
14.	Shri Dharmendra Pratap Singh Gautam	Technician (T-1)	25.04.2024
15.	Shri Sabbavarapu V.S. Narayana	Technician (T-1)	15.05.2024
16.	Shri Venkateshwar Pandey	Technician (T-1)	24.05.2024
17.	Shri Ankush Kumar	Technician (T-1)	05.06.2024

Promotions

S.No.	Name, Promoted to, w.e.f.
1.	Dr. Laxman M. Ahire, Chief Technical Officer, w.e.f. 23.09.2022
2.	Shri M. Ravi, Chief Technical Officer, w.e.f. 11.12.2022
3.	Shri Danam Murahari, Sr. Technical Assistant, w.e.f. 21.06.2023
4.	Shri Sham Bahadur, Chief Technical Officer, w.e.f. 29.06.2023
5.	Shri G. Raj Reddy, Asst. Administrative Officer, w.e.f. 12.03.2024 (AN)
6.	Dr. P. Vijender Reddy, CTO has been granted one advance increment, w.e.f. 10.08.2022
7.	Dr. Sanjiv Kumar has been promoted to next higher grade of Sr. Scientist, w.e.f. 02.07.2020
8.	Dr.P.Namdev, CTO has been granted one advance increment w.e.f. 27.07.2022
9.	Dr. (Mrs.) Sona Yengkokpam promoted under CAS as Principal Scientist, w.e.f. 07.01.2023
10.	Dr. (Mrs.) B. Nirmala promoted under CAS as Principal Scientist, w.e.f. 08.01.2023
11.	Dr. Dipesh Debnath promoted under CAS as Principal Scientist, w.e.f. 26.02.2023

Financial Upgradations under MACPS

S. No.	Name	Designation	1 st /2 nd /3 rd Financial Upgradation with GP/Level of Pay	Effective Date
1.	Smt. V. Saroja	Multi-tasking Staff	1 st Financial Upgradation with Level #2 (GP: Rs.1900/-)	25.06.2024
2.	Shri S.N. Rasool	Multi-tasking Staff	1 st Financial Upgradation with Level #2 (GP: Rs.1900/-)	25.06.2024
3.	Shri B. Premdas	Multi-tasking Staff	1 st Financial Upgradation with Level #2 (GP: Rs.1900/-)	25.06.2024
4.	Smt. S. Shanthi	Private Secretary	3 rd Financial Upgradation with Level #9 (GP: Rs.5400/-)	25.02.2023
5.	Smt. V. Shailaja	Personal Assistant	3 rd Financial Upgradation with Level #9 (GP: Rs.5400/-)	26.02.2023

LIST OF IMPORTANT COMMITTEES

VIII Quinquennial Review Team (QRT)

S.No.	Name & Designation	Role
1.	Dr. S. Ayyappan, Former Secretary, DARE & DG, ICAR	Chairman
2.	Dr. Anil Kumar Singh, Former Vice-Chancellor, Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior	Member
3.	Dr. T. Balaguru, Former Acting Director, ICAR-NAARM, Hyderabad	Member
4.	Dr. P. Sateesh Kumar, Executive Director, M/s. Nuziveedu Seeds Limited, Hyderabad	Member
5.	Dr. V.K. Unni, Professor, Public Policy Management, Indian Institute of Management, Calcutta	Member
6.	Dr. P.D. Kamala Jayanthi, ICAR National Professor, Division of Crop Protection, ICAR-IIHR, Bengaluru	Member
7.	Dr. U.S. Dash, Director, IRMA, Anand	Member
8.	Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM, Hyderabad	Member Secretary

IX Research Advisory Committee (RAC)

S.No.	Name & Designation	Role
1.	Dr. Chintala Govindarajulu, Former Chairman, NABARD	Chairman
2.	Dr. Arvind Kumar, Ex-VC, RLBCAU, Jhansi	Member
3.	Dr. P.K. Joshi, Former Director, ICAR-NAARM and Former Regional Director, IFPRI	Member
4.	Shri Mohan Kumar Mishra, Former Secretary, National Council for Cooperative Training	Member
5.	Dr. Gopala Krishna, Vice President, M/s. Nagarjuna Fertilizers Pvt. Ltd.	Member
6.	Dr. R.C. Agrawala, DDG (Edn.), ICAR, New Delhi	Member
7.	Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, Hyderabad	Member
8.	Sri V. Asha Kiran, Farmer Representative	Member
9.	Sri. M.M. Kumareshan, Farmer Representative	Member
10.	Dr. Tavva Srinivas, Head, ABM Division, ICAR-NAARM, Hyderabad	Member Secretary

Institute Management Committee (IMC)

S.No.	Name & Designation	Role
1.	Director, ICAR-NAARM, Hyderabad	Chairman
2.	Director of Agriculture, Department of Agriculture, Government of Telangana	Member
3.	Spl. Commissioner and Director, Department of Agriculture, Government of Andhra Pradesh	Member
4.	Director (Research), Prof. Jayashankar Telangana Agricultural University, Rajendranagar, Hyderabad	Member
5.	Dr. Sheikh N. Meera, Director, ICAR-ATARI, ZONE-X, Hyderabad	Member
6.	Dr. G. Byju, Director, ICAR-CTCRI, Thiruvananthapuram, Kerala	Member
7.	Dr. C.N. Ravishankar, Director, ICAR-CIFE, Mumbai	Member
8.	Dr. O.P. Yadav, Director, ICAR-CAZRI, Jodhpur	Member
9.	Dr. (Mrs) Seema Jaggi, ADG (HRD), ICAR, New Delhi	Member
10.	AS & FA, DARE/ICAR	Member
11.	Sri V. Asha Kiran, Farmer Representative	Member
12.	Sri. M.M. Kumareshan, Farmer Representative	Member
13.	CAO (SG), ICAR-NAARM, Hyderabad	Member Secretary





भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

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